

Wages, amenities and the value of a formal employment:

A structural approach for Mexico

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Abstract

Workers move frequently between formal and informal jobs, yet the motives behind these transitions remain debated. Using household survey data from Mexico, I document that most informal-to-formal transitions take place within the same firm, consistent with a foot-in-the-door route to formality, and that sustained formalisation raises hourly wages only modestly. The wage gain is concentrated among low-skilled workers, even though skilled workers formalise at higher rates. These limited wage gains suggest that workers accept formal employment for reasons beyond current wage. I estimate a two-sector search model in which informal workers receive formal offers both from outside firms and from their current employer, and decompose the value of formal employment into a sector-specific non-wage amenity, future job opportunities, and employment stability. The estimated formal-sector amenity is worth 29.4% of the median informal wage for skilled men and 15.2% for skilled women, is negative for low-skilled women, and is not statistically significant for low-skilled men. Future formal job opportunities also contribute substantially to acceptance of formality, the within-firm foot-in-the-door channel is worth about 16% of the median informal wage for skilled men and about 4% for low-skilled women and is quantitatively important for employment. In absence of this channel formal employment drops by roughly 8–14 percentage points for men and about 1–2 points for women while informality and non-employment increase.

JEL Codes: J24, J31, J46, J64, O17.

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1. Introduction

Across low- and middle-income economies, informal wage employment accounts for an average 56% of total wage employment (ILO, 2026). Its prevalence, is often viewed by policymakers as a development concern because informal jobs tend to pay less, lack social and labour protection, erode the tax base and it is assumed that workers would prefer a formal alternative if available. Yet, workers move frequently between formal and informal employment and the motives behind these transitions are still subject to debate, recent work emphasizes that exclusion from formal employment and worker choice may coexist to explain informal employment (Maloney et al., 2026). For example, in Mexico informal wage workers move into formal employment more frequently than formals move to informal employment, and most of these moves take place within the firm. Despite this, the wage gains from informal-to-formal transitions are modest. What makes formal employment valuable to workers if the wage motive is limited? The answer matters for policy: if workers value formal employment partly for its non-wage attributes, the attractiveness of formal jobs and mobility between jobs cannot be inferred from wage gains alone.

The literature interprets mobility with limited wage gains using the standard hedonic framework (Rosen, 1974; Rosen, 1986). In the framework, workers in an environment of perfect competition choose a job based on a combination of wage and non-wage attributes as if selecting from a menu and may forgo wage when accepting. This interpretation, faces two main challenges. First, jobs are bundles of attributes, some of which may only be learned after acceptance, so the relevant trade-off is between entire job packages rather than isolated characteristics (Sullivan and To, 2014). Second, search frictions and information asymmetries restrict the set of jobs available to workers, so observed mobility reflects both preferences and opportunity constraints (Burdett and Mortensen, 1998; Sorkin, 2018).¹

I use the revealed-preference principle from observed job transitions and a search friction model with non-wage attributes to infer the value of formal employment. I start by using the rotating-panel structure of Mexico's National Survey of Occupation and Employment (ENOE) to study transitions between informal and formal wage employment and the effect on wages and hours of work. The estimates show that informal workers move into formal employment more frequently than formal workers move into informality, and 77% of those transitions take place within the firm. This dominant within-firm margin suggests that informal employment can operate as a foot in the door to formality, firms may initially employ workers informally and later offer them formal status, while workers

¹In addition, evidence from the United States suggests that the standard wage-amenity trade-off need not hold throughout the wage distribution, as high-wage jobs may also offer high amenity values (Mas, 2025b).

decide whether to accept the formal offer. Separating within- from between-firm transitions allows me to distinguish changes of formal status from wage changes that coincide with movement to a different employer.

I use the within-firm margin to compare workers who formalise with observably similar workers who remain informal, while holding the employer-employer match fixed using the [Callaway and Sant'Anna, 2021](#) estimation strategy. The results suggest that hourly wages rise by approximately 5% on average, with the gain concentrated among low-skilled workers. By contrast, between-firm formalisation produces no statistically significant hourly-wage effect. This pattern is consistent with [Ulyssea 2018](#), who shows for Brazil that the formal-informal wage gap largely disappears after controlling for firm fixed effects. Monthly wages and hours begin rising before formalisation, however, suggesting that the transition may occur along a career scheme, employer-learning, or match-quality process. The reduced-form evidence suggests that wages alone are unlikely to explain why workers accept formal jobs, but it does not identify which non-wage attributes generate the remaining value.

I rationalize these facts by estimating a two-sector search model in partial equilibrium following [Burdett and Mortensen 1998](#) and [Bradley et al. 2017](#) and including offers from the same employer. The model allows me to recover the workers' willingness to accept a formal job and decompose it into four components: the formal-sector amenity, the formal option, the informal option, and job stability. In the setting, workers search both off and on the job and receive take-it-or-leave-it wage offers drawn from exogenous formal and informal wage-offer distributions. While employed, utility is linear and additive in wages and a sector-specific non-wage component to describe the wage and formal amenity trade-off. Following [Sullivan and To 2014](#), this component captures the combined value of tangible and intangible features of the employment relationship, such as social-security coverage, schedule flexibility, commuting conditions, and other non-wage attributes. Holding future job options and employment stability fixed, workers may accept a lower wage in exchange for a more valuable non-wage component.

The estimates suggest that workers' willingness to accept a formal job differs sharply by skill and gender, and that the underlying sources of value also differ across groups. For low-skilled workers, the formal-sector amenity is small or negative. In particular, low-skilled women place a negative value on the formal-sector amenity at the median wage, implying that the informal-sector amenity is valued at around 14% of the median informal wage. By contrast, the formal-sector amenity is positive and sizeable for high-skilled workers, amounting to 29% of the median informal wage for men and 15% for women. In all cases

its contribution becomes smaller as wages rise. Future job opportunities also matter. The formal job option is valuable for all groups, at around 10–17% of the median informal wage. At the same time, the within-firm foot-in-the-door channel raises the value of remaining informal because workers can wait for a future formal offer from their current employer; this option is worth around 4–16% of the median informal wage. Finally, the contribution of job stability changes over the wage distribution. At low wages, the estimated value of non-employment is high compared to employment in either sector, so the lower separation rate of formal jobs adds nothing to the value of formality for low wage earners and its contribution is negative. As wages rise, remaining employed becomes more valuable and the stability advantage of formal jobs becomes increasingly important. The counterfactual confirms the quantitative importance of the within-firm route: shutting down within-firm formal offers reduces formal employment by about 8–14 percentage points for men, compared with roughly 1–2 percentage points for women. The model therefore shows that workers’ willingness to formalise reflects not only the immediate wage change, but also the non-wage value of formal status and differences in future job opportunities.

Related literature: This paper first relates to the literature on employment transitions and wage dynamics in economies with large informality ([Maloney, 1999](#); [Bosch and Maloney, 2010](#)).² Recent research shows that high worker mobility in developing economies does not translate into wage progression.³ [Donovan et al. 2023](#), for example, document substantially larger labour-market flows in developing than in advanced economies, but show that much of this mobility is informal-to-informal transitions that imply little wage growth. By contrast, [Menezes-Filho and Narita 2025](#) show for a set of Latin American countries that transitions across sectors, particularly from informal to formal employment, produce wage gains on average, although a substantial share of workers experience wage losses. Similarly, [Engbom 2022](#) find for Brazil that wage gains are concentrated in cross-sector transitions rather than mobility within sectors. This literature leaves three issues that are central to my analysis.

First, much of the evidence compares wages before and after a transition without constructing an explicit counterfactual for how wages would have evolved in the absence of formalisation. Second, transitions into formality do not distinguish transitions within and between firms. This distinction matters because between-firm transitions combine a change in formality status with changes in employer characteristics. Third, the litera-

²A related strand studies worker reallocation between formal and informal employment over the business cycle ([Bosch and Esteban-Pretel, 2012](#); [Leyva and Urrutia, 2020](#)).

³A broad literature also finds that informal workers earn less on average, even after conditioning on observable characteristics and worker fixed effects ([Ulyssea et al., 2025](#)). However, the wage dynamics shows a different view.

ture focuses primarily on earnings, with less attention to hours worked and therefore to whether earnings gains arise from a higher hourly wage or from longer working hours. I complement this literature by distinguishing within- and between-firm formalisation and estimating the wage and hours consequences of the within-firm transition relative to comparable workers who remain informal.

A second strand of literature studies how workers value the non-wage attributes of formal and informal employment.⁴ Experimental and discrete-choice studies find that workers are willing to forgo substantial fractions of their wages for attributes of formal jobs, including training opportunities, unemployment benefits, shorter commutes, job stability, and schedule flexibility (Felix et al., 2026; Mahmud et al., 2021; Maloney et al., 2026).⁵ Maloney et al. 2026, for example, provide systematic evidence for several Latin American countries using discrete-choice methods and find substantial heterogeneity in the valuation of formal and informal job attributes. I depart from this approach in three respects. First, much of the existing evidence relies on stated or experimentally elicited choices over hypothetical job bundles, whereas I recover the value of formal employment from realised transitions and wage changes observed in the labour market. Second, these studies typically price individual attributes separately, while I recover the value of the formal employment bundle as a whole. Third, much of the evidence concerns self-employment, where autonomy and flexibility are particularly salient, whereas I focus on salaried workers moving between informal and formal employment within and across firms.

Finally, the paper relates to structural search models that recover job values in the presence of labour-market frictions.⁶ The paper closest to my within-firm mechanism is Bobba et al. 2021, who study within-job formalisation in Mexico through on-the-job human-capital accumulation and wage renegotiation. I instead model within-firm formalisation as a distinct wage-offer channel in a wage-posting framework, where workers can receive formal offers from both their current employer and outside firms. This makes the

⁴A much larger literature studies non-wage job attributes in advanced economies; see, among others, Mas and Pallais 2020; Mas 2025a.

⁵A related policy literature studies how social-protection programs alter the relative attractiveness of formal and informal employment. Non-contributory health insurance (Camacho et al., 2014), old-age pensions (Joubert, 2015), and cash transfers (De Brauw et al., 2015; Garganta and Gasparini, 2015; Bergolo and Cruces, 2021) can affect informal employment, while schedule flexibility (Berniell et al., 2021) and commuting costs (Zárate, 2022) provide additional non-wage motives for accepting informal jobs.

⁶In advanced economies, related work studies desired hours (Bloemen, 2008), occupational choice (Sullivan, 2010; Sullivan and To, 2026), and non-wage amenities. Sullivan and To 2014 extend Burdett and Mortensen 1998 with match-specific non-wage utility and find that amenities account for a substantial share of the gains from mobility; Hall and Mueller 2018 use rejected offers from unemployed workers; and Taber and Vejlín 2020 embed amenities in a Roy-search model with worker and firm heterogeneity. In these models, amenities are generally properties of the firm or worker-firm match.

within-firm “foot-in-the-door” margin part of the worker’s continuation value. My second departure is to model the non-wage component as sector-specific rather than firm- or match-specific. It therefore captures the residual value associated with being formally rather than informally employed after accounting for wages, future job opportunities, and employment stability. Related structural work includes [Ginja et al. 2025](#), who use a household search framework to study the value of social protection and formal employment. This approach is also related to evidence from Mexico showing that the expansion of universal health insurance (*Seguro Popular*) generated relatively modest reallocations between formal and informal employment ([Juarez, 2012](#); [Bosch and Campos-Vazquez, 2014](#); [Del Valle, 2021](#)).

This paper is organised as follows. Section 2 describes the data and the empirical findings. Section 3 describes the model and Section 3.2 reports the estimates. Section 4 presents the counterfactual that closes the within-firm route and Section 5 concludes.

2. Data and empirical findings

I use micro-data from the National Survey of Occupation and Employment (ENOE), collected by Mexico’s national statistics office (INEGI) over 2005–2025. The survey covers all 32 states and both urban and rural areas, sampling about 150,000 dwellings and 380,000 individuals per quarter, and follows a rotating panel in which each individual is interviewed for five consecutive quarters, with responses self-reported by household members.⁷ Following the legal definition of informality ([Perry et al., 2007](#); [OIT, 2013](#)), I classify a worker in their main job as informal if employed in a private business in a subordinate relationship while not enrolled in the social security system, and I assign this status in each of up to five interviews using the panel structure (see Appendix A for question flow).⁸ I focus on private wage workers because formalisation in this setting is a change in the status for the same employer–employee relationship. The survey carries no firm-side information, so I cannot observe whether the firm is either informal or formal.

Table 1 summarises worker characteristics. Near a half of the the working-age population is employed, formal workers are 16.3%, informal 12.9%, other employment (public, self-employment and unpaid work) 24.6%, and the non-employed 46.1%; among private wage workers, 55.9% are formal and 44.1% informal. The groups differ sharply. Informal

⁷Before 2005 the National Survey of Urban Employment (ENEU) and the National Survey of Employment (ENE) preceded the ENOE, and the number of cities rose from 32 to 39 between 2005 and 2020. Workers observed in all five interviews are 70% of the sample before the COVID-19 pandemic and 60% after.

⁸Workers with two jobs represent 3% of the sample.

workers are younger, less educated and less concentrated in the industrialised North and Central regions (36% versus 54% of formal workers), consistent with higher informality in the south documented for Mexico (Perez, 2020). Women are far more likely to be in other employment (43%) or non-employed (69%) than in private wage work. In the raw data, formal monthly wages are about 56% higher than informal (MXN 5,865 versus 3,751) and formal hours longer (48.6 versus 43.7 per week).⁹

Moreover, informality is concentrated in small firms and workers exhibit lower tenure in the firm. The data suggest that, near 95% of informal workers report to work in firms with fewer than 50 employees, against 49% of formal workers, closely related to the evidence that small firms are more likely to operate undetected by enforcement authorities (Ulyssea, 2018). Moreover, the survey records the worker's start date with the current employer, from which I construct tenure as the difference between the date of the interview and the reported start date, following the approach of Lancaster, 1990 and Babington P. and Cano-Urbina, 2018 for details).¹⁰ Workers in informal employment exhibit lower years of tenure, three years compared to five for formal.

Using the quarterly rotating panel structure of the data, I can follow the employment formality status and non-employment of each worker in each interview for up to five periods. I focus on workers aged 20 to 60 years old observed in at least two consecutive interviews that report three distinct and mutually exclusive states: formal employment (f), informal employment (i), and not employed (n) where formal and informal employment relate to private business only and not employed includes unemployed and inactive workers.¹¹ A quarterly transition probability is defined as the share of the number of N workers in the current state $s' = \{f, i, n\}$ over the sum of workers in all states in the previous state s .

$$\pi_{ss'} = \frac{N_{s'}}{\sum_s N_s} \quad s = \{f, i, n\}$$

⁹The earnings question does not specify clearly whether the amount is net or gross; the interviewer and procedures manual indicate that net wages are the intended measure. Because earnings are self-reported by the worker rather than the firm, they are not subject to the firm-side underreporting that Kumler et al. 2020 document in the social security administrative records, where compliance rises with firm size; but they are net of tax and contributions, so the gross formal–informal gap is wider than the take-home gap observed here.

¹⁰The question asks the year of first entry, not an elapsed duration: “¿En qué año [nombre] entró a trabajar por primera vez para su actual empresa, institución o patrón?” The month of entry is recorded only when that year is the current or the previous one. This information is collected only in the extended questionnaire, administered in the first quarter of each year. See Appendix B for more detail.

¹¹Although public employment involves a dependent relationship, the wage termination is determined by a set of specific rules, thus I exclude them from the employment states. In addition, I exclude, self-employment with and without employees because this represents entrepreneurship, a distinct margin involving capital constraints and risk preferences. Domestic and non paid work are excluded from the analysis.

TABLE 1. Summary of characteristics by employment state (pooled), 2005–2025

	Formal	Informal	Other employment	Not employed
Working age pop., (%)	16.3	12.9	24.6	46.1
Total employment, (%)	28.8	22.7	43.4	5.0
Wage employment, (%)	55.9	44.1		
Aged 18–25, (%)	22.1	27.2	9.0	16.8
Aged 26–54, (%)	71.3	55.1	67.3	30.2
Aged 55+, (%)	7.0	9.9	22.2	29.7
Years of education	11.2	8.9	10.3	7.8
North/Central, (%)	53.6	35.8	32.8	40.4
Female, (%)	36.5	28.3	43.0	68.7
Average real wage (monthly)	5865.4	3751.3	5010.8	
Average weekly hours	48.6	43.7	39.1	
Small firm (<50 empl.), (%)	49.4	94.9	55.8	
Tenure at the firm (years)	4.9	3.2	7.6	
Individuals	4,111,609	3,297,806	6,270,904	13,333,035

Source: Authors' estimates based on data from ENOE 2005–2025. Pooled observation-level summary, each worker-quarter is counted, so a worker observed across multiple interviews contributes once per interview.

Wage employment shares are only for private formal and informal wage workers in the denominator. Other-employment includes, public employment, self-employment and non paid work and not-employed includes unemployment and inactive.

2.1. The majority of informal-to-formal transitions take place within the firm.

I classify the transition between informal and formal as within and between firms using tenure data. For example, if a worker reports an informal-to-formal transition between interviews and the reported tenure is more than one quarter; this transition is a within firm transition. And if tenure is less than one quarter, this is a transition to another firm.¹² Table 2 Panel A and B show the transitions across states for the aggregate data and for gender and skill level group. The data shows that, 77% of informal-to-formal quarterly transitions occur within the firm and 23% in another firm, while the share of formal to informal transitions within the firm is 79% and across firms 21%. This pattern is similar across the demographic groups. Moreover, the within-firm informal to formal rate is 7.8% compared to 2.3% between firms. An informal-to-formal transition probability is 10.1% (7.8% within) per quarter while a formal-to-informal 7.5% (5.9%), suggesting that there is a higher probability of formalisation.¹³ This pattern of transition probability is similar

¹²This implies that $\pi_{if} = \pi_{if}^{within} + \pi_{if}^{between} = \frac{N_i^{within}}{\sum N_s} + \frac{N_i^{between}}{\sum N_s}$. About half of the observed employment-to-employment transition pairs cannot be classified, 52% of informal-to-formal and 54% of formal-to-informal pairs, and the within/between split is computed on the available pairs. Because tenure is collected once every year, the classification is driven by the survey design rather than by the worker or the transition behavior.

¹³[Samaniego and Fernández, 2024](#) report similar within-firm formalisation rates for 2005–2016, and [Bobba et al., 2021](#) document approximately 13% quarterly formalisation rates for 2015–2016. Differences across studies arise from the definition of the non-employment group and whether self-employment is included as

across groups except for low skilled males where the probability of transiting from formal to informal is 8.1% and informal-to-formal 7.4%.

The within and between classification results are preserved to various measurement error aspects. First, the tenure data show bunching at five-year intervals; this does not affect the definition, since the tenure threshold separating within-firm from between-firm changes is one quarter. Second, the classification is also robust to the threshold; reclassifying with a six-month threshold the shares of transition types is little changed, informal-to-formal 77% to 71%) and of formal-to-informal 79% to 73% as well as the quarterly probabilities (see Appendix Table B4). Third, the within-firm shares do not depend on how non-employment is defined. Restricting non-employment to workers who report being in non-employment and available for work leaves the within-firm share of informal-to-formal transitions at 77.1% against 77.2% here, and the formal-to-informal share at 79.3% against 78.7%; no demographic group moves by more than half a percentage point. Restricting it further to the unemployed alone gives 76.9% and 78.8%.

In addition, a transition classified as within-firm should leave the worker's recorded industry unchanged, since it is a change of status at the same firm, and a high rate of industry change would indicate that the rule is misclassifying moves between firms as within-firm ones. The data show that 71.2% of within-firm informal-to-formal transitions report no change of two-digit sector, and the same holds for formal-to-informal transitions.¹⁴ Workers who do not change status report even less change (around 80% no change), so a within-firm transition adds only a small amount of recorded sector change on top of reporting noise (Appendix Table B5). Meanwhile, a within-firm formalisation could be interpreted as a change of role inside the firm rather than a change of contract on the same job. Occupation records speak directly to this margin. Using the survey's eleven-group classification, aligned with the international ISCO standard, 71.5% of within-firm transitions report no change of occupation, and 83.2% report no change between manual and cognitive work; the latter is almost identical to the 83.4% no-change rate of workers who never change status, so a within-firm formalisation involves no change in the type of work performed.¹⁵ The evidence is consistent with substantial continuity in the worker-firm

a separate state.

¹⁴ENOE codes self-reported industry with SCIAN, the Mexican version of the North American Industry Classification System (NAICS); only the two-digit sector is comparable across firms. At the finer four-digit the survey has 178 subgroups and the no-change share falls to about 60%, but this level records the activity description rather than the firm's sector; for example, within construction (sector 23) the same firm may be reported as residential building (2361) or other stone work in the sector (2382), so within-firm changes here reflect coding granularity rather than reallocation (Appendix Table B5).

¹⁵At the four-digit occupation level the no-change share falls to about 49%, but the SINCO unit group identifies the product rather than the skill content of the job: growing maize (6111) versus growing vegetables (6112), or bricklaying (7121) versus house painting (7135). Movement across these four-digit groups is close to

match and suggests that most classified transitions do not involve a major change in job type.

Moreover, separation rates and job-finding rates differ sharply by formality status and across gender. Workers in informal job have a probability of separating from their job of 40% and formal 20%. The data does not include questions about the reasons behind the separations such as quit or layoff, the data simply tracks the stated across interviews. This is also true for job-to-job separations. This is a limitation for the structural model since voluntary moves may be correlated to the valuation of jobs ([Bonhomme and Jolivet, 2009](#)). Women exhibit substantially higher job separation from both informal employment (54–58% versus 29–39% for men) and formal employment (28–30% versus 15–18%).¹⁶ On the other hand, the most likely entry to employment is informal employment; the job-finding from non-employment to the informal sector with a probability of 6.6% per quarter compared to 3.8% for formal. This is higher for men, high-skilled males re-enter at a 21% quarterly rate and low-skilled males at 38%, compared with 9% for high-skilled females and only 5% for low-skilled females. Women's higher exit rates to non-employment, combined with lower finding rates, are consistent with their high level of inactivity.

The data describe a labour market that is not segmented. Across all demographic groups workers move frequently between formal and informal employment. In particular, informality is characterised by high job-to-job transitions and higher finding and separation rates than formal employment, and serves as the principal route into formal work. The informal-to-formal route splits in two, within the same firm and between firms. The transitions within the firm can be explained through several channels of firm-selection, in which informality can act as a period where the firm checks the quality of the worker-firm match ([Jovanovic, 1979](#)), or the firm learns about the worker's general skill from observed performance ([Farber and Gibbons, 1996](#)) or informality may be the entry point of a designed internal hiring scheme in which workers are promoted ([Lazear, 1979](#); [Lazear and Moore, 1984](#); [Lazear, Altmann, et al., 2011](#)).

A second set of channels relate to a worker-selection effect, in which workers accept jobs with a particular mix of wage and non-wage attributes. In the context of informality, non-wage amenities take several forms such as social protection benefits provided to both formal and informal workers ([Bergolo and Cruces, 2021](#)), employment protection ([Chaurey et al., 2025](#)), schedule flexibility ([Berniell et al., 2021](#); [Berniell et al., 2023](#)), commuting time

lateral and carries substantial coding noise (Appendix Table B6).

¹⁶Most separations lead to inactivity rather than unemployment; if non-employment is restricted to the unemployed only, separation rates to unemployment are 9.3% from formal employment and 15.1% from informal employment which are close to the estimates of [Bosch and Maloney, 2010](#) (see Appendix Table B3).

TABLE 2. Quarterly transitions probabilities

	<i>Next period state, %</i>					
	Formal		Informal		Not em- ployed	Shares
	<i>(Within)</i>	<i>(Between)</i>	<i>(Within)</i>	<i>(Between)</i>		
Panel A. Aggregate						
Formal	65.7	6.3	5.9	1.6	20.6	15.5
Informal	7.8	2.3	36.4	13.7	39.8	11.5
Not employed		3.8		6.6	89.6	73
Panel B. By characteristics						
Female and High Skill						
Formal	59.7	6.8	4.4	1.3	27.7	13.5
Informal	9.4	3.3	22.9	10.1	54.4	6.8
Not employed		4.2		4.8	91	79.8
Female and Low Skill						
Formal	58.1	6.8	4.2	1.4	29.5	6.1
Informal	5	1.8	23.9	11.2	58.1	5.3
Not employed		1.7		3.5	94.8	88.6
Male and High Skill						
Formal	69.7	6.2	5.1	1.5	17.5	32.1
Informal	11.3	3.4	34	12.3	39	14.6
Not employed		9.6		11.9	78.6	53.3
Male and Low Skill						
Formal	70	5.7	8.1	1.9	14.4	31.3
Informal	7.4	1.9	45.6	16	29	32.8
Not employed		10.2		27.4	62.5	35.9

Source: Authors' estimates based on data from ENOE 2005–2025. The sample includes workers aged 20 to 60.
Note: A worker is *skilled* if they report twelve or more completed years of education and *unskilled* otherwise.
Twelve years corresponds to completed upper-secondary schooling.

to the workplace (Zárate, 2022) and overall well-being (Falco et al., 2015).¹⁷ In the hedonic benchmark model (Rosen, 1974), workers accept lower pay in exchange for amenities. However, under search frictions, as in Burdett and Mortensen, 1998, workers take the wage and non-wage bundle as it arrives and cannot move to other jobs instantaneously, so an accepted job may include a low wage with a low amenity value (Hwang et al., 1998).¹⁸ In this literature, the wage–amenity trade-off is revealed through between-firm transitions, where workers who experience a wage cut reveal how workers value those amenities (Bonhomme and Jolivet, 2009; Sullivan and To, 2014; Sorkin, 2018).

These competing sets of channels imply different predictions for the wage consequences of informal-to-formal transitions. Matched-quality, employer-learning, and incentive models predict that workers should experience wage gains upon informal-to-formal transitions. By contrast, search models with non-wage amenities allow workers to accept jobs even when wages fall if the amenity gain compensates the wage loss. The data shows that this could be the case. Figure 1A shows that the formal and informal log real wage distributions overlap substantially. In particular, Figure 1B shows that the share of informal wage employment is present across deciles, from about 88% in the bottom decile to about 21% in the top. Moreover, Figure 1C shows larger dispersion of hours of work in the informal sector than in the formal. Weekly hours of informal workers range from 20 to 80, consistent with greater variation in working-time arrangements, while in the formal case hours of work are tightly concentrated around 48 hours per week (Mexico’s legal standard workweek). The share of informal employment is larger at the lower and top deciles, 65% and 40% respectively, as shown in Figure 1D. The overlap in the wage and hours of work distributions suggests that a transition between jobs can cause either an hourly wage increase or a cut. The next section estimates the causal effect of informal-to-formal transitions on wages and hours to distinguish between these competing mechanisms.

2.2. Within-firm informal-to-formal transitions raise the hourly wage by 5%.

The main challenge of estimating the causal effect is the presence of several potential sources of selection, as discussed in the previous section. The firm, the worker, and the evolution of the worker–firm match jointly determine both the informal-to-formal transition and the wage increase (or cut). On the firm side, firms differ in productivity, their wage-setting and compliance policies that correlate with formality status and wages. I

¹⁷In the context of Mexico, Table B7 summarises the institutional benefits and costs that differ between contributory and non-contributory schemes.

¹⁸Even in the context of no frictions, there is also evidence that jobs with higher wages also report higher value of amenities which invalidates the hedonic framework (Mas and Pallais, 2017). This typically relates to high ranking employees (e.g. CEOs) that report high wages and receiving high value amenities.

FIGURE 1. Distribution of wages and hours of work



Source: Author's estimates based on data from ENOE 2005–2025. The data for hours of work and wages trims the top and bottom 1%.

consider differences in firm characteristics by focusing on the within firm informal to formal transitions data sample. This allows me to partially address time-invariant firm characteristics of the same firm and use the between firms data sample as a complementary estimate which involve both formalisation and potentially firm effects.

On the worker side, informal workers who change to formality may differ systematically from those who stay informal in their observable characteristics and their underlying wage trajectories, the more educated, experienced, or career-oriented are more likely to receive and accept formal offers, and workers sort across sectors on their preferences for amenities such as schedule flexibility or social protection, even within the same firm (Ulyssea, 2018). I consider worker selection by comparing two groups with similar observable characteristics: informal workers that eventually change to formality in the same firms and workers who remain informal.

However, even after conditioning on observable worker characteristics and firm characteristics with the specific data sample, workers may remain on different trajectories because the change in status may follow gradual improvements in productivity or match quality that gradually increases wages. I use the lagged dependent variable adjustment and include the lagged outcome as a covariate to proxy for these unobserved channels (Fan and Peng, 2019). In addition, I do a sensitivity analysis of the estimates to provide bounds where the parallel trend assumption can be valid (Rambachan and Roth, 2023).

Data sample and event-study conditional on observables: I focus on workers who report only wage employment across all available interviews.¹⁹ Approximately 1.1 million workers are informal in their first interview of which 10% change to formality eventually, while the remaining 90% remain informal in all interviews. Two types of informal-to-formal transitions are observed, permanent transition to formality (6%) and those who switch back to informality (4%). Only 8% of permanent informal-to-formal transitions lack a tenure record (see Appendix Table C2).²⁰ Following the Callaway and Sant’Anna, 2021 strategy, I use permanent transitions, workers that change to formality and remain formal until the last interview. This sample ensures the irreversibility assumption while reduces noise from transitions between jobs. On the other hand, the sample of workers reporting to be formal on the first interview is near 1.7 million, of which 6.2% change to informality (see Appendix Table C2). As a robustness check, I estimate the effect including treatment reversion, that is workers transitioning to formality and then back to informality, using the Chaisemartin and D’Haultfoeuille, 2020 approach.

I use the doubly robust strategy that combines an outcome regression (Heckman, 1997) with matching (Hirano and Imbens, 2001; Abadie, 2005) in a multi-period design (Callaway and Sant’Anna, 2021). In the main setting, the treated are workers moving from informal to formal, and the control group is workers who remain informal in every interview.²¹ The parameter of interest is the average treatment effect on the treated (Callaway and Sant’Anna, 2021), recovered by reweighting the control group to match the pre-treatment characteristics of the treated. The weights come from a propensity score, estimated from a logit of eventual formal status on covariates measured before the event, so that treated and control workers are comparable on observables at baseline. Reweighting delivers balance

¹⁹This implies to drop workers that report non-employment or other employment in one or more interviews.

²⁰Conditioning on permanence uses information from later interviews, and it does so unevenly across cohorts: a worker who formalises at the second interview must stay formal for three more waves to qualify, whereas one who formalises at the last interview faces no such requirement, since there are no later waves to observe. Early cohorts are therefore filtered more strictly than late ones. The reversibility estimates reported below address this directly, since they impose no persistence requirement at all.

²¹In addition, I use not-yet-treated workers as the control group in both settings to confirm the results.

on the covariates; it does not deliver identification. Identification rests on a separate and untestable assumption, that the two groups would have followed parallel paths conditional on those covariates, stated formally in Equation (2) below. Balance makes the assumption more plausible by removing composition differences, but conditioning on observables cannot rule out that formalisation is timed to an unobserved shock to the wage path. Flat pre-treatment estimates are evidence consistent with the assumption rather than a test of it, which is why the sensitivity analysis below asks how large a violation would have to be before the conclusions change.

With short rotating panel data the information set for the estimates is limited. The time period t is the interview number rather than calendar time and takes values $t = 1, \dots, 5$.²² For some workers transition occurs in the second interview while others transition in the last interview. This implies that workers who transition in the second interview $t = 2$ are in the second group $g = 2$ and those who transition in the last interview in $g = 5$. Let k denote the event time index observed for a group relative to the transition interview, so that $k = t - g$ with $k = \{-4, \dots, 3\}$ where $k = -4$ is the group of workers in period $t = 1$, that transitioned in the last interview $g = 5$ and $k = 3$ is the group of workers transitioning in the second interview $g = 2$ in period $t = 5$ (see Table C1 in Appendix for a full description of the information set). Because interview number is determined by the survey rotation schedule rather than by labour-market events or worker characteristics, the timing of the treatment is plausibly as good as random.²³ The object estimated is not a regression coefficient, and in particular the estimates are not obtained from a two-way fixed-effects specification. The building block is the group-time average treatment effect $ATT(g, t)$: for each group g and period t , the change in the outcome between the last pre-treatment interview $g - 1$ and interview t for workers in group g is contrasted with the same change for comparison workers, who are reweighted by the propensity score and residualised by an outcome regression, which is what makes the estimator doubly robust (Callaway and Sant'Anna, 2021). The event-study profile reported below aggregates these group-time

²²Indexing on the interview rather than the calendar quarter is without loss here, and it is worth being precise about why. Each household is interviewed once per quarter for five consecutive quarters, and the sample retains only workers whose interview number advances by exactly one throughout, so within a worker the interview index and the calendar quarter advance one for one and differ only by the quarter in which that worker entered the panel. Event time $k = t - g$ is a difference taken within a worker, so that entry offset cancels exactly: one unit of k is one calendar quarter for every worker, and an implementation on calendar quarters would return the same k by construction. What the interview index does not do is absorb shocks common to a calendar quarter, since two workers at the same interview number need not have entered the panel in the same one. With entry cohorts spread across 2005–2025 such shocks average out across cohorts rather than being differenced out within them, which is a consequence of pooling cohorts rather than of the choice of clock, and the sensitivity analysis below admits departures from parallel trends of exactly this kind.

²³In longer panels this would not be the case, since the informal share of employment fluctuates with the business cycle (Bosch and Esteban-Pretel, 2012).

effects to event time k , using the unbalanced aggregation of [Callaway and Sant'Anna, 2021](#):

$$(1) \beta_k = \sum_{g \in \mathcal{G}} \omega(g, t) \cdot ATT(g, t) = \sum_{g \in \mathcal{G}} \mathbb{1}\{t-g = k\} \cdot \mathbb{1}\{g+k \leq 5\} \cdot P(D^g = 1 \mid g+k \leq 5) \cdot ATT(g, t),$$

where the outcome y_{it} is log wages, log hours of work per week, or log hourly wage for worker i at interview $t = 1, \dots, 5$, and g_i is the interview number at which worker i first transitioned.²⁴ With at most five interviews per worker the event-time index $k = t - g_i$ spans -4 to 3 , and the weight $\omega(g, t)$ is the share of the treated accounted for by group g among the groups observed at event time k . Inference is by wild bootstrap, clustered on the worker, which is the level at which treatment is assigned and at which observations repeat: a worker contributes up to five interviews and their outcomes are not independent across them. Appendix C.3 gives the full derivation.

Identification relies on the assumption that, absent formalisation (or informalisation), treated and control workers would have followed parallel trends conditional on observable characteristics. Formally,

$$(2) \quad \mathbb{E}\left[y_{it}(0) - y_{i,t-1}(0) \mid D_i^g = 1, X_i\right] = \mathbb{E}\left[y_{it}(0) - y_{i,t-1}(0) \mid D_i^g = 0, X_i\right],$$

for $t \geq g$, where $D_i^g = 1$ indicates that worker i belongs to group g , $y_{it}(0)$ is the untreated potential outcome, and X_i are the pre-treatment covariates measured at $k = -1$. I start the estimates by computing the propensity score $P(D = 1 \mid X)$ using a logit model in which the outcome is a dummy for eventual treatment and the covariates X are pre-treatment worker characteristics observed at $k = -1$:

$$(3) \quad P(D = 1 \mid X) = \Lambda(X' \beta),$$

where $\Lambda(z) = e^z / (1 + e^z)$ is the logistic function. The covariates X include years of education, age and age squared, a dummy for female, household size and headship, a dummy for marriage and dummies for the presence of children below five and of children aged five to seventeen in the household, each interacted with marriage and with gender, occupational skill dummies (elementary, intermediate, and professional/managerial), and dummies for industry, firm size and region. Figures C2A–C2B compare the propensity-score densities of treated and control workers at $k = -1$, before and after reweighting; the densities overlap substantially in all cases once the reweighting is applied. Tables C3 and C4 report covariate means for treated and control groups at $k = -1$, before and after reweighting, across the

²⁴The hourly wage is wage per hour of work, $hr = w/h$, so $\log(hr) = \log(w) - \log(h)$; monthly wages and weekly hours are reported in the data. I compute hourly real wages as monthly take-home pay divided by 4.33 times reported weekly hours, deflated by the National Statistics Office consumer price index.

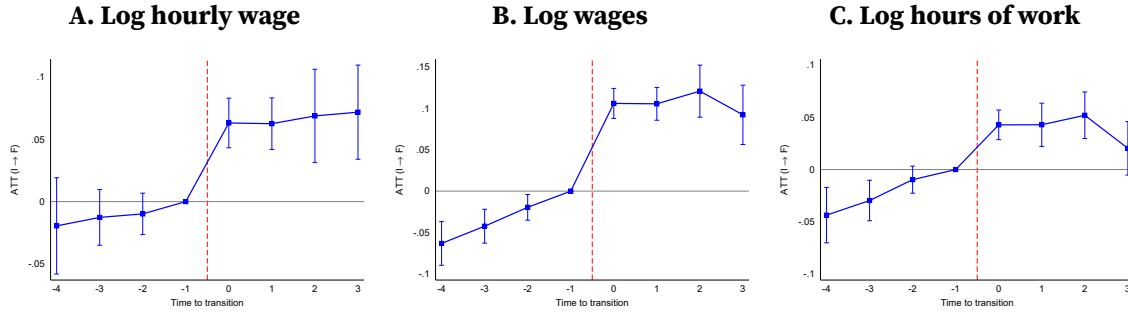
pooled, within-firm, and between-firm transition samples. Before reweighting, the raw groups differ on almost every covariate. In Table C3 Panel A, formalisers are on average 1.7 years younger and have 1.8 more years of schooling than always-informal workers. They are twice as likely to hold a high-skill occupation (0.14 against 0.07) and to work in the north of the country (0.22 against 0.11), and substantially less likely to work in a small firm (0.61 against 0.84) or in agriculture (0.07 against 0.21). After reweighting, Panel B shows that these differences collapse: the age gap falls from 1.7 years to 0.7, the schooling gap from 1.8 years to 0.1, and the occupation, region and firm-size gaps to 0.02 or less.²⁵

Informal to formal transitions: Figures 2A–4C report the treatment effects of transitions from informal to permanent formal employment. The pooled estimates indicate a positive and persistent effect on log hourly wages of approximately 5.3% over the three post-transition periods. Monthly wages increase by around 10%, while hours worked increase by 4.8% that partially compensates the increase in log monthly wage. These effects are largely driven by transitions occurring within the same firm, which account for 77% of all informal-to-formal transitions. Figures 3A–3C present the within-firm estimates and show that hourly wages increase by 5%, monthly wages by 9%, and hours worked by about 4%. Note that the hourly-wage leads are small relative to the post-treatment change and are not jointly statistically distinguishable from zero. For log of wages and hours of work is less likely to hold. This pattern is found in the within firm transition data sample. On the other hand, Figures 4A–4C report between firm estimates which show no statistical significance on hourly wages. This is mostly driven by hours of work that fully compensate for the increase in log wages.²⁶

²⁵Table C4 Panels A and B show a similar pattern for informalising workers

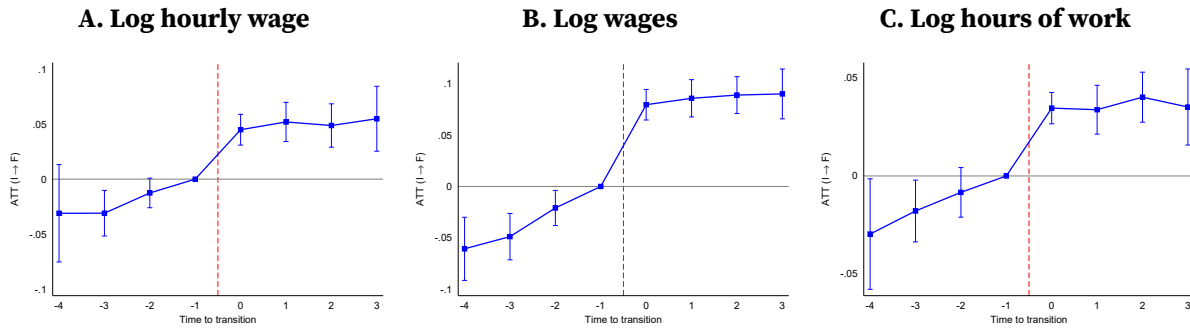
²⁶Using the not yet treated control group, adds workers who formalise at a later interview to the controls of earlier cohorts, leaves the estimates essentially unchanged, with differences below 0.3 percentage points at every event time and no change in significance. The choice of comparison group does not drive the results; Appendix C.7 states the not-yet-treated estimand.

FIGURE 2. Average treatment effect of informal to formal transition ($I \rightarrow F$)



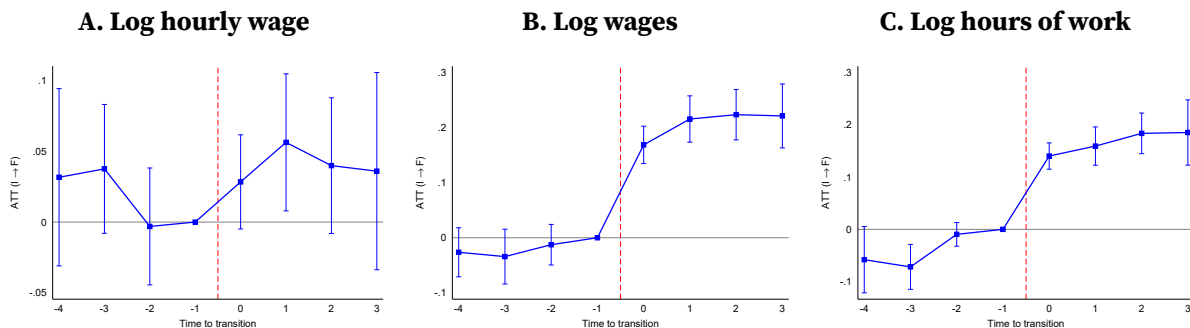
Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; the horizontal axis is event time in interviews relative to the transition; vertical bars are 95% confidence intervals from a wild-cluster bootstrap with 999 replications.

FIGURE 3. Average treatment effect of informal to formal transition ($I \rightarrow F$) within firm



Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; the horizontal axis is event time in interviews relative to the transition; vertical bars are 95% confidence intervals from a wild-cluster bootstrap with 999 replications.

FIGURE 4. Average treatment effect of informal to formal transition ($I \rightarrow F$) between firms



Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; the horizontal axis is event time in interviews relative to the transition ; vertical bars are 95% confidence intervals from a wild-cluster bootstrap with 999 replications.

I extend the event study analysis in several ways to study the implications of such pre-trend behaviour on the parallel trend assumption and the estimates. First, I include the lagged outcome as covariate because formalisation is likely to depend on recent worker performance and match quality, lagged wages and hours provide informative proxies for otherwise unobserved determinants of both treatment and post-event outcomes. This is closely related to the literature emphasising the relationship between difference-in-differences and lagged dependent variable adjustment (Fan and Peng, 2019).²⁷ Second, I do a sensitivity analysis and allow for deviations in the parallel trend assumption to offer bounds where it holds (Rambachan and Roth, 2023).

The estimate above is the main result of this section, and the four exercises that follow are checks on it. The first two, ask whether the within-firm hourly gain survives conditioning on pre-event levels and bounded departures from parallel trends. The third runs the same design in the opposite direction, on workers losing formal status, where measurement error would produce a gain of the same sign and the mechanism predicts the opposite. The fourth removes the persistence requirement from the treatment definition.

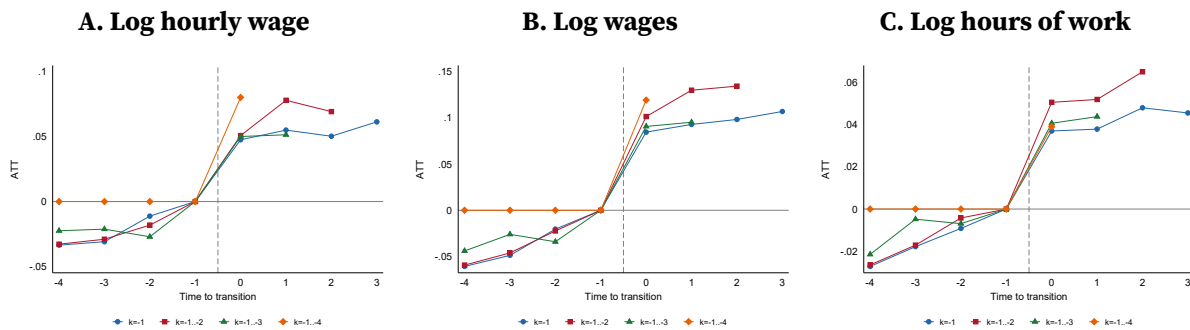
Lagged outcome: I add the pre-event outcome levels at each interview $k = \{-1, -2, -3, -4\}$ to the conditioning set of the doubly-robust estimator, entering them cumulatively: the first specification conditions on the level at $k = -1$ only, the second on $k = -1$ and $k = -2$, and so on up to the fourth, which conditions on all four leads. This is the lagged-dependent-variable adjustment of Fan and Peng, 2019, and it does not change the form of the identifying assumption: the estimand remains the conditional parallel-trends restriction in Equation 2, with the covariate vector X enriched by the pre-period outcome levels. Conditioning on the lagged outcome probes a level-based selection restriction, that formalisation is unrelated to a worker’s wage and hours level before the event (Fan and Peng, 2019). Two requirements are important: the overlap must continue to hold with the new covariate set, and because the number of pre-event interviews a worker contributes equals her cohort $g - 1$, each additional lead mechanically restricts the sample to workers observed that many periods before the transition; the fourth specification retains only the $g = 5$ cohort and loses the $k = 3$ horizon, so deeper specifications trade coverage for a tighter match on levels.

Conditioning on the pre-period levels leaves the effect on the hourly wage essentially unchanged. Figure 5 overlays the within-firm event studies across the four specifications: the jump at formalisation stays close to the baseline 5% through the first three, and moves

²⁷I do not condition directly on pre-treatment outcome growth in the main specification because recent wage or hours changes may either proxy for latent productivity growth or already reflect an anticipatory formalisation process. Conditioning on such changes could therefore absorb economically relevant pre-treatment dynamics or induce bias if they are jointly determined by latent productivity and anticipation.

only in the fourth, where the $g = 5$ -only sample is small and noisier. The stability of the estimate suggests that observed pre-treatment wage and hours levels are not the primary source of the hourly-wage jump. The pre-treatment leads, however, do not flatten. Matching on levels removes a difference in levels, not a difference in slopes, so the gradual upward path in monthly wages and hours ahead of formalisation survives every specification. That the leads do not flatten is itself the informative result, the pre-trend is a trajectory rather than a level gap, so I address it with a sensitivity analysis.

FIGURE 5. Lagged-outcome adjusted ATT within-firm informal-to-formal transition ($I \rightarrow F$)



Source: Author's estimates based on data from ENOE 2005–2025. Each panel overlays the within-firm event study conditioning cumulatively on the pre-event outcome level at $k = -1$ (l1) through $k = -1, \dots, -4$ (l4). Doubly-robust estimates.

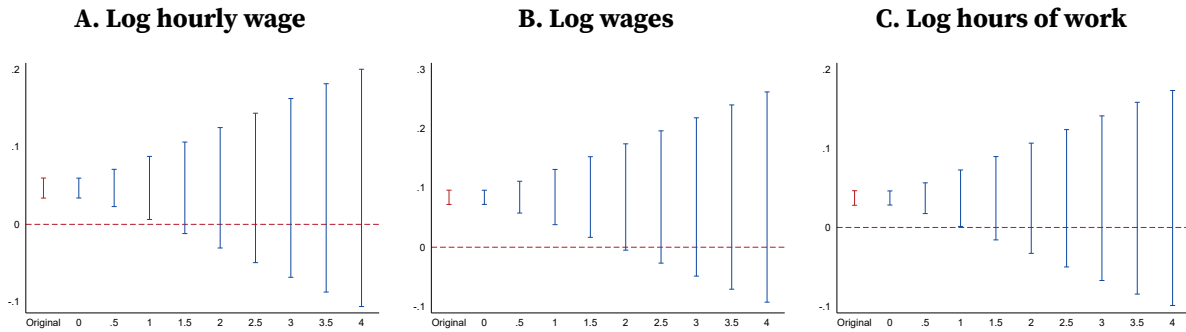
Sensitivity to violations of parallel trends: Empirically, the leads are flat for the hourly wage but slope upward for hours and monthly log wages, so the anticipatory margin is the intensive margin of labour supply rather than the hourly price. Workers raise their hours as the relationship deepens and formalisation approaches, while the hourly rate does not move until the change takes place. The pre-treatment dynamics for monthly wages and hours exhibit a clear upward trend prior to formalisation, suggesting that part of the estimated post-treatment effects may reflect continuation of pre-existing trajectories rather than the causal effect of formalisation itself. Such patterns are consistent with the evolution of the match. These mechanisms are expected to generate gradual rather than abrupt departures from parallel trends. I implement the procedure proposed by [Rambachan and Roth, 2023](#); rather than imposing exact parallel trends, it allows to construct robust confidence intervals under bounded departures from parallel trends.²⁸

Figures 6A–6C report the sensitivity analysis under the relative-magnitude restriction. Moving to the right along the horizontal axis corresponds to allowing progressively larger deviations from parallel trends. The estimated effect on hourly wages remains statistically

²⁸Appendix C.1 states the restrictions and the breakdown values formally.

distinguishable from zero for deviations up to approximately one time the magnitude of the pre-treatment violation, while the corresponding threshold is approximately 1.5 for monthly wages. The estimated effect on hours worked is substantially more sensitive to violations of the identifying assumption. These results suggest that the positive effects on wages are reasonably robust to moderate departures from parallel trends, whereas the estimated increase in hours worked is more likely to reflect pre-existing trends.²⁹

FIGURE 6. Honest-DiD sensitivity (relative-magnitude restriction)



Source: Author's estimates based on data from ENOE 2005–2025.

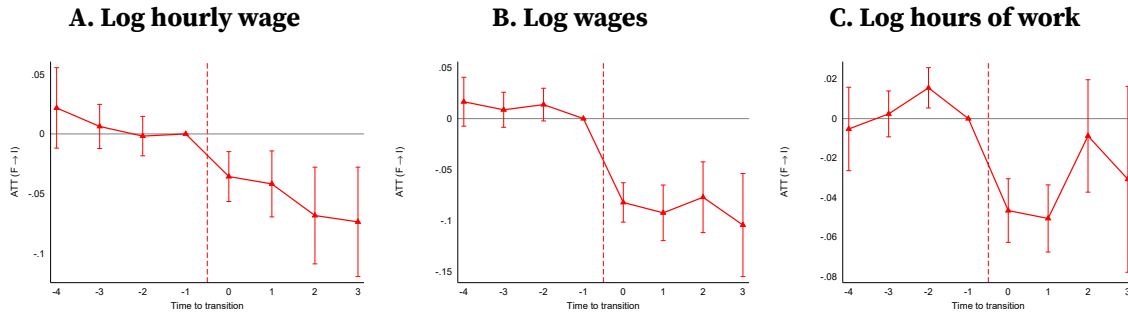
The estimates show two distinct features. The hourly wage shows a discrete jump at formalisation, indicating a causal effect. The monthly and hours gains instead depict a pre-existing upward trend: the event-study pre-coefficients are flat for the hourly wage but rising for monthly wages and weekly hours, so workers raise their hours and log wages ahead of formalisation while the hourly rate moves modestly. This anticipatory hours path suggests that the transition could be a reward for productivity change, the internal promotion scheme, or employer learning, in which a worker is offered a formal job. I do not rule out that channel; instead, I claim that what formalisation delivers is the discrete and modest hourly gain. Formalisation also moves the worker toward the standardised 48-hour week, raising and regularising hours and giving up the schedule flexibility related to informal work. A modest hourly gain, set against the loss of flexible hours, points to a non-wage value of formality.

Formal-to-informal transitions: This setting confirms that the wage change is not measurement error and operates in the opposite direction almost symmetrically to the previous case. Figures 7A–7C show that informalisation lowers the hourly wage by 5% when changing status, and drops further to 8% three periods later. This is driven by a 9% fall in monthly wages and a 4% fall in weekly hours. Within-firm informalisation implies a

²⁹Two further exercises confirm this split, the same strategy calibrated to the pre-period rather than to relative magnitudes, and a linear extrapolation of the pre-trend netted out of the estimates; both are reported in Appendix C.1.

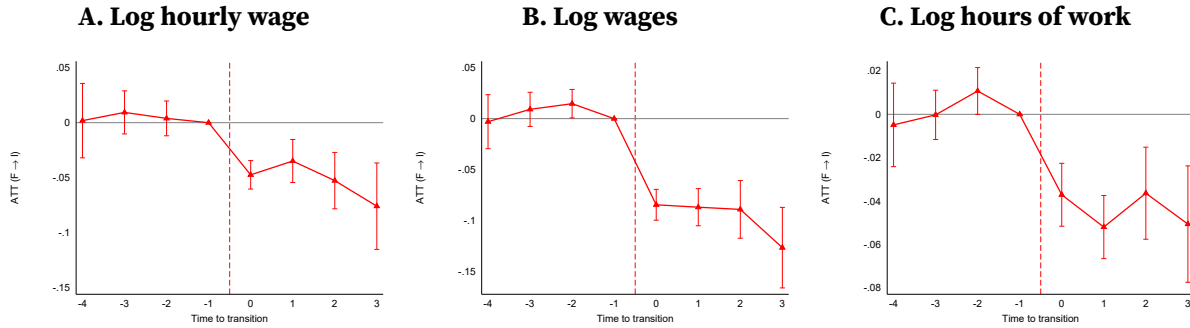
5% drop in hourly wages, composed of a 9% fall in monthly wages and a 4% fall in hours of work. Conversely, between-firm informalisation shows a larger monthly drop of 12% and an hours fall of 8%. The symmetry of the within-firm hourly effect across the two directions, a 5% gain on formalising and a 5% loss on informalising, indicates that the premium moves with formal status rather than with fixed worker characteristics. In this setting there is no clear pre-trend on any outcome, suggesting that dynamic selection or a career path operates only for informal-to-formal transitions, with no evidence of informality as a career path within the firm. In addition, the symmetry of the within-firm effects is inconsistent with a pure compensating-differentials channel, under which informalisation would restore hours flexibility; in this case, hours of work drop to gain more flexibility in informal work.

FIGURE 7. Average treatment effect of formal to informal transition ($F \rightarrow I$)



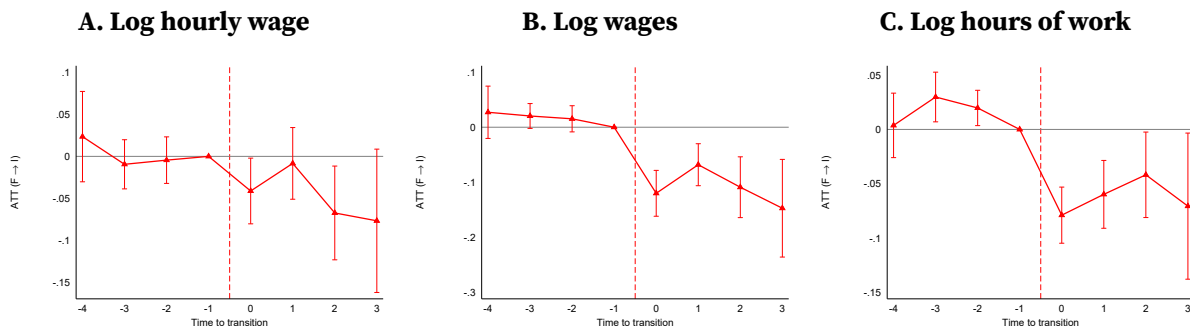
Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; the horizontal axis is event time in interviews relative to the transition; vertical bars are 95% confidence intervals from a wild-cluster bootstrap with 999 replications.

FIGURE 8. Average treatment effect of formal to informal transition ($F \rightarrow I$) within firm



Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; the horizontal axis is event time in interviews relative to the transition; vertical bars are 95% confidence intervals from a wild-cluster bootstrap with 999 replications.

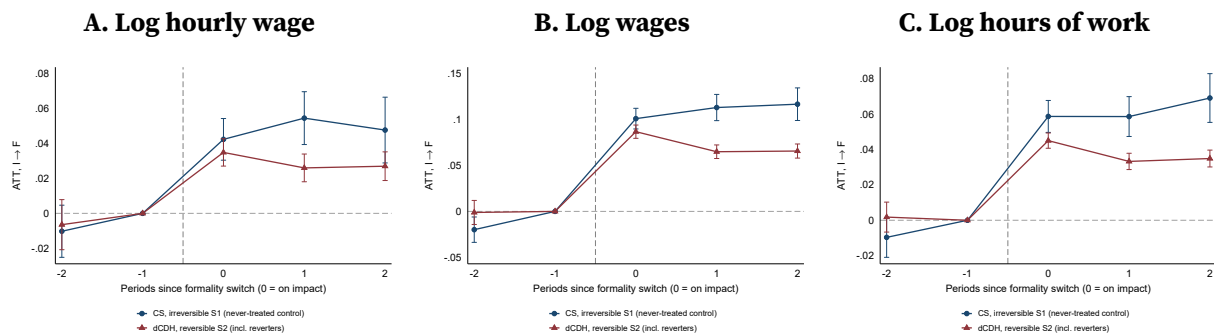
FIGURE 9. Average treatment effect of formal to informal transition ($F \rightarrow I$) between firms



Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; the horizontal axis is event time in interviews relative to the transition; vertical bars are 95% confidence intervals from a wild-cluster bootstrap with 999 replications.

Treatment reversibility: The baseline conditions on permanent transitions, excluding workers who switch formality status more than once, so treatment status is assigned using interviews that come after the transition. This is the estimate that removes that dependence: I re-estimate on the reversible sample with the [Chaisemartin and D’Haultfoeuille, 2020](#) estimator, which allows the binary treatment to switch on and off across interviews and therefore requires no persistence and no information about a worker’s later status. A switcher is a worker whose formal status changes between two interviews: a joiner moving informal to formal, or a leaver moving formal to informal. At each switching date the estimator compares the outcome change of switchers with that of stable groups whose status does not change, the always-informal for joiners and the always-formal for leavers; it aggregates these comparisons with non-negative weights and reports dynamic effects after a switch together with placebos before it. Identification requires common trends on both the treated and the untreated potential outcomes, stronger than the baseline, and a stable group at each switching date, which the large always-formal and always-informal populations satisfy; Appendix C.8 states the assumptions and derivation. Figure 10 shows that the reversible-sample effects are lower, about 4% on hourly wages, 6% on monthly wages and 4% on hours of work, against 5.3%, 10% and 4.8% in the baseline. The permanence restriction therefore accounts for part of the measured effect but not for its existence: roughly a quarter of the hourly gain and a larger share of the monthly one, with the hours effect essentially unchanged. The reading that formalisation raises the hourly wage modestly, and raises monthly earnings mainly through hours, holds under a treatment definition that uses no information about what the worker does later.

FIGURE 10. Reversible-treatment robustness: informal-to-formal transition ($I \rightarrow F$)



Source: Author’s estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Each panel overlays the baseline Callaway–Sant’Anna estimate on the pooled data against the reversible-treatment estimate (maroon), aligned at the on-impact period.

Heterogeneity across gender and skill:³⁰ Formalisation wage premia differ sharply across workers. Table 3 reports, for each group, the average effect over the interviews that follow the transition. The within-firm hourly gain is concentrated among the low-skilled, 6.6% for men and 4.2% for women, against 3.4% and 1.0% for high-skilled men and women, neither distinguishable from zero; the two low-skilled paths differ in shape, the male premium holding across the window while the female one fades from 5.5% on impact to 2.7% by the last interview. Between firms only high-skilled women record an hourly gain, and only at the ten per cent level, while monthly earnings and hours rise in every group on both margins. That high-skilled workers formalise at the highest rates while obtaining no hourly premium suggests that, for these workers, formalisation is driven by the non-wage value of a formal job rather than by a higher wage. The full set of event studies by gender and skill is reported in Appendix C.9.

TABLE 3. Average effect of formalisation by gender and skill

	Within firm			Between firms		
	Hourly wage	Monthly wage	Hours	Hourly wage	Monthly wage	Hours
Panel A. Average effect after the transition						
High-skilled men	0.034 (0.031)	0.079*** (0.031)	0.045** (0.018)	0.091 (0.081)	0.359*** (0.086)	0.268*** (0.078)
High-skilled women	0.010 (0.030)	0.060** (0.030)	0.050** (0.022)	0.120* (0.063)	0.296*** (0.069)	0.176*** (0.046)
Low-skilled men	0.066*** (0.010)	0.106*** (0.009)	0.040*** (0.007)	0.021 (0.023)	0.170*** (0.022)	0.148*** (0.019)
Low-skilled women	0.042*** (0.013)	0.073*** (0.011)	0.031*** (0.010)	-0.004 (0.040)	0.182*** (0.037)	0.186*** (0.034)

Source: Authors' calculations using ENOE 2005–2025. Sample covers workers aged 20–60 years old. Each cell is a separate Callaway–Sant'Anna estimate on the group named in the row, estimated with the doubly robust method and the same controls as the aggregate estimates. The reported average runs from the interview of the transition to the third interview after it. Within-firm transitions are those where the worker stays with the same employer and between-firm transitions those where the employer changes. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The previous results show that the wage return to formalisation is positive and modest and different across demographic groups. Within the firm the hourly gain is about 5% and concentrated among low-skilled men; for the other groups it is small or not statistically distinguishable from zero, and between-firm formalisation raises monthly earnings almost entirely through longer hours rather than a higher hourly rate. The within-firm transition gain is also hard to attribute to the formal status change alone: because the firm is held fixed, part of it may reflect career progression or employer learning that would have raised

³⁰ Looking at the formalisation heterogeneity over the wage distribution. About 41% of within-firm and 43% of between-firm formalisers take an hourly wage cut, and Table C5 reports the distribution: among within-firm formalisers who take an hourly cut the median loss is about 20%, while among those who gain the median rise is about 26%. Moreover, the treated and control distributions are close. The median hourly change is about 7% for within-firm formalisers against 2% for stayers, a gap of roughly the 5% average effect of the previous section, and a comparable share of stayers (47%) also record an hourly cut. The prevalence of cuts is therefore largely ordinary wage volatility.

wages regardless of registration, a concern I bound with the pre-trend and sensitivity analysis above and cannot fully rule out. At the same time, formalisation bundles a move toward the standard 48-hour workweek and a gain of formal sector benefits, so workers who formalise give up the schedule flexibility associated with informal jobs while gaining formal job benefits.

Taking stock, the two facts point to a value of formality that wages alone do not capture. Formalisation is overwhelmingly a within-firm event, a foot-in-the-door in which the current firm registers the worker, so the relevant margin is not a move up a job ladder but a change of contract at a given match. This change raises the hourly wage only modestly and unevenly—and not at all for the high-skilled workers who formalise at the highest rates—and it comes bundled with a move toward the standard workweek and the loss of the schedule flexibility of informal work. That workers formalise for little or no hourly gain implies that the formal job carries a non-wage value, through the social-security bundle, job stability, or schedule, that compensates for the wage. The reduced-form estimates measure the wage and hours consequences of formalisation but cannot separate this non-wage value from the search frictions and selection that also move workers across sectors. In the next section, I build and estimate a two-sector search model that prices the within-firm and between-firm transition margins jointly and recovers the non-wage value of a formal job as the residual that rationalises the observed transitions and wages.

3. Model

3.1. Environment

I follow the wage-posting approach of [Burdett and Mortensen, 1998](#) and add a sector-level valuation of amenities as in [Sullivan and To, 2014](#) and [Bradley et al., 2017](#) in partial equilibrium.³¹ The model follows the modelling strategy of [Bradley et al., 2017](#), but in place of the public sector I use formal and informal employment similar to [Meghir et al., 2015](#). The approach uses wage posting, rather than wage bargaining, as there is little evidence that workers hold bargaining power over wages ([Armangué-Jubert et al., 2025](#)) and instead monopsonistic wage-setting is documented for Mexico ([Estefan et al., 2024](#)) and for similar economies ([Amodio and Roux, 2024](#)). As in [Burdett and Mortensen, 1998](#), search frictions prevent workers from moving instantly to better-paying jobs, so a low offered wage does not trigger immediate quits while a high wage attracts more workers; the resulting upward-sloping labour-supply curve is the source of firms' monopsony power. Because I abstract

³¹Similar set-ups in partial equilibrium can be found in [Jolivet et al., 2006](#), [Dey and Flinn, 2008](#) and [Bloemen, 2008](#).

from the firm side of wage-setting, this explicit monopsony channel is absent from my model.

Time is continuous and the economy is in steady state. I abstract from the firm side and assume the formal and informal wage offer distributions are exogenous. Workers are risk-neutral and ex-ante homogeneous. They search on and off the job, and wage offers are of the type take-it-or-leave-it arrive sequentially according to a Poisson process from the known exogenous distributions. In this set-up, workers can receive offers from outside firms and from their current firm; where the last is the foot-in-the-door channel. The worker's decision is to accept or reject according to a reservation-wage strategy. When on-the-job, workers derive flow utility from wages and a linear additive non-wage amenity component. The additive component depicts the trade-off between wage and the non-wage amenity. In addition, workers face an exogenous job destruction shock that implies non-employment. When workers are non-employed they receive a flow value b , assumed homogeneous within a demographic group.

Moreover, I estimate the model across sub-samples of data for four different groups, high skilled males and females, and low skilled males and females and explore the quantitative implications of shutting down the arrival of formal wage offers from the same firm when the worker is holding an informal job. For simplicity, estimating the model separately by gender and skill treats these four cells as segmented labor markets: workers receive offers only from the offer distributions estimated for their own group.

Worker employment and non-employment: Risk-neutral workers can be in one of three states, employed formally (f), employed informally (i), or not employed (n), where the not-employed pool includes both unemployed and inactive workers.³² The number of workers in each state in steady state is denoted N_s for $s \in \{f, i, n\}$. Workers receive take-it-or-leave-it formal and informal wage offers while on and off-the-job and decide whether to accept or not. Offers arrive at Poisson rate $\lambda_{ss'}$, where s is the current state and s' the destination state. When employed in the informal state, the worker receives informal

³²The survey separates the non-employed into the unemployed, who are not working and are actively searching; the *available*, who are not searching but report an interest in working and no impediment to it, essentially discouraged workers; and the *not available*, who either report no interest, or an urgent and non-transferable responsibility that prevents them working, or a permanent incapacity. The not-available are the great majority of the inactive, 86% nationally and slightly more among women. Structural search models of informality usually exclude the inactive altogether, on the grounds that a search model does not account for their behaviour (Meghir et al., 2015; Bobba et al., 2021); descriptive treatments instead keep them as a separate state (Bosch and Maloney, 2010). I retain them inside n because in Mexico they are close to half of the working-age population and the large majority of non-employed women, so excluding them would remove the gender margin that the heterogeneity in this paper is about. Section 3.2 reports what the choice costs.

and formal wage offers from other firms at rates λ_{ii} and λ_{if} , as well as formal wage offers from the same firm at rate η_{if} . The arrival rate η_{if} describes the foot-in-the-door channel, in which informal workers change to formality within the same firm. By symmetry, a worker currently in the formal state receives formal and informal wage offers from other firms at rates λ_{ff} and λ_{fi} , and an informal wage offer from the same firm at rate η_{fi} , the within-firm de-registration of an existing job. Moreover, a worker may receive an offer when not-employed at rate $\lambda_{ns'}$ with $s' \in \{f, i\}$. The exogenous job destruction occurs at rate δ_s for $s \in \{f, i\}$, after which the worker transitions to non-employment n .³³

Wage offers from the formal and informal sector (I use state and sector interchangeably) are draws from an offer distribution $F_s(\cdot)$ for $s = \{f, i\}$. However, the distribution will be parameterised to a specific exogenous distribution with no optimising behaviour of firms. The flow utility derived from working in $s \in \{f, i\}$ is given by $w + a_s$, where w is the hourly wage and a_s is a sector-specific non-wage attribute. The additive specification follows the search and amenities literature (Hwang et al., 1998; Bonhomme and Jolivet, 2009; Sorkin, 2018; Taber and Vejlín, 2020), where quasi-linearity in the utility function reduces the worker's reservation policy to scalars and keeps the on-the-job search problem tractable. The approach has caveats: Mas and Pallais, 2017 and Hall and Mueller, 2018 find evidence in the United States that amenities increase with wages and suggest potential complementarities between wage and non-wage attributes rather than perfect substitution. In this set-up, this is not considered since all workers receive the same benefits, either formal or informal, and this is independent of wages.³⁴ The parameter is closer to the approach of Sullivan and To, 2014, where it is an aggregate of all attributes of formal and informal work, including those related to social security as well as job attributes such as flexibility and proximity to the workplace. Without loss of generality, a_i is normalised to zero, so a_f reflects the relative utility surplus (or loss) of holding a formal job relative to an informal one. This is the two-sector amenity of Bradley et al., 2017, applied to the formal–informal margin rather than the public–private one.

Values and reservation wages: A worker's utility is the present discounted sum of future wages and the non-wage component. The total value of employment is given by the utility flow and the expected value of outside options, within or across sectors and from the same or another firm. For a given worker, the arrival rates are unchanged if they move jobs within a sector. The acceptance decision for an offer within the worker's current sector is therefore determined by the worker's current wage and the new wage being

³³There is no information on the reason for job separation, such as quit or layoff. Thus, I cannot distinguish them from δ .

³⁴There is no data in the Employment survey about non-wage attributes other than those related to the social security records.

offered. If a new wage offer, x , is higher than the worker's current wage, w , the worker will accept and otherwise reject. However, since a change in sector is not only associated with a different wage but also with a change in transitional parameters, the acceptance decision is not so trivial when the job offer is from another sector. Thus, depending on the two sets of transition parameters, an individual may accept a job offer from a different sector with a wage cut, or conversely, require a higher wage in order to accept. These acceptance decisions can be characterised by a set of reservation wages. There will be four corresponding reservation wages, R_{nf} , R_{ni} , $R_{if}(w)$, $R_{fi}(w)$.

Let r denote the discount rate assumed constant across workers. The value function for a formal worker is:

$$\begin{aligned}
 (4) \quad rV_f(w) = & \underbrace{w + a_f}_{\text{wage + amenity}} + \underbrace{\delta_f [V_n - V_f(w)]}_{\text{value of separation}} + \underbrace{\lambda_{ff} \int_w^{+\infty} [V_f(x) - V_f(w)] dF_f(x)}_{\text{formal-to-formal job ladder}} \\
 & + \underbrace{(\lambda_{fi} + \eta_{fi}) \int_{R_{fi}(w)}^{+\infty} [V_i(x) - V_f(w)] dF_i(x)}_{\text{option value of informalisation (between + within)}} .
 \end{aligned}$$

The first two components are the flow payoff, the wage plus the amenity, and the loss of value the worker suffers on separation, weighted by the rate of exogenous job destruction δ_f . The formal worker receives outside formal wage offers x , drawn from the distribution $dF_f(x) = f(x)dx$ at rate λ_{ff} ; this is the formal job ladder. If the offer is greater than the current wage, the worker accepts. Thus, the integral denotes all the accepted offers x that are above the lower bound w . In the case of informal-sector offers, x is drawn from the informal distribution and is accepted if it is above the reservation wage $R_{fi}(w)$. This term denotes the informalisation of formal workers, which may occur either at other firms (λ_{fi}) or within the firm (η_{fi}).

The value of an informal worker employed at wage w is:

$$\begin{aligned}
 (5) \quad rV_i(w) = & \underbrace{w}_{\text{wage}} + \underbrace{\delta_i [V_n - V_i(w)]}_{\text{value of separation}} + \underbrace{\lambda_{ii} \int_w^{+\infty} [V_i(x) - V_i(w)] dF_i(x)}_{\text{informal-to-informal job ladder}} \\
 & + \underbrace{(\lambda_{if} + \eta_{if}) \int_{R_{if}(w)}^{+\infty} [V_f(x) - V_i(w)] dF_f(x)}_{\text{option value of formalisation (between + within)}} .
 \end{aligned}$$

Equation 5 values an informal job at wage w as the wage flow, minus the expected loss from

separation into non-employment at rate δ_i , plus three option values: climbing the informal ladder to a better-paying informal job, moving into formal employment at another firm, and being formalised within the current firm. The formal option pools two routes that the data keep distinct, a between-firm offer at rate λ_{if} and a within-firm reclassification of the same job at rate η_{if} ; both draw a formal wage that is accepted above the threshold $R_{if}(w)$.

Moreover, the value of non-employment is defined as:

$$(6) \quad rV_n = \underbrace{\text{flow value of } n}_b + \underbrace{\lambda_{nf} \int_{R_{nf}}^{+\infty} [V_f(x) - V_n] dF_f(x)}_{\text{formal offer option value}} + \underbrace{\lambda_{ni} \int_{R_{ni}}^{+\infty} [V_i(x) - V_n] dF_i(x)}_{\text{informal offer option value}}$$

Equation 6 is value of non-employment, the flow return rV_n equals the flow value of non-employment b plus the expected gain from search formal and informal jobs. This state n is the value of not holding a private wage job, so it collects the value of searching and the time in inactivity; and since it is recovered from the reservation-wage condition rather than measured, it is the value that rationalises the rate at which arriving offers are refused. A non-employed worker samples formal and informal offers at rates λ_{nf} and λ_{ni} and accepts one only when it raises lifetime value, that is, when the offered wage clears the reservation threshold R_{nf} or R_{ni} . This is the two-sector analogue of the unemployed worker's problem in [Burdett and Mortensen, 1998](#), with a single search channel replaced by sector-specific arrival rates.

The value functions given by Equations 4, 5, and 6 allows to obtain the reservation wage required to move from informal to formal employment and vice versa as a function of the transition parameters. The reservation wages can be pinned down using the indifference condition, which states that there is a wage that makes the worker indifferent between staying informal or changing to formal. This reservation wage $R_{if}(w)$ denotes the informal-to-formal threshold satisfying $V_i(w) = V_f(R_{if}(w))$. The two schedules are inverses of one another, so for every wage w :

$$(7) \quad R_{fi}(R_{if}(w)) = w.$$

The same logic applies to the non-employed worker: the reservation wages R_{nf} and R_{ni} solve the indifference condition $V_n = V_f(R_{nf}) = V_i(R_{ni})$. Applying the inverse identity to this equality yields:

$$(8) \quad R_{fi}(R_{nf}) = R_{ni}, \quad R_{if}(R_{ni}) = R_{nf}.$$

The last expression links the reservation wages across states: the wage at which a non-employed worker accepts a formal job maps, through $R_{fi}(\cdot)$, into the wage at which they accept an informal one. Combining the indifference condition $V_i(w) = V_f(R_{if}(w))$ with the Bellman equations (5)–(4) the reservation wage equation $R_{if}(w)$ is:

$$\begin{aligned}
 (9) \quad R_{if}(w) = & -a_f + \frac{r + \delta_f}{r + \delta_i} w + \left[\frac{(\delta_i - \delta_f)}{r + \delta_i} \right] rV_n \\
 & + \left[\frac{r + \delta_f}{r + \delta_i} \lambda_{ii} - (\lambda_{fi} + \eta_{fi}) \right] \int_w^\infty V'_i(x) [1 - F_i(x)] dx \\
 & + \left[\frac{r + \delta_f}{r + \delta_i} (\lambda_{if} + \eta_{if}) - \lambda_{ff} \right] \int_{R_{if}(w)}^\infty V'_f(x) [1 - F_f(x)] dx.
 \end{aligned}$$

Equation 9 decomposes the motives of an informal worker to accept a formal job (the details are derived in Appendix D.1, including $R_{fi}(w)$). The non-wage component enters with a negative sign, $-a_f$, so a positive estimate reduces the reservation wage and raises the likelihood of accepting a formal offer. The current wage enters with a positive coefficient, since a higher wage in an informal job reduces the likelihood of accepting a formal wage offer. The stability component depends on how workers value non-employment, V_n in Equation 6. When the flow value of non-employment is high, a worker rather be in non-employment. In such case, if informal jobs are destroyed at a higher rate $\delta_i > \delta_f$; the worker values short term employment in the informal sector raising $R_{fi}(w)$.

The second term is the formal option value which is decomposed into two channels. The first is the formal job ladder while holding a formal job λ_{ff} the reduces the reservation wage. The second is the arrival of formal jobs while holding an informal job from other firms λ_{if} and the same firm η_{if} , the foot-in-the-door channel. Higher rates imply a higher volume of offers that the worker can choose from and pick the best; workers can reject less attractive offers to move from informal to formal and this the reservation wage increases.³⁵

Steady-state distributions: In steady state, the flows in and out of any given sector are equal. Steady-state stocks satisfy a single accounting identity, the flow out of non-employment equals the flow into non-employment from both employment states:

$$(11) \quad (\lambda_{nf} + \lambda_{ni}) N_n = \delta_f N_f + \delta_i N_i,$$

³⁵Assuming differentiability, the reservation wage is an ordinary differential equation (ODE):

$$(10) \quad \frac{dR_{if}(w)}{dw} = \frac{r + \delta_f + (\lambda_{fi} + \eta_{fi})\bar{F}_i(w) + \lambda_{ff}\bar{F}_f(R_{if}(w))}{r + \delta_i + \lambda_{ii}\bar{F}_i(w) + (\lambda_{if} + \eta_{if})\bar{F}_f(R_{if}(w))},$$

where $\bar{F}_s(\cdot) = 1 - F_s(\cdot)$ is the survival probability.

together with the adding-up constraint $N_n + N_f + N_i = N$. Because workers are homogeneous within a group, firms never post wages below a worker's reservation wage, so $F_f(R_{nf}) = 0$ and $F_i(R_{ni}) = 0$.³⁶ The steady-state assumption restricts the data as much as the model, since it requires the shares of workers in the three states to be roughly constant over the estimation window: once the transition rates are matched, the employment shares the model implies are the stationary shares of that transition matrix, and they can only reproduce the observed ones if the observed shares are themselves stationary. [Bradley et al., 2017](#) satisfy this by estimating on a five-year window chosen for that property. The twenty years pooled here are a weaker basis for it, and Section 3.2 reports where the assumption fails, which is on the female employment shares rather than on the transitions.

Moreover, Equation 12 depicts the flow balance of workers into and out of informal work. The left-hand side describes the inflow of workers from non-employment and from formal employment, equal to the outflows on the right-hand side: to non-employment, to a formal job (through either between-firm search or within-firm transitions), and to other informal jobs:

$$\begin{aligned}
 (12) \quad & \underbrace{N_n \lambda_{ni} F_i(w)}_{\text{from } n} + \underbrace{N_f (\lambda_{fi} + \eta_{fi}) \int_{R_{nf}}^{R_{if}(w)} [F_i(w) - F_i(R_{fi}(x))] dG_f(x)}_{\text{from formal employment}} \\
 & = \underbrace{N_i \delta_i G_i(w)}_{\text{to } n} + \underbrace{N_i (\lambda_{if} + \eta_{if}) \int_{R_{ni}}^w \bar{F}_f(R_{if}(x)) dG_i(x)}_{\text{to formal job}} + \underbrace{N_i \lambda_{ii} \bar{F}_i(w) G_i(w)}_{\text{to informal job}}.
 \end{aligned}$$

Differentiating the flow balance with respect to w gives the steady-state density condition: at every wage, the density of informal workers balances the inflow of accepted offers against the total exit hazard:

$$(13) \quad \frac{dG_i(w)}{dw} = \frac{f_i(w) \left[N_n \lambda_{ni} + N_f (\lambda_{fi} + \eta_{fi}) G_f(R_{if}(w)) + N_i \lambda_{ii} G_i(w) \right]}{N_i \left[\delta_i + (\lambda_{if} + \eta_{if}) \bar{F}_f(R_{if}(w)) + \lambda_{ii} \bar{F}_i(w) \right]},$$

where $f_i(\cdot)$ is the density of informal offers. The numerator collects everyone who accepts an informal offer at w : the non-employed, who accept any offer above R_{ni} ; formal workers whose informal reservation wage lies below w , the share $G_f(R_{if}(w))$; and informal workers

³⁶With exogenous offer distributions this restriction is imposed rather than derived, since there is no firm problem in the model to deliver it. It is nonetheless the restriction a wage-posting equilibrium would deliver: with a positive vacancy-posting cost and homogeneous workers within a group, an offer below R_{ns} is accepted by no one and is therefore strictly dominated by not posting the vacancy at all. This truncation is what makes the reservation wage the location parameter of F_s in the estimation (Section 3.2).

climbing their own ladder from wages below w , the share $G_i(w)$. The denominator is the rate at which workers at w flow out of their informal job, through destruction, formalisation, or a better informal offer. The term $G_f(R_{if}(w))$ is the share of formal workers whose informal reservation wage lies below w , and it ties the two accepted-wage distributions together: the informal schedule cannot be solved without the formal one, and vice versa. The companion expression for $G_f(R_{if}(w))$, obtained by combining the informal and formal flow balances, closes the system (Appendix D.2, Equation (D5)).

Two features of the flow equations map directly to the reduced-form facts. First, the informal-to-formal flow enters at the combined rate $\lambda_{if} + \eta_{if}$, splitting formalisation into a between-firm draw (λ_{if}) and a within-firm formalisation of the same match (η_{if}); symmetrically, $\lambda_{fi} + \eta_{fi}$ on the formal side. The first fact, that 77% of informal-to-formal transitions occur within the firm, disciplines this split and makes η_{if} the dominant channel and λ_{if} the minor one. Second, whether a formalising worker's wage rises or falls is governed by the position of the drawn formal wage from the distribution $F_f(w)$ relative to reservation wage $R_{if}(w)$ and by the overlap $G_f(R_{if}(w))$. Fact 2, that within-firm formalisation raises the hourly wage only modestly and leaves a large share unchanged or lower, disciplines exactly these objects: a small wage change requires $R_{if}(w)$ close to w and heavy overlap of the two distributions, which in turn informs the amenity value a_f and the options values.

The tuple of objects $\{R_{nf}, R_{ni}, R_{if}(w), R_{fi}(w), G_i(w), G_f(w), N_f, N_i, N_n\}$ and primitives $\{\lambda_{ss'}, \eta_{ss'}, \delta_s, F_f(w), F_i(w), a_f, b, r\}$ for $s \in \{f, i, n\}$ and $s' \in \{f, i\}$, describe the steady-state equilibrium: the four reservation wages $R_{nf}, R_{ni}, R_{if}(w), R_{fi}(w)$ satisfy their reservation equations; the accepted-wage distributions $G_i(w)$ and $G_f(w)$ satisfy the flow-balance conditions; and the employment stocks satisfy the stock-flow condition.

3.2. Estimates

The estimation is based on minimum-distance matching of certain descriptive moments of the data.³⁷ The structural parameters to estimate are:

$$(14) \quad \Theta = \{\delta_i, \delta_f, \lambda_{nf}, \lambda_{ni}, \lambda_{ff}, \lambda_{fi}, \eta_{fi}, \lambda_{if}, \eta_{if}, \lambda_{ii}, F_f(\cdot), F_i(\cdot)\}.$$

The wage offer distributions $F_s(w)$ for $s = \{i, f\}$ are assumed to follow a Generalised Pareto distribution (GPD). Each distribution has scale parameter $\sigma_s > 0$ and shape parameter $\xi_s \in (-\infty, \infty)$ and the support is bounded below by the reservation wage R_{ns} and above by the maximum observed wage $\bar{w}_s$:

³⁷The two-step structure follows the logic of indirect inference (Gourieroux et al., 1993), with the transition probabilities and wage quantiles serving as the auxiliary statistics.

$$F_s(w) = \begin{cases} 1 - \left(1 + \frac{\xi_s(w - R_{ns})}{\sigma_s}\right)^{-\frac{1}{\xi_s}} & \text{if } w \in [R_{ns}, \bar{w}_s] \text{ and } \xi_s \neq 0, \\ 1 - \exp\left(-\frac{w - R_{ns}}{\sigma_s}\right) & \text{if } w \in [R_{ns}, \bar{w}_s] \text{ and } \xi_s = 0, \\ 0 & \text{if } w < R_{ns}, \\ 1 & \text{if } w > \bar{w}_s. \end{cases}$$

The Generalised Pareto distribution reflects the support accurately because its location parameter is the lower bound of the support, which the model requires to be the reservation wage, and because it nests exponential ($\xi_s = 0$) and Pareto ($\xi_s > 0$) tails within a single two-parameter form. Because $F_s(w)$ for $s = \{f, i\}$ places no mass above $\bar{w}_s$, the upper limit of integration in the value and flow equations is effectively $\bar{w}_s$. On the other hand, the lower bound of each offer distribution is therefore the reservation wage of the corresponding sector, and both are estimated. [Meghir et al., 2015](#) instead fix the formal lower bound at the legal hourly minimum wage for Brazil and El Salvador. In the case of Mexico, the legal minimum is in any case a weak floor on pay in this setting. The data shows that 6.7% of formal and 14.5% of informal hourly wage observations in the estimation sample fall below the contemporaneous legal minimum.

Identification: The model is identified from two sets of moments, ten quarterly transition probabilities $\hat{\pi}_{ss'}$ described in Table 2 and the quantiles of the wage distribution of the employment survey data. For the first set of moments, I derive the theoretical transition probabilities of the model to estimate the arrival rates. I assume that job offers arrive as independent Poisson events, thus, an informal worker currently earning wage w exits the informality state due to a job destruction shock, a better informal offer, or a formal offer above the reservation wage from the same or another firm. This implies that the total exit (or hazard) rate can be described as follows: $h_i(w) = \delta_i + \lambda_{ii}\bar{F}_i(w) + (\lambda_{if} + \eta_{if})\bar{F}_f(R_{if}(w))$, with survival probability $\bar{F}_s(w) = 1 - F_s(w)$, that is the probability that an offer draw exceeds the current wage, and $\bar{F}_f(R_{if}(w))$ is the probability that an arriving formal offer exceeds the informal-to-formal reservation wage $R_{if}(w)$. Note that the probability of transition is conditional on the current wage in informal employment; thus the exit probability is weighted by the accepted-wage density $g_i(w)$ and integrated over the full support of the informal distribution. The theoretical one-quarter transition rates take the form:

$$\pi_{ii} = \int_{R_{ni}}^{\infty} \underbrace{\frac{\lambda_{ii}\bar{F}_i(w)}{h_i(w)}}_{\text{conditional move}} \cdot \underbrace{\left(1 - e^{-h_i(w)}\right)}_{\text{exit probability}} \cdot \underbrace{g_i(w)}_{\text{accepted-wage density}} dw,$$

and analogously for the other seven employment-origin moments (see Appendix D.3). For

non-employment:

$$\pi_{ni} = \frac{\lambda_{ni}}{h_n} (1 - e^{-h_n}), \quad \pi_{nf} = \frac{\lambda_{nf}}{h_n} (1 - e^{-h_n}),$$

where $h_n = \lambda_{nf} + \lambda_{ni}$. Note that there is no integration over w since non-employment only yields the flow value b . The arrival and destruction rates are derived from the observed and theoretical probabilities and the distribution parameters of $F_s(\cdot)$ from the quantiles of the observed data.

Algorithm: I implement this identification numerically via a two-step minimum-distance procedure. Let $\theta = \{\xi_f, \sigma_f, \xi_i, \sigma_i, R_{ni}\}$ and $\lambda = \{\delta_i, \delta_f, \lambda_{nf}, \lambda_{ni}, \lambda_{ff}, \lambda_{fi}, \eta_{fi}, \lambda_{if}, \eta_{if}, \lambda_{ii}\}$:

- (a) Given θ , update λ by minimising $Q_2(\lambda \mid \theta) = \sum_{s,s'} ((\hat{\pi}_{ss'} - \pi_{ss'})/\hat{\pi}_{ss'})$, the sum relative residuals across the ten transitions.
- (b) Given λ , update θ by minimising $Q_1(\theta \mid \lambda) = (\mathbf{q}(\hat{G}(\theta, \lambda)) - \mathbf{q})' (\mathbf{q}(\hat{G}(\theta, \lambda)) - \mathbf{q})$, the squared distance between model and data wage quantiles.

The model is estimated separately on each of the four demographic groups. It matches the targeted moments closely: all ten transition probabilities are reproduced for every group (see Appendix Table D1 Panel A). The employment and non-employment rates are untargeted, and comparing them with the observed shares tests the steady-state assumption. It reproduces the male shares within one percentage point for the formal share and two to four points elsewhere; for women it understates both employment states, the informal share by about 43% in both groups, and overstates non-employment correspondingly.

The failure of stationarity is concentrated among women, and its source is that the model treats non-participation as search. State n pools the unemployed with the inactive, of whom 86% nationally report being unavailable for work, and a search model can only rationalise their staying put by driving the arrival rates towards zero and the flow value b towards the wage, which is what the female estimates show. Restricting non-employment to those reporting availability makes the point, the quarterly rate of entering either sector rises from 9.0% to 35.0% for skilled women and from 5.2% to 26.6% for the unskilled, while persistence in non-employment falls from 91.0% to 65.0% and from 94.8% to 73.4%. I do not adopt the restriction, because it discards roughly half of women's separations and would leave formal and informal jobs looking more durable than they are. The reduced-form facts are unaffected either way, the within-firm share of informal-to-formal transitions is 77.1% under the restricted definition against 77.2% here, with no group moving by more than half a percentage point.

Moreover, Figures D1A–D2D compare the empirical and model-implied cumulative distribution function of wages and show a substantial close fit. Given the model moments target well the data moments, I retrieve the arrival and destruction rates and the distribution parameters.

TABLE 4. Arrival, finding and destruction rates estimates

	Female		Male	
	Skilled	Unskilled	Skilled	Unskilled
<i>Formal offer arrival</i>				
λ_{ff}	0.256 [0.242, 0.269]	0.256 [0.244, 0.271]	0.293 [0.280, 0.310]	0.408 [0.384, 0.419]
λ_{if}	0.112 [0.103, 0.122]	0.062 [0.058, 0.070]	0.113 [0.103, 0.120]	0.091 [0.087, 0.098]
η_{if}	0.318 [0.305, 0.335]	0.172 [0.165, 0.187]	0.377 [0.363, 0.388]	0.356 [0.341, 0.366]
<i>Informal offer arrival</i>				
λ_{ii}	0.534 [0.511, 0.563]	0.546 [0.524, 0.569]	0.683 [0.655, 0.714]	1.009 [0.981, 1.035]
λ_{fi}	0.117 [0.108, 0.131]	0.123 [0.104, 0.128]	0.169 [0.153, 0.182]	0.215 [0.198, 0.227]
η_{fi}	0.396 [0.373, 0.417]	0.369 [0.335, 0.377]	0.576 [0.548, 0.605]	0.918 [0.865, 0.943]
<i>Destruction rates</i>				
δ_f	0.355 [0.350, 0.362]	0.383 [0.376, 0.388]	0.209 [0.204, 0.213]	0.172 [0.169, 0.175]
δ_i	1.046 [1.025, 1.065]	1.095 [1.082, 1.119]	0.642 [0.631, 0.656]	0.422 [0.417, 0.429]
<i>Offer arrival from non-employment</i>				
λ_{ni}	0.050 [0.049, 0.050]	0.036 [0.035, 0.036]	0.134 [0.132, 0.135]	0.344 [0.339, 0.348]
λ_{nf}	0.044 [0.044, 0.045]	0.017 [0.017, 0.018]	0.108 [0.106, 0.109]	0.128 [0.125, 0.129]

Source: Author's estimates based on data from ENOE 2005–2025. All rates are quarterly. Brackets report 95% bootstrap percentile confidence intervals from 100 replications. Each replication resamples workers with replacement from the pooled panel, holding the number of workers the same, and re-estimates the model in full; a worker drawn twice enters as two separate individuals.

Structural rates estimates: Table 4 shows the estimated structural arrival rates in quarterly units; the estimates determine the expected waiting time until each type of shock.³⁸

³⁸For a Poisson process, the waiting time until the first event is exponentially distributed with expected value $E(T) = 1/4\lambda$ or $E(T) = 1/(4\delta_s)$.

First, the informal destruction rate δ_i exceeds its formal counterpart δ_f by a factor of 2.5 to 3.1 across groups. The expected waiting time until exogenous formal destruction is about eight months for skilled and unskilled women, 14 months for skilled men, and 17 months for unskilled men. Meanwhile, when holding informal jobs is three months for skilled women and unskilled women, five months for skilled men, and seven months for unskilled men. These magnitudes are consistent with the higher labour-market churning documented for Mexico relative to Brazil where [Meghir et al., 2015](#) report Brazilian waiting times of roughly 1.3 years for informal workers and over 5 years for formal workers, two to three times the durations estimated here. In addition, similar patterns of high turnover are reported in Mexico ([Maloney, 1999](#); [Bosch and Esteban-Pretel, 2012](#)) .

Informal workers have a foot in the door: formal offers arrive from their own employer between 2.8 and 3.9 times more often than from other employers. The waiting time is shorter for men, who receive a formal offer from their employer every eight months in both skill groups, against nine months for skilled women and 1.5 years for unskilled women. The gender gap in access to within-firm formalisation is therefore wide and least favourable for unskilled women. A formal offer from another employer, by contrast, takes between two and four years to arrive in every group. Within-firm informalisation is faster still, arriving between 1.2 and 2.6 times more often than the foot-in-the-door channel: a formal male worker receives an informal offer from his own employer every three to five months, and a formal female worker every seven to eight months.

Note that offers from the same employer are not necessarily accepted. The asymmetry suggests that formal status inside a match is a reversible margin rather than a milestone: formalising a worker follows screening, or waiting until the match has proved valuable, which is the foot-in-the-door mechanism itself, while returning to informality is a contribution-cost adjustment the firm can make without ending the match. The door opens, but it can also close. Although the arrival rates favour informalisation, the observed probabilities still favour informal-to-formal transitions, with 7.8% of informal workers formalising within the firm each quarter against 5.9% moving the other way. The difference is acceptance: a share of the informal offers reaching formal workers fall below $R_{fi}(w)$ and are rejected, while the formal offers reaching informal workers clear $R_{if}(w)$.

The turnover in the informal sector is larger compared to the formal; in line with the transitions probabilities. Informal workers receive other informal job offers every three to six months on average, while formal workers wait about seven to 12 months for a formal offer. These results differ markedly from estimates for Brazil, where formal male workers receive formal offers every 4.8 years and formal females every 7.4 years, while informal

male workers receive informal offers every 14 months and informal females every 16 months (Meghir et al., 2015). Non-employed workers receive job offers less frequently, and informal offers arrive first in every group. Unskilled men receive an informal offer on average every nine months and skilled men every 1.9 years; skilled women wait approximately five years and unskilled women about seven years. Formal offers arrive roughly every two years for unskilled men and every 2.3 years for skilled men, but every 5.7 years for skilled women and 14.3 years for unskilled women. Informal employment is therefore the main entry route out of non-employment.

Multiple arrivals: The results show that when considering multiple arrivals: (i) the expected spell lengths are strictly increasing in wages for both job types, since low-wage workers face a large pool of acceptable outside offers; and (ii) formal spells exceed informal spells at every wage point, reflecting that lower destruction rates and slower within-firm transitions to informality more than offset the differences in job-to-job arrival rates across job types (see Appendix D.6 for details). The non-employment spell reveals a sharp gender gap. For females it sits well above the formal ceiling, skilled women remain non-employed for 2.7 years and unskilled women for 4.7 years on average, longer than any formal or informal spell in their respective distributions. For males this is the opposite, unskilled men exit non-employment in six months and skilled men in 12 months, similar to the ceiling of the formal spell length. This means that for men, non-employment is a transitory state that resolves faster than typical employment spells terminate; for women it is the longest expected state in the labour market, consistent with the substantially lower offer arrival rates λ_{nf} and λ_{ni} documented in Table 4.

Estimates of the flow value of non-employment b and the formal sector amenity a_f : Once the structural parameters are estimated, the flow value of non-employment b and the formal-sector amenity a_f are recovered from the reservation wages as residuals.³⁹ The amenity value has two important characteristics: (i) setting $a_i \equiv 0$ makes a_f the difference

³⁹From the reservation wage equation of non-employment to informal employment R_{ni} :

$$(15) \quad b = R_{ni} - (\lambda_{ni} - \lambda_{ii}) \int_{R_{ni}}^{\infty} V'_i(x) [1 - F_i(x)] dx - (\lambda_{nf} - (\lambda_{if} + \eta_{if})) \int_{R_{nf}}^{\infty} V'_f(x) [1 - F_f(x)] dx.$$

And from the reservation wage of non-employment to formal employment R_{nf} , the amenity value of formality is:

$$(16) \quad a_f = -R_{nf} + b + (\lambda_{nf} - \lambda_{ff}) \int_{R_{nf}}^{\infty} V'_f(x) [1 - F_f(x)] dx + (\lambda_{ni} - (\lambda_{fi} + \eta_{fi})) \int_{R_{ni}}^{\infty} V'_i(x) [1 - F_i(x)] dx.$$

between the two sectors' non-wage flows, not the level of the formal one, so a negative estimate says the formal bundle is worth less than the informal bundle to that group, not that formality amenity is not valuable and (ii) if the result is zero both formal and informal employment have similar amenity values that cancel each other. Table 5 shows the recovered values of the flow value of non-employment b and the amenity a_f surplus and the reservations wages. The recovered flow value of non-employment b is high, at 26.8 pesos per hour for skilled women, 14.3 for unskilled women, 31.0 for skilled men and 18.0 for unskilled men, close to the median hourly wage in every group. The magnitudes map directly onto the, destruction and finding structural rates, the transitions probabilities and the steady state rates of non-employment. Non-employed workers waiting time to receive offers is between two years and 14 years between, there is a probability of 53% and 76% of all moves out of informal employment go to non-employment, against 2% to 5% that go to a formal job at another firm, and non-employment accounts for 79.8% and 88.6% of skilled and unskilled women and 53.3% of skilled men, though only 35.9% of unskilled men.

On the other hand, the formal amenity surplus is positive and substantial for the two high-skilled groups, about 15% of the median informal wage for skilled women and 29% for skilled men. It is negative only for unskilled women, at -14%; for unskilled men the interval spans zero, from -0.29 to 0.31 pesos per hour, so the data do not distinguish their valuation of the formal contract from zero. The negative value indicates that informality is valuable for women, suggesting that flexibility and non-contributory benefits play a role for this group. The results relate to empirical findings on hourly wage gains and the model setup. Equation 4 describes a substitution relation between w and a_f , a positive a_f means workers are willing to accept a lower w . The reduced-form estimates and the recovered amenities line up group by group. The empirical estimates show that the two high-skilled groups see no significant rise in log hourly wages on formalising within the firm and both hold a positive amenity, so what they accept the formal offer for is the non-wage value rather than the wage increase. Unskilled men and women are the opposite where the gains are near 5%, and their amenity is either negative or null, so for them the value of the formal job is in the wage.

Moreover, for both female groups the non-employment-to-informal threshold R_{ni} lies below its formal counterpart, 6.9 against 9.5 pesos per hour for unskilled women and 9.0 against 11.3 for skilled women. For both male groups the two thresholds nearly coincide, 9.2 against 9.3 pesos for the skilled and 8.1 against 7.9 for the unskilled, suggesting that non-employed men are close to indifferent between accepting a formal and an informal

$$\text{where } V'_i(x) = \frac{1}{r+\delta_i+\lambda_{ii}\bar{F}_i(x)+(\lambda_{if}+\eta_{if})\bar{F}_f(R_{if}(x))} \text{ and } V'_f(x) = \frac{1}{r+\delta_f+\lambda_{ff}\bar{F}_f(x)+(\lambda_{fi}+\eta_{fi})\bar{F}_i(R_{fi}(x))}.$$

job. They reach that indifference for opposite reasons. Skilled men hold the largest amenity in the sample, 7.59 pesos per hour, which is almost exactly offset by the option value of informality as a route into formality; for unskilled men both terms are under half a peso, and it is there rather than for the skilled that the near-coincidence of the two thresholds goes with an amenity indistinguishable from zero.

A negative estimate admits two readings: that the informal bundle is worth more than the formal one to that group, or that the formal contract carries little non-wage value in itself. The two are observationally equivalent within the model, and the reduced-form evidence favours the first. Formalisation moves workers towards the standard 48-hour week, so the informal job being given up carries a schedule flexibility the formal job does not, and access to non-contributory social security narrows the gap from the other side by supplying outside the formal contract part of what the contract is meant to provide. The -2.56 pesos per hour recovered for unskilled women is therefore better read as that flexibility being worth more to them than the contributory bundle than as an aversion to formal employment. Separating the two would require a direct measure of non-wage attributes, which the survey does not collect.

A second qualification concerns the wage measure. The survey records take-home pay, so the formal wage entering the flow payoff is already net of the worker-side contribution and income tax, a wedge of approximately 11% (OECD, 2025). The amenity is thus measured after the worker has paid for the bundle, a_f is what the formal job is worth over and above the price charged for it, which is the object an acceptance decision prices. Expressed in gross terms the formal wage would be higher by the tax wedge, so the gross formal-informal gap is wider than the take-home gap observed here, and a decomposition run on gross wages would attribute more of the value of formality to the wage and correspondingly less to the amenity. All four estimates are upper bounds on the gross-wage amenity.

Informal-to-formal reservation wage: In the previous reservation wage estimates, R_{nf} and R_{ni} , do not depend on the wage level. However, workers searching on the job compare the cross-sector reservation wages $R_{if}(w)$ and $R_{fi}(w)$, each a function of the wage currently earned, the formal amenity valuation, the job-stability difference and the job option prospects for each sector. The reservation wage schedule $R_{if}(w)$ is the formal wage that makes an informal worker earning w indifferent between staying informal and changing to formality, expressed in formal-wage units.

Figure 11 plots the indifference schedule $R_{if}(w)$ against the current informal wage

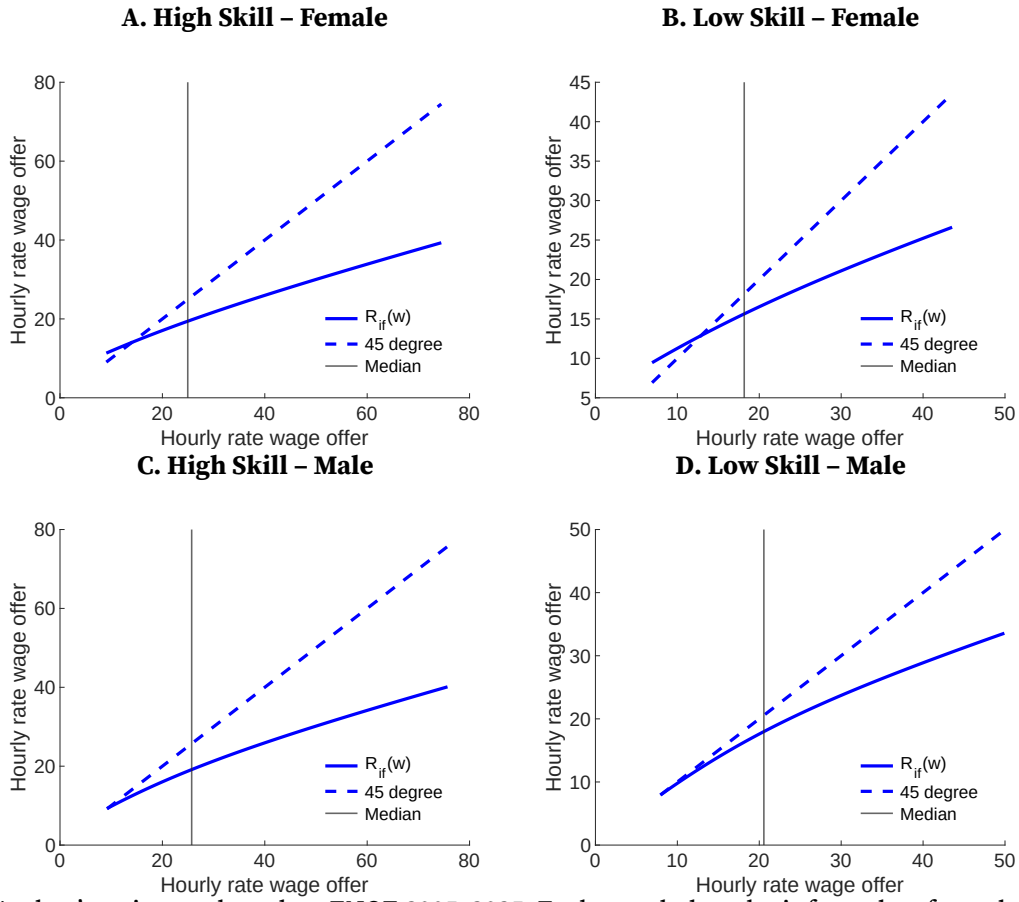
TABLE 5. Model parameter estimates

	Female		Male	
	Skilled	Unskilled	Skilled	Unskilled
Median informal hourly wage (reference)				
<i>Pesos per hour</i>	25.00	18.14	25.78	20.58
Flow value of non-employment (b)				
<i>Pesos per hour</i>	26.755	14.346	30.950	18.046
	[26.272, 27.538]	[14.149, 14.591]	[30.370, 31.771]	[17.957, 18.337]
Amenity (a_f)				
<i>Pesos per hour</i>	3.795	-2.563	7.588	-0.158
	[2.866, 4.335]	[-3.086, -2.281]	[6.624, 8.299]	[-0.281, 0.332]
<i>% of median informal wage</i>	15.2%	-14.1%	29.4%	-0.8%
Reservation wage (R_{nf})				
<i>Pesos per hour</i>	11.330	9.450	9.266	7.935
	[11.151, 12.091]	[9.340, 10.137]	[9.265, 9.566]	[7.932, 8.194]
Reservation wage (R_{ni})				
<i>Pesos per hour</i>	9.047	6.915	9.185	8.119
	[8.556, 9.048]	[6.573, 6.915]	[9.165, 9.475]	[8.004, 8.263]
Other parameters				
<i>Shape</i>				
ξ_f	0.085	-0.051	0.151	0.130
ξ_i	0.069	-0.108	0.116	0.112
<i>Scale</i>				
σ_f	16.798	11.180	15.065	7.139
σ_i	14.058	11.739	11.089	6.541

Source: Author's estimates based on data from ENOE 2005–2025. The flow value of non-employment b and the amenity a_f are recovered from the estimated reservation wages, not estimated directly. Brackets report 95% bootstrap percentile confidence intervals from the 100 replications described in Table 4, propagated through the recovery step; the shape and scale parameters are reported without intervals. The reference row is the median informal hourly wage, which is the denominator used by the percentage row; b and the two reservation wages are reported in pesos per hour only. Because the amenity is recovered as a residual using the estimate of R_{nf} , Appendix D.5 reports the same results fixing R_{nf} to the minimum wage instead of estimating it freely. The ordering of the amenity across groups is unchanged; the level moves, substantially so for low-skilled men.

w , along with the 45-degree line. Any offer above the thick $R_{if}(w)$ line is accepted and any offer below is rejected. Note that for the high-skilled groups, almost all offers in the support are accepted except at the very bottom of the distribution where the worker is close to indifference. Conversely, for the low-skilled a sizable share of offers below the median informal wage are rejected.

FIGURE 11. Reservation wage $R_{if}(w)$



Source: Author's estimates based on ENOE 2005–2025. Each panel plots the informal-to-formal reservation wage $R_{if}(w)$ against the current informal wage; the 45-degree line marks equal pay and formal offers above $R_{if}(w)$ are accepted.

Willingness to pay and the value of formality: I approximate the value of formality by using the difference of the wage the worker is currently earning and the informal-to-formal reservation wage $w - R_{if}(w)$. The difference computes the pay cut or the wage increase the worker is willing to accept/require in order to accept a formal job. A positive value means the informal wage is higher than the reservation wage and thus the informal worker is willing to experience a wage cut to change to a formal job whereas a negative value is the required wage increase to accept a formal job. The difference can be decomposed into formal amenity, job stability and formal and informal job option channels.

$$\begin{aligned}
(17) \quad WTP = w - R_{if}(w) = & a_f + \overbrace{\frac{(\delta_i - \delta_f)}{(r + \delta_i)}}^{\text{Stability}} [w - rV_n] \\
& + \left[\overbrace{\frac{(\lambda_{fi} + \eta_{fi})}{(r + \delta_i)}}^{\text{Inf. fallback}} - \overbrace{\frac{r + \delta_f}{r + \delta_i} \lambda_{ii}}^{\text{Inf. job ladder}} \right] \int_w^\infty V'_i(x) [1 - F_i(x)] dx \\
& + \left[\underbrace{\lambda_{ff}}_{\text{For. job ladder}} - \underbrace{\frac{r + \delta_f}{r + \delta_i} (\lambda_{if} + \eta_{if})}_{\text{Foot-in-the-door}} \right] \int_{R_{if}(w)}^\infty V'_f(x) [1 - F_f(x)] dx
\end{aligned}$$

Figure 12 plots the decomposition as a share of the wage across the informal wage support. The value of a formal job is not a single number, both its size and its composition change systematically as the wage rises. The total rises in every group: at the median informal wage it is 21.9% of that wage for skilled women, 13.2% for unskilled women, 25.6% for skilled men and 13.3% for unskilled men, and by the 90th percentile it lies between 27.9% and 42.3%. The composition changes because the channels do not scale alike. The amenity and the two option channels shrink toward zero as informal wage increases, while the stability differential scales with $w - rV_n$ and grows with wage.

The non-wage formal amenity. The estimated formal-sector amenity differs sharply by skill. At the median informal wage, it is worth 15.2% of earnings for skilled women and 29.4% for skilled men, compared with -14.1% for low-skilled women and -0.8% for low-skilled men. Skilled men have the largest positive amenity in the sample, at 7.59 pesos per hour, whereas low-skilled women are the only group with a clearly negative estimate, at -2.56 pesos per hour. For low-skilled men, the amenity is statistically indistinguishable from zero, implying that the value of formal employment for this group must come primarily from wages, future job opportunities, or employment stability rather than from the formal-sector amenity itself. Because the amenity enters utility additively in levels, its importance declines as wages increase. For skilled women, for example, the amenity is equivalent to 32.4% of wages at the tenth percentile but only 6.7% at the ninetieth percentile. For low-skilled women, the amenity remains negative throughout the wage distribution, although its magnitude relative to earnings becomes smaller, falling from -28.7% at the tenth percentile to -7.1% at the ninetieth percentile.

The job-stability differential. Formal jobs are more durable than informal jobs, but greater stability raises formal job value only when wages are sufficiently high relative to non employment. Since informal jobs are destroyed between 2.5 and 3.1 times as fast as formal ones in every group, the differential is always positive but $w - rV_n$ can be negative or positive. A positive difference suggest that, losing the job is a loss and the durability of a formal job insures against it, thus there is a positive valuation of formality. A negative difference, suggest that the worker would rather be searching from non-employment than hold a job paying low w .

Across groups, job stability is valuable at the top of the informal wage distribution. Workers there are willing to accept a wage cut and take a formal job that provides them with insurance against job loss. Women in particular value this at 32% and 38% of the wage at the top of the distribution, for the skilled and the unskilled respectively, and men at 24% and 25%; conversely, at the bottom of the distribution the valuation turns negative and reduces the willingness to accept a wage cut, since there is an incentive for workers to remain searching in non-employment and receive b . For skilled women the channel moves from -94.9% of the wage at the tenth percentile to 32.4% at the ninetieth, and for skilled men from -135.2% to 23.9%.

The informal job option and informal fallback: The informal job option is positive for all groups, although its contribution declines rapidly as the informal wage increases. At the median informal wage, it is worth 5.4% for skilled women, 4.4% for low-skilled women, 8.2% for skilled men, and 9.4% for low-skilled men. This component captures the difference between access to informal offers while currently informal and access to informal offers after becoming formal. It depends on the informal job ladder, λ_{ii} , and on informal offers received while formally employed, λ_{fi} and η_{fi} . Table 4 shows that this expression is positive for every group: 0.33 and 0.30 for skilled and low-skilled women, and 0.52 and 0.71 for skilled and low-skilled men. Importantly, this is not because informal offers necessarily arrive faster while workers are formally employed. For both female groups, $\lambda_{fi} + \eta_{fi}$ is slightly below λ_{ii} : 0.51 versus 0.53 for skilled women, and 0.49 versus 0.55 for low-skilled women. The positive contribution instead comes from the longer expected duration of formal jobs. Because formal jobs are destroyed 2.5 to 3.1 times less frequently than informal jobs, the factor $\frac{r+\delta_f}{r+\delta_i}$ lies between 0.34 and 0.42 across groups. An informal worker therefore holds access to the informal job ladder only for as long as the informal match survives, whereas a formal worker retains access to informal offers for substantially longer. This makes access to future informal opportunities more valuable from a formal job than from an informal one.

The channel is quantitatively more important for men. At the median informal wage it is worth 8.2% for skilled men and 9.4% for low-skilled men, compared with 5.4% and 4.4% for skilled and low-skilled women. This ordering follows the number of informal offers expected over the duration of a job spell, which ranges from about 2.5 for low-skilled women to 3.3 for skilled men. The contribution is positive at every wage, but, like the amenity component, its importance declines sharply with earnings. For skilled women, for example, it falls from 23.5% of wages at the tenth percentile to 0.4% at the ninetieth percentile. The channel is driven primarily by within-firm reversibility rather than by the outside informal job ladder. A worker who accepts formal status retains the possibility of subsequently returning to an informal arrangement with the same employer. Formalisation is therefore less irreversible than it would be in a model without same-employer status changes.

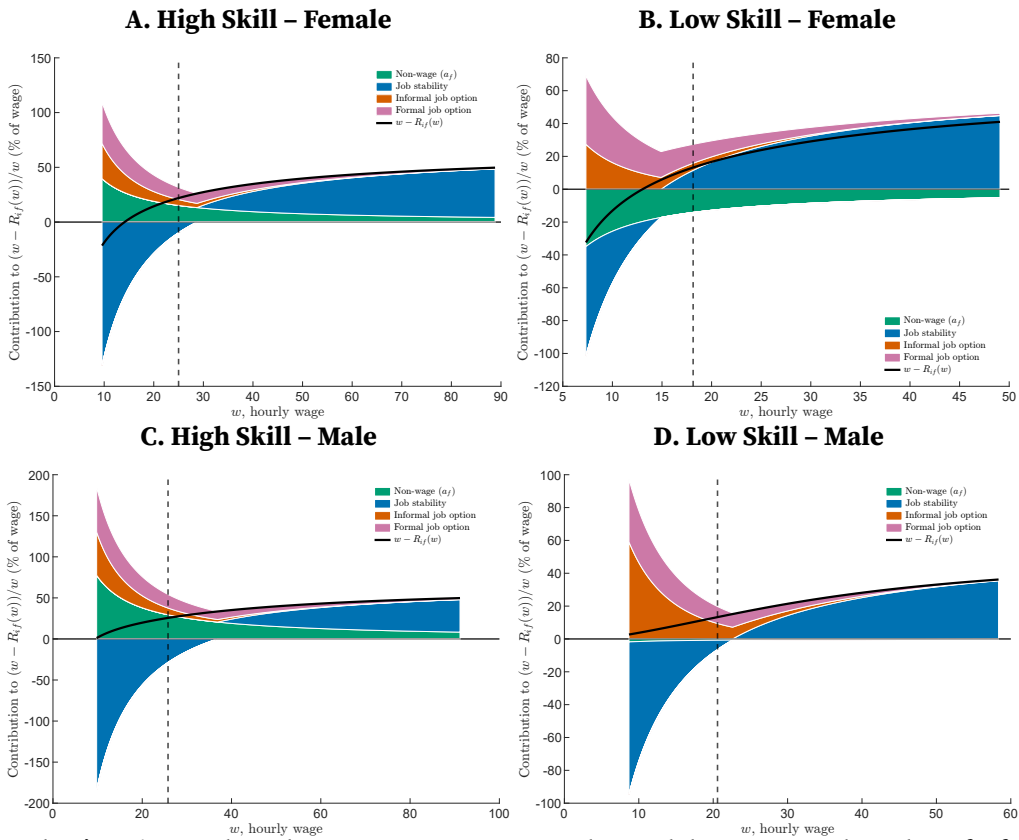
The formal job option and the foot-in-the-door: The formal job option contributes positively to the value of formal employment in every group. At the median informal wage, it is worth 11.1% for skilled women, 11.5% for low-skilled women, 16.6% for skilled men, and 10.3% for low-skilled men. The first term, λ_{ff} , captures the formal job ladder available once a worker is formally employed. The second term captures the formal opportunities already available while the worker is informal, either from outside firms, λ_{if} , or from the current employer, η_{if} . The formal job ladder itself is sizeable. At the median informal wage, it is worth 26.4% for skilled women, 17.0% for low-skilled women, 37.9% for skilled men, and 19.2% for low-skilled men. This value is partly offset because informal workers already have access to formal offers which arrive frequently. The within-firm route is the larger offset in every group. For skilled women, the foot-in-the-door component is worth 11.3% of the median informal wage, compared with 4.0% for outside formal offers; for skilled men, the corresponding values are 16.4% and 4.9%. On its own, the within-firm channel offsets between 24% and 43% of the value of the formal job ladder.

The intuition is straightforward. An informal worker who expects to receive a formal offer from the current employer has less reason to accept a lower wage in order to obtain formal status immediately. The foot-in-the-door channel therefore raises the value of remaining informal by preserving access to future formalisation within the current match. At the median informal wage, this channel is worth 2.8 and 4.2 pesos per hour for skilled women and men, respectively, and 0.7 and 1.5 pesos for low-skilled women and men.⁴⁰ The net formal job option remains positive because the value of the formal job ladder exceeds the value of the formal opportunities already available while informal. As the

⁴⁰Monthly figures scale the hourly amount by the standard formal workweek (4.33×48 hours per month); dollar figures use the 2015–2025 average exchange rate of about 19 pesos per US dollar.

informal wage rises, however, the contribution of this option becomes smaller relative to current earnings.

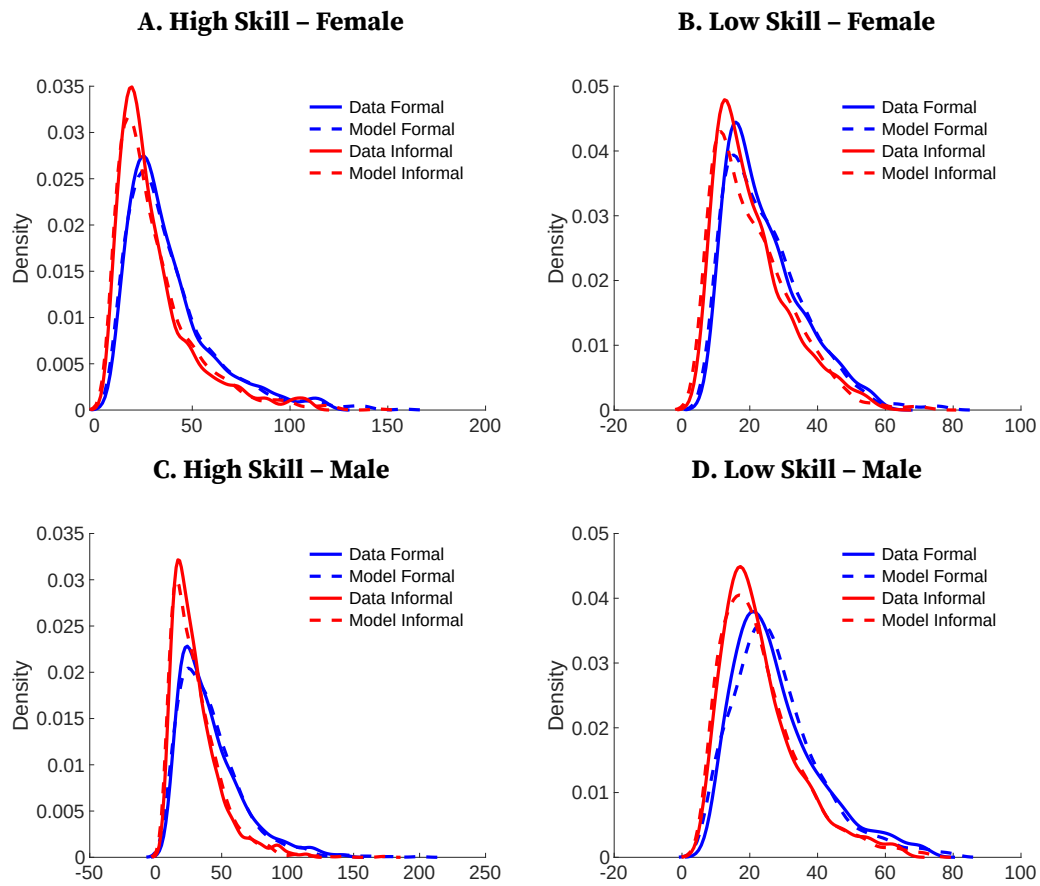
FIGURE 12. Decomposition of $w - R_{if}(w)$



Source: Author's estimates based on ENOE 2005–2025. Each panel decomposes the value of a formal job $w - R_{if}(w)$ into its additive channels across the informal wage support; the dashed line marks the median informal wage.

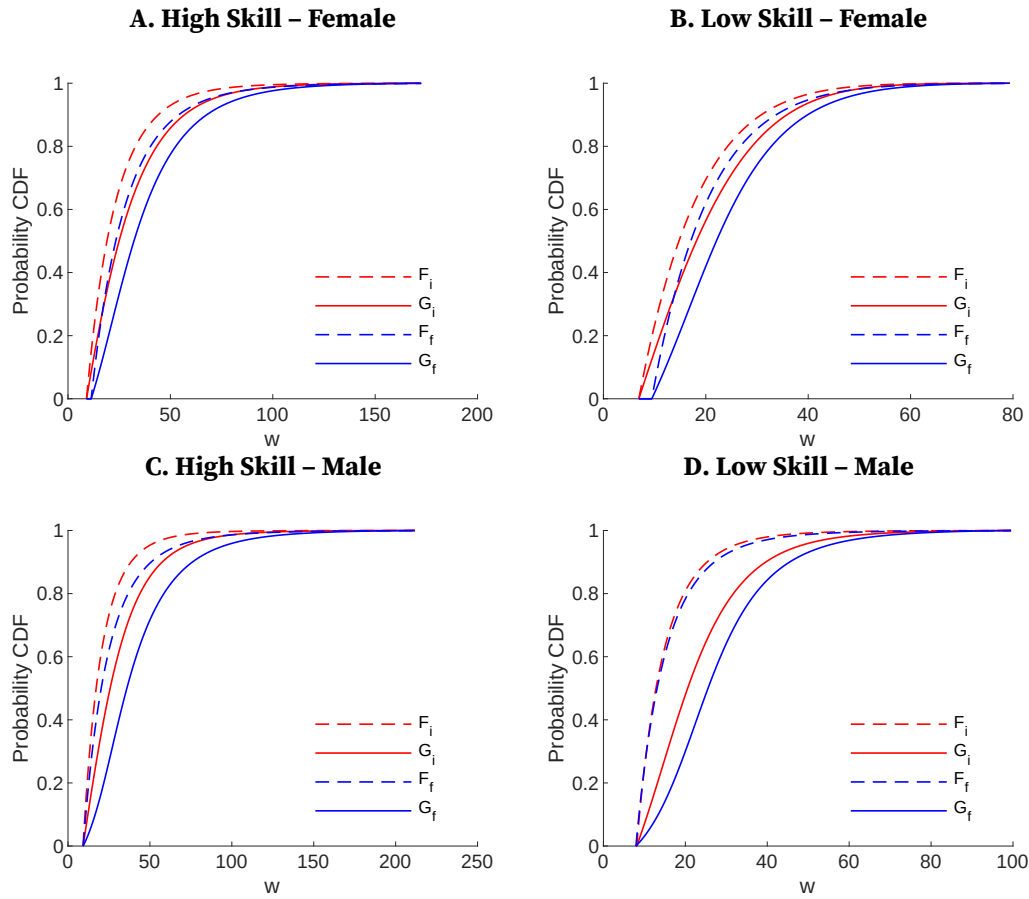
Diagnostics: Figures 13 and 14 report the fit of the estimated model across gender and skill. Figure 13 shows a good fit for both the formal and informal wage distributions, although the model does not match the upper-tail outliers in the data. Figure 14 depicts the estimated offered and accepted distributions. The accepted-wage distribution dominates the offered distribution in every case, reflecting the high estimated arrival rates and the resulting selection on the job ladder. The formal accepted distribution dominates the informal accepted distribution in every group except unskilled women, where the two distributions overlap closely; this reflects the longer non-employment spell of unskilled women, which makes their accepted informal wages closer to the reservation wage and compresses the formal-informal gap.

FIGURE 13. Wage-density fit



Source: Author's estimates based on ENOE 2005–2025. Each panel overlays the model-implied and empirical wage densities for the formal and informal sectors.

FIGURE 14. Offered and accepted wage distributions $G(w)$



Source: Author's estimates based on ENOE 2005–2025. Each panel plots the offered and accepted wage distributions; the accepted distribution dominates the offered one, reflecting selection on the job ladder.

4. Counterfactual: closing the within-firm route

Informal employment is the main route out of non-employment in every demographic group, and among those in informal employment, roughly the expected time to receive a formal offer from their employer is three quarters. For a substantial share of workers the path into formal employment therefore takes the form of non-employment-informality and informality-formality within the same firm. The empirical section showed that this route explains 22.8% of all quarterly entries into formal employment, ranging from 14.2% for unskilled women to 36.2% for unskilled men. Moreover, the informal-to-formal path is consistent with several firm-side channels: the firm may learn about the quality of the match, learn about the worker's general skill from observed performance, or operate an internal hiring scheme. The model does not distinguish between them, however, each generates a rate at which an existing informal match is converted into a formal one at the same employer and the value for the worker. I use a counterfactual experiment to explore, what would be the employment effect of changing the route informal to formal route to non-employment to formality only?

I use a counterfactual that removes the foot-in-the-door channel by setting $\eta_{if} = 0$. Preferences (b and a_f), the discount rate, every remaining arrival and destruction rate, and the two wage-offer distributions are held at their estimated baseline values, the last of these including their lower support points R_{nf} and R_{ni} , which the estimation recovers as the location parameters of F_f and F_i . What the experiment re-solves is the worker's on-the-job problem: the cross-sector reservation schedules $R_{if}(w)$ and $R_{fi}(w)$, the value functions, the accepted-wage distributions and the employment stocks. Informal jobs continue to arrive at the same rate and workers may continue to accept them; what disappears is the conversion of an existing informal match into a formal one. The comparison is across steady states, so the results describe a worker entering the labour market under the new regime rather than the transition path of incumbent workers.

Employment effects. Table 6 shows that closing the channel reduces formal employment in every demographic group. The effect is modest among women, at 2.1 percentage points for skilled women and 0.8 percentage points for unskilled women, and much larger among men, at 8.4 points for skilled men and 13.7 points for unskilled men. The contrast in percentage points overstates the difference in exposure, because women hold far fewer private wage jobs to begin with. Measured against each group's own baseline formal stock, the losses are 18.0% and 17.8% for skilled and unskilled women against 26.6% and 44.9% for skilled and unskilled men: the channel matters less to women than to unskilled men, but by a factor of two rather than the factor of six the level comparison suggests.

The decline in formal employment is absorbed through both higher informality and higher non-employment. Among unskilled men informal employment rises by 9.8 percentage points and non-employment by 3.9; among skilled men the corresponding increases are 4.3 and 4.1 points. The adjustment is substantially smaller for women, at 0.8 and 0.3 points of informality and 1.3 and 0.6 points of non-employment. Closing the route therefore does not merely keep workers informal for longer. Some remain in informal employment, while others no longer find employment sufficiently valuable and withdraw from it altogether.

TABLE 6. Employment composition when the within-firm route is closed

	Female		Male	
	Skilled	Unskilled	Skilled	Unskilled
Formal employment share (%)				
Baseline	11.5	4.6	31.7	30.5
No within-firm transition	9.4	3.8	23.2	16.8
Δ (pp)	-2.07	-0.82	-8.43	-13.69
	[-2.19, -2.00]	[-0.91, -0.79]	[-8.73, -8.16]	[-14.04, -13.31]
Informal employment share (%)				
Baseline	3.9	3.0	11.2	30.8
No within-firm transition	4.7	3.3	15.5	40.7
Δ (pp)	+0.77	+0.27	+4.30	+9.84
	[+0.74, +0.81]	[+0.26, +0.29]	[+4.11, +4.48]	[+9.56, +10.05]
Non-employment share (%)				
Baseline	84.6	92.4	57.1	38.7
No within-firm transition	85.9	92.9	61.2	42.5
Δ (pp)	+1.30	+0.55	+4.13	+3.85
	[+1.24, +1.39]	[+0.53, +0.62]	[+3.97, +4.29]	[+3.72, +3.99]

Source: Author's estimates based on data from ENOE 2005 – 2025. The flow value of non-employment b and the amenity a_f are held fixed at their baseline values; the counterfactual re-solves the worker-side equilibrium with $\eta_{if} = 0$. Shares are percentages of the working-age population in the three modelled states and Δ is in percentage points. 95% bootstrap percentile confidence intervals in brackets.

What the experiment does and does not identify. Several distinct institutional changes would close the within-firm route, and they differ in what else they change. Stronger enforcement requiring registration from the start of a match would prevent firms from using unregistered employment as a probationary period, but would also reduce the rate at which informal jobs arrive at all. A statutory probationary contract, a temporary reduction in dismissal costs, or a contribution rebate for newly registered workers would each relocate the screening inside formal employment, raising the rate at which formal jobs are offered directly from non-employment. Each of these acts on η_{if} , and each also moves at least

one other primitive that the model holds fixed. Which of them is the relevant comparison depends on which of the channels behind the transitions generates the route.

The experiment holds those additional margins fixed, and therefore measures the component common to all of them: the contribution of the transition technology itself to the observed allocation. It is not an evaluation of any of these policies, and a model without a firm side cannot supply the missing terms.⁴¹ What it does supply is the magnitude at stake. The omitted responses work in the opposite direction in every case, since each of these policies closes the route by making direct formal hiring cheaper or compulsory rather than by removing it outright. The estimates are best read as the maximum contraction in formal employment attributable to this margin, and as the benchmark against which any policy acting on it would have to be assessed.

That magnitude is large, particularly for unskilled men, for whom the route supplies over a third of all entries into formal employment and whose formal employment stock falls by nearly half when it is removed. Whether the screening function the informal spell currently performs can be relocated inside a sufficiently flexible formal contract is, on this evidence, the question a formalisation policy has to answer.

⁴¹The firm side does not respond: vacancy creation, wage posting, and job destruction are invariant. Nor is the government budget priced, so the transfer between workers, firms and the social security system that a formalisation policy implies is outside the experiment.

5. Conclusion

This paper studies the value of formal employment in a setting where workers move frequently between formal and informal jobs and where a large share of formalisation occurs within an existing employer–worker relationship. Using the rotating panel of Mexico’s National Survey of Occupation and Employment, I document that 77% of informal-to-formal transitions take place within the same firm, and that formalisation which persists raises hourly wages by only about 5%. The gain is concentrated among low-skilled workers, at 6.6% for men and 4.2% for women, while skilled workers see no statistically significant hourly-wage increase despite formalising at the highest rates. Monthly earnings rise by more, but largely because hours rise: formalisation moves workers towards the standard 48-hour week and away from the more dispersed hours of informal work. Workers therefore accept formal status for a modest hourly gain, or none, and give up schedule flexibility in doing so. The value of formal employment cannot be inferred from the immediate wage change alone.

I use these facts to discipline a two-sector search model in which workers receive formal and informal wage offers both from outside firms and from their current employer. The estimated model implies that the willingness to accept a formal job is the sum of four components whose relative importance differs sharply across groups. The formal-sector non-wage amenity is positive and sizeable for skilled workers, worth 29.4% of the median informal wage for men and 15.2% for women; it is negative for low-skilled women, at –14.1%, and statistically indistinguishable from zero for low-skilled men. Because the amenity is a difference between the two sectors’ non-wage flows, a negative estimate says that the informal bundle is worth more to that group than the formal one, not that formal employment carries no value.

The two remaining components are ones the reduced-form estimates cannot recover. Future job opportunities matter on both sides: the formal job ladder raises the value of becoming formal, while the prospect of a formal offer from the current employer gives informal employment an option value of its own, worth 11.3% and 16.4% of the median informal wage for skilled women and men. A worker who expects to be registered by their current firm has less reason to pay for formal status today. Employment stability, in turn, works against the usual presumption at the bottom of the distribution: for workers earning little, holding a job is worth barely more than searching for another, so the greater durability of a formal job adds nothing and can subtract. The insurance value of formal employment turns positive only higher up the wage distribution, where it becomes the largest of the four components.

The within-firm channel is therefore economically important for two reasons. First, it provides a route into formality that does not require changing employer. Second, because workers can wait for a formal offer from their current employer, it also changes the value of remaining informal. In a partial-equilibrium counterfactual that closes this route while holding preferences and all other rates at their estimated values, formal employment falls by 13.7 percentage points among low-skilled men and 8.4 among skilled men, against roughly one to two points among women — a gap that narrows to a factor of two, rather than six, once each group is measured against its own formal stock. The displaced workers do not simply stay informal for longer; part of the adjustment is a withdrawal from employment altogether. Preventing firms from hiring informally and registering the same match later is usually read as closing a leak from formality; in this model it removes a route into it. Because the firm side does not respond, the estimate is best read as the maximum contraction attributable to this margin rather than as the effect of any particular reform.

More broadly, formalisation is not a single wage transition. Workers value the formal contract differently across demographic groups and across the wage distribution, and their decisions depend on the future opportunities and the stability attached to the job as much as on the wage itself. For skilled workers the non-wage value of formal employment is a substantial part of the decision to formalise. For low-skilled workers the case rests more heavily on wages and on future job opportunities, because the formal-sector amenity is small or negative. Policies that seek to raise formality should therefore be evaluated not only by their effect on wages or registration rates, but by how they change the broader value of the employment relationship.

The paper leaves open what generates the within-firm route in the first place. The model treats the arrival of a formal offer from the current employer as a rate, and is silent on why a firm converts an existing informal match into a formal one: the firm may be learning about the quality of the match or about the worker's general ability, the match itself may be improving through on-the-job accumulation, or registration may be a step in an explicit internal promotion scheme. These channels are observationally similar in the data used here but carry different predictions — learning implies that formalisation follows a period of observed performance, match improvement that pay and productivity rise together, and promotion that registration coincides with a change of role or responsibility. Distinguishing them would require firm identifiers or richer information on job content than the survey provides. It would also complement rather than replace the amenity channel priced here: what a worker is willing to give up for a formal job and what leads a firm to offer one are separate questions, and the observed pattern of formalisation depends

on both.

References

- Abadie, A. (2005). "Semiparametric Difference-in-Differences Estimators". eng. In: *The Review of economic studies* 72.1, pp. 1–19. ISSN: 0034-6527. DOI: [10.1111/0034-6527.00321](https://doi.org/10.1111/0034-6527.00321).
- Amodio, F. and N. de Roux (Oct. 2024). "Measuring Labor Market Power in Developing Countries: Evidence from Colombian Plants". In: *Journal of Labor Economics* 42.4, pp. 949–977. ISSN: 0734-306X. DOI: [10.1086/725248](https://doi.org/10.1086/725248).
- Armangué-Jubert, T., N. Guner, and A. Ruggieri (June 2025). "Labor Market Power and Development". In: *American Economic Review: Insights* 7.2, pp. 177–195. ISSN: 2640-205X. DOI: [10.1257/aeri.20230564](https://doi.org/10.1257/aeri.20230564).
- Babington P., M. and J. Cano-Urbina (2018). "Stock Sampling with Interval-Censored Elapsed Duration: A Monte Carlo Analysis". Florida.
- Bergolo, M. and G. Cruces (2021). "The anatomy of behavioral responses to social assistance when informal employment is high". eng. In: *Journal of public economics* 193, p. 104313. ISSN: 0047-2727. DOI: [10.1016/j.jpubeco.2020.104313](https://doi.org/10.1016/j.jpubeco.2020.104313).
- Berniell, I., L. Berniell, D. de la Mata, M. Edo, and M. Marchionni (2021). "Gender gaps in labor informality: The motherhood effect". eng. In: *Journal of development economics* 150, p. 102599. ISSN: 0304-3878. DOI: [10.1016/j.jdeveco.2020.102599](https://doi.org/10.1016/j.jdeveco.2020.102599).
- (July 2023). "Motherhood and flexible jobs: Evidence from Latin American countries". In: *World Development* 167, p. 106225. ISSN: 0305750X. DOI: [10.1016/j.worlddev.2023.106225](https://doi.org/10.1016/j.worlddev.2023.106225).
- Bloemen, H. G. (Jan. 2008). "Job Search, Hours Restrictions, and Desired Hours of Work". In: *Journal of Labor Economics* 26.1, pp. 137–179. ISSN: 0734-306X. DOI: [10.1086/522069](https://doi.org/10.1086/522069).
- Bobba, M., L. Flabbi, S. Levy, and M. Tejada (Aug. 2021). "Labor market search, informality, and on-the-job human capital accumulation". In: *Journal of Econometrics* 223.2, pp. 433–453. ISSN: 0304-4076. DOI: [10.1016/J.JECONOM.2019.05.026](https://doi.org/10.1016/J.JECONOM.2019.05.026).
- Bonhomme, S. and G. Jolivet (Aug. 2009). "The pervasive absence of compensating differentials". In: *Journal of Applied Econometrics* 24.5, pp. 763–795. ISSN: 0883-7252. DOI: [10.1002/jae.1074](https://doi.org/10.1002/jae.1074).
- Bosch, M. and R. Campos-Vazquez (2014). "The Trade-Offs of Welfare Policies in Labor Markets with Informal Jobs: The Case of the "Seguro Popular" Program in Mexico". In: *American Economic Journal: Economic Policy* 6.4, pp. 71–99. ISSN: 1945774X. DOI: [10.1257/POL.6.4.71](https://doi.org/10.1257/POL.6.4.71). URL: <https://EconPapers.repec.org/RePEc:aea:aejpol:v:6:y:2014:i:4:p:71-99>.
- Bosch, M. and J. Esteban-Pretel (July 2012). "Job creation and job destruction in the presence of informal markets". In: *Journal of Development Economics* 98.2, pp. 270–286. ISSN: 0304-3878. DOI: [10.1016/J.JDEVECO.2011.08.004](https://doi.org/10.1016/J.JDEVECO.2011.08.004).

- Bosch, M. and W. F. Maloney (Aug. 2010). “Comparative analysis of labor market dynamics using Markov processes: An application to informality”. In: *Labour Economics* 17.4, pp. 621–631. ISSN: 0927-5371. DOI: [10.1016/J.LABECO.2010.01.005](https://doi.org/10.1016/J.LABECO.2010.01.005).
- Bradley, J., F. Postel-Vinay, and H. Turon (2017). “Public sector wage policy and labor market equilibrium: A structural model”. eng. In: *Journal of the European Economic Association* 15.6, pp. 1214–1257. ISSN: 1542-4766. DOI: [10.1093/jeea/jvw026](https://doi.org/10.1093/jeea/jvw026).
- Burdett, K. and D. T. Mortensen (1998). “Wage Differentials, Employer Size, and Unemployment”. eng. In: *International economic review (Philadelphia)* 39.2, pp. 257–273. ISSN: 0020-6598. DOI: [10.2307/2527292](https://doi.org/10.2307/2527292).
- Callaway, B. and P. H. Sant’Anna (2021). “Difference-in-Differences with multiple time periods”. eng. In: *Journal of econometrics* 225.2, pp. 200–230. ISSN: 0304-4076. DOI: [10.1016/j.jeconom.2020.12.001](https://doi.org/10.1016/j.jeconom.2020.12.001).
- Camacho, A., E. Conover, and A. Hoyos (Jan. 2014). “Effects of Colombia’s social protection system on workers’ choice between formal and informal employment”. In: *World Bank Economic Review* 28.3, pp. 446–466. ISSN: 1564698X. DOI: [10.1093/WBER/LHT028](https://doi.org/10.1093/WBER/LHT028).
- Chaisemartin, C. de and X. D’Haultfoeuille (2020). “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects”. eng. In: *The American economic review* 110.9, pp. 2964–2996. ISSN: 0002-8282. DOI: [10.1257/aer.20181169](https://doi.org/10.1257/aer.20181169).
- Chaurey, R., G. Chiplunkar, and V. Soundararajan (Dec. 2025). “Employment Protection Legislation and Informality: Theory and Evidence from India”. In: *Journal of Political Economy Microeconomics*. ISSN: 2832-9368. DOI: [10.1086/740101](https://doi.org/10.1086/740101).
- De Brauw, A., D. Gilligan, J. Hoddinott, and S. Roy (Apr. 2015). “Bolsa Família and Household Labor Supply”. In: *Economic Development and Cultural Change* 63.3, pp. 423–457. ISSN: 00130079. DOI: [10.1086/680092](https://doi.org/10.1086/680092). URL: <https://www.journals.uchicago.edu/doi/10.1086/680092>.
- Del Valle, A. (May 2021). “The effects of public health insurance in labor markets with informal jobs: Evidence from Mexico”. In: *Journal of Health Economics* 77, p. 102454. ISSN: 01676296. DOI: [10.1016/j.jhealeco.2021.102454](https://doi.org/10.1016/j.jhealeco.2021.102454).
- Dey, M. and C. Flinn (2008). “Household search and health insurance coverage”. eng. In: *Journal of econometrics* 145.1, pp. 43–63. ISSN: 0304-4076. DOI: [10.1016/j.jeconom.2008.05.013](https://doi.org/10.1016/j.jeconom.2008.05.013).
- Donovan, K., W. J. Lu, and T. Schoellman (Sept. 2023). “Labor Market Dynamics and Development”. In: *The Quarterly Journal of Economics* 138.4, pp. 2287–2325. ISSN: 0033-5533. DOI: [10.1093/qje/qjad019](https://doi.org/10.1093/qje/qjad019).
- Engbom, N. (2022). “Labor Market Fluidity and Human Capital Accumulation”. eng. In: *NBER Working Paper Series*. ISSN: 0898-2937. DOI: [10.3386/w29698](https://doi.org/10.3386/w29698).

- Estefan, A., R. Gerhard, J. Kaboski, I. Kondo, and W. Qian (Jan. 2024). *Outsourcing Policy and Worker Outcomes: Causal Evidence from a Mexican Ban*. Tech. rep. Cambridge, MA: National Bureau of Economic Research. DOI: [10.3386/w32024](https://doi.org/10.3386/w32024).
- Falco, P., W. F. Maloney, B. Rijkers, and M. Sarrias (Mar. 2015). “Heterogeneity in subjective wellbeing: An application to occupational allocation in Africa”. In: *Journal of Economic Behavior & Organization* 111, pp. 137–153. ISSN: 01672681. DOI: [10.1016/j.jebo.2014.12.022](https://doi.org/10.1016/j.jebo.2014.12.022).
- Fan, L. and D. Peng (2019). “A Bracketing Relationship between Difference-in-Differences and Lagged-Dependent-Variable Adjustment”. In: *Political Analysis* 27.4, pp. 605–615.
- Farber, H. S. and R. Gibbons (1996). “Learning and Wage Dynamics”. eng. In: *The Quarterly journal of economics* 111.4, pp. 1007–1047. ISSN: 0033-5533. DOI: [10.2307/2946706](https://doi.org/10.2307/2946706).
- Felix, M., B. Marcoje, I. Matavelli, and M. C. Rodrigues (May 2026). “Employment Preferences of Favela Residents”. In: *AEA Papers and Proceedings* 116, pp. 156–161. ISSN: 2574-0768. DOI: [10.1257/pandp.20261045](https://doi.org/10.1257/pandp.20261045).
- Garganta, S. and L. Gasparini (2015). “The impact of a social program on labor informality: The case of AUH in Argentina”. eng. In: *Journal of development economics* 115, pp. 99–110. ISSN: 0304-3878. DOI: [10.1016/j.jdeveco.2015.02.004](https://doi.org/10.1016/j.jdeveco.2015.02.004).
- Ginja, R., G. Conti, and R. Narita (Dec. 2025). “The Value of Health Insurance: A Household Job Search Approach”. In: *Journal of Labor Economics*. ISSN: 0734-306X. DOI: [10.1086/740191](https://doi.org/10.1086/740191).
- Gourieroux, C., A. Monfort, and E. Renault (1993). “Indirect inference”. eng. In: *Journal of applied econometrics (Chichester, England)* 8.S1, S85–S118. ISSN: 0883-7252. DOI: [10.1002/jae.3950080507](https://doi.org/10.1002/jae.3950080507).
- Hall, R. and A. Mueller (2018). “Wage Dispersion and Search Behavior: The Importance of Nonwage Job Values”. eng. In: *The Journal of political economy* 126.4, pp. 1594–1637. ISSN: 0022-3808. DOI: [10.1086/697739](https://doi.org/10.1086/697739).
- Heckman, J. (1997). “Instrumental variables: A study of implicit behavioral assumptions used in making program evaluations”. In: *Journal of Human Resources* 32.3. ISSN: 0022166X. DOI: [10.2307/146178](https://doi.org/10.2307/146178).
- Hirano, K. and G. W. Imbens (2001). “Estimation of Causal Effects using Propensity Score Weighting: An Application to Data on Right Heart Catheterization”. eng. In: *Health services and outcomes research methodology* 2.3-4, p. 259. ISSN: 1387-3741. DOI: [10.1023/a:1020371312283](https://doi.org/10.1023/a:1020371312283).
- Hwang, H.-s., D. T. Mortensen, and W. R. Reed (1998). “Hedonic Wages and Labor Market Search”. eng. In: *Journal of labor economics* 16.4, pp. 815–847. ISSN: 0734-306X. DOI: [10.1086/209907](https://doi.org/10.1086/209907).
- ILO (2026). “Employment and social trends 2026”. Geneva. DOI: [10.54394/AHRT2681](https://doi.org/10.54394/AHRT2681).
- Jolivet, G., F. Postel-Vinay, and J.-M. Robin (May 2006). “The empirical content of the job search model: Labor mobility and wage distributions in Europe and the US”. In:

- European Economic Review* 50.4, pp. 877–907. ISSN: 00142921. DOI: [10.1016/j.euroecorev.2006.02.005](https://doi.org/10.1016/j.euroecorev.2006.02.005).
- Joubert, C. (May 2015). “Pension Design with a Large informal Labor Market: Evidence from Chile”. In: *International Economic Review* 56.2, pp. 673–694. ISSN: 00206598. DOI: [10.1111/iere.12118](https://doi.org/10.1111/iere.12118).
- Jovanovic, B. (1979). “Job Matching and the Theory of Turnover”. eng. In: *The Journal of political economy* 87.5, pp. 972–990. ISSN: 0022-3808. DOI: [10.1086/260808](https://doi.org/10.1086/260808).
- Juarez, L. (2012). “Are Informal Workers Compensated for the Lack of Fringe Benefits? Free Health Care as an Instrument for Formality”. Mexico DF.
- Kaplan, D. and S. Joyce (Jan. 2008). “Enforceability of Labor Law: Evidence from a Labor Court in Mexico”.
- Kumler, T., E. Verhoogen, and J. Frías (Dec. 2020). “Enlisting Employees in Improving Payroll Tax Compliance: Evidence from Mexico”. In: *The Review of Economics and Statistics* 102.5, pp. 881–896. ISSN: 0034-6535. DOI: [10.1162/restf-1a-000907](https://doi.org/10.1162/restf-1a-000907).
- Lancaster, T. (1990). *The econometric analysis of transition data*. eng. Econometric Society monographs ; no. 17. Cambridge: Cambridge University Press. ISBN: 0521265967.
- Lazear, E., S. Altmann, and K. Zimmermann (2011). *Inside the firm : contributions to personnel economics*. Oxford University Press, p. 539. ISBN: 9780198779964.
- Lazear, E. and R. Moore (May 1984). “Incentives, Productivity, and Labor Contracts”. In: *Quarterly Journal of Economics* 99.2, p. 275. ISSN: 00335533. DOI: [10.2307/1885526](https://doi.org/10.2307/1885526).
- Lazear, E. (Dec. 1979). “Why Is There Mandatory Retirement?” In: *Journal of Political Economy* 87.6, pp. 1261–1284. ISSN: 0022-3808. DOI: [10.1086/260835](https://doi.org/10.1086/260835).
- Leyva, G. and C. Urrutia (2020). “Informality, labor regulation, and the business cycle”. eng. In: *Journal of international economics* 126, p. 103340. ISSN: 0022-1996. DOI: [10.1016/j.jinteco.2020.103340](https://doi.org/10.1016/j.jinteco.2020.103340).
- Mahmud, M., I. A. Gutierrez, K. B. Kumar, and S. Nataraj (Apr. 2021). “What Aspects of Formality Do Workers Value? Evidence from a Choice Experiment in Bangladesh”. In: *The World Bank Economic Review* 35.2, pp. 303–327. ISSN: 0258-6770. DOI: [10.1093/wber/lhz046](https://doi.org/10.1093/wber/lhz046).
- Maloney, W. F. (1999). “Does Informality Imply Segmentation in Urban Labor Markets? Evidence from Sectoral Transitions in Mexico”. eng. In: *The World Bank economic review* 13.2, pp. 275–302. ISSN: 0258-6770. DOI: [10.1093/wber/13.2.275](https://doi.org/10.1093/wber/13.2.275).
- Maloney, W. F., G. Beylis, N. Gonzalez-Prieto, and A. Zambrano (June 2026). *Rationalizing Informality: Social Protection, Job Quality, and Growth*. Washington, DC: World Bank. ISBN: 978-1-4648-2322-0. DOI: [10.1596/978-1-4648-2322-0](https://doi.org/10.1596/978-1-4648-2322-0).
- Mas, A. (Apr. 2025a). *Non-Wage Amenities*. Tech. rep. Cambridge, MA: National Bureau of Economic Research. DOI: [10.3386/w33643](https://doi.org/10.3386/w33643).

- Mas, A. (2025b). “Non-wage amenities”. In: *Handbook of Labor Economics*. Ed. by C. Dustmann and T. Lemieux. Vol. 6. Chap. 5, pp. 373–446. DOI: [10.1016/bs.heslab.2025.07.002](https://doi.org/10.1016/bs.heslab.2025.07.002).
- Mas, A. and A. Pallais (2017). “Valuing Alternative Work Arrangements”. eng. In: *The American economic review* 107.12, pp. 3722–3759. ISSN: 0002-8282. DOI: [10.1257/aer.20161500](https://doi.org/10.1257/aer.20161500).
- (2020). “Alternative Work Arrangements”. eng. In: *Annual review of economics* 12.1, pp. 631–658. ISSN: 1941-1383. DOI: [10.1146/annurev-economics-022020-032512](https://doi.org/10.1146/annurev-economics-022020-032512).
- Meghir, C., R. Narita, and J.-M. Robin (2015). “Wages and Informality in Developing Countries”. eng. In: *The American economic review* 105.4, pp. 1509–1546. ISSN: 0002-8282. DOI: [10.1257/aer.20121110](https://doi.org/10.1257/aer.20121110).
- Menezes-Filho, N. and R. Narita (Feb. 2025). “Labor market turnover and inequality in Latin America”. In: *Oxford Open Economics* 4.Supplement_1, pp. i349–i375. ISSN: 2752-5074. DOI: [10.1093/ooec/odae027](https://doi.org/10.1093/ooec/odae027).
- OECD (Apr. 2025). *Taxing Wages 2025*. OECD Publishing. ISBN: 9789264394131. DOI: [10.1787/b3a95829-en](https://doi.org/10.1787/b3a95829-en).
- (Apr. 2026). *Taxing Wages 2026*. Tech. rep. Paris: OECD Publishing. DOI: [10.1787/3a5169ef-en](https://doi.org/10.1787/3a5169ef-en).
- OIT (2013). *La medición de la informalidad: Manual estadístico sobre el sector informal y el empleo informal Organización Internacional del Trabajo*. Tech. rep. Geneva: International Organization of Labor.
- Perez, J. (2020). “The minimum wage in formal and informal sectors: Evidence from an inflation shock”. eng. In: *World development* 133, p. 104999. ISSN: 0305-750X. DOI: [10.1016/j.worlddev.2020.104999](https://doi.org/10.1016/j.worlddev.2020.104999).
- Perry, G., W. Maloney, O. Arias, P. Fajnzylber, A. Mason, and J. Saavedra-Chanduvi (2007). *Informality: Exit and Exclusion*. Washington, DC: World Bank Latin American and Caribbean Studies.
- Rambachan, A. and J. Roth (Sept. 2023). “A More Credible Approach to Parallel Trends”. In: *Review of Economic Studies* 90.5, pp. 2555–2591. ISSN: 0034-6527. DOI: [10.1093/restud/rdad018](https://doi.org/10.1093/restud/rdad018).
- Rosen, S. (1974). “Hedonic prices and implicit markets: product differentiation in pure competition”. ger. In: *The Journal of political economy* 82.1, pp. 34–55. ISSN: 0022-3808.
- (Jan. 1986). “Chapter 12 The theory of equalizing differences”. In: *Handbook of Labor Economics* 1.C, pp. 641–692. ISSN: 1573-4463. DOI: [10.1016/S1573-4463\(86\)01015-5](https://doi.org/10.1016/S1573-4463(86)01015-5).
- Samaniego, B. and L. B. Fernández (2024). “Increasing the Cost of Informal Employment: Evidence from Mexico”. eng. In: *American economic journal. Applied economics* 16.1, pp. 377–411. ISSN: 1945-7782. DOI: [10.1257/app.20200763](https://doi.org/10.1257/app.20200763).

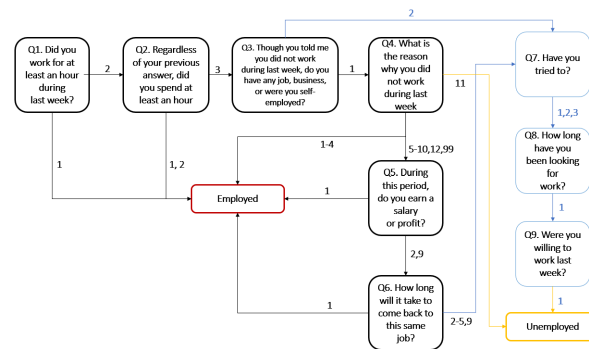
- Sant'Anna, P. H. and J. Zhao (2020). "Doubly robust difference-in-differences estimators". eng. In: *Journal of econometrics* 219.1, pp. 101–122. ISSN: 0304-4076. DOI: [10.1016/j.jeconom.2020.06.003](https://doi.org/10.1016/j.jeconom.2020.06.003).
- Sorkin, I. (2018). "Ranking Firms Using Revealed Preference". eng. In: *The Quarterly journal of economics* 133.3, pp. 1331–1393. ISSN: 0033-5533. DOI: [10.1093/qje/qjy001](https://doi.org/10.1093/qje/qjy001).
- Sullivan, P. (Feb. 2010). "A Dynamic Analysis of Educational Attainment, Occupational Choices, and Job Search". In: *International Economic Review* 51.1, pp. 289–317. ISSN: 0020-6598. DOI: [10.1111/j.1468-2354.2009.00580.x](https://doi.org/10.1111/j.1468-2354.2009.00580.x).
- Sullivan, P. and T. To (2014). "Search and Nonwage Job Characteristics". eng. In: *The Journal of human resources* 49.2, pp. 472–507. ISSN: 0022-166X. DOI: [10.3368/jhr.49.2.472](https://doi.org/10.3368/jhr.49.2.472).
- (Mar. 2026). "Job Dispersion and Compensating Wage Differentials". In: *ILR Review* 79.2, pp. 280–309. ISSN: 0019-7939. DOI: [10.1177/00197939251383858](https://doi.org/10.1177/00197939251383858).
- Taber, C. and R. Vejlín (2020). "Estimation of a Roy/Search/Compensating Differential Model of the Labor Market". eng. In: *Econometrica* 88.3, pp. 1031–1069. ISSN: 0012-9682. DOI: [10.3982/ECTA14441](https://doi.org/10.3982/ECTA14441).
- Ulyssea, G. (2018). "Firms, Informality, and Development". eng. In: *The American economic review* 108.8, pp. 2015–2047. ISSN: 0002-8282. DOI: [10.1257/aer.20141745](https://doi.org/10.1257/aer.20141745).
- Ulyssea, G., M. Bobba, and L. Gadenne (Apr. 2025). "Informality".
- Zárate, R. D. (Mar. 2022). *Spatial Misallocation, Informality, and Transit Improvements: Evidence from Mexico City*. World Bank, Washington, DC. DOI: [10.1596/1813-9450-9990](https://doi.org/10.1596/1813-9450-9990).

Appendix A. Definition of type of employment and informality

A.1. Classification of employment, unemployment and informality

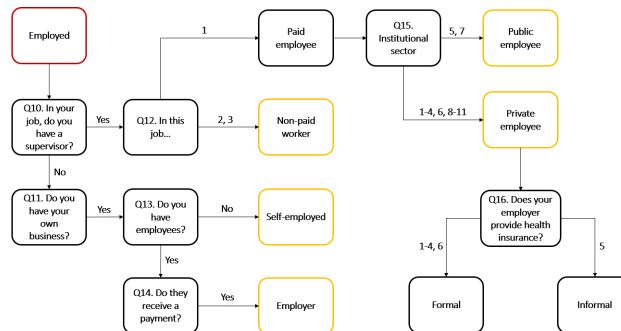
Figures A1 and A2 depict the flowchart of questions used to classify employment and informality. The first describes how workers are classified as employed and unemployed. Those who worked in the week before the interview or did not work, but have a job and will return eventually are classified as employed. On the other hand, unemployed did not worked, are willing to work and actively searched for a job for some period. The second figure describes who employed are classified to as informal using two main conditions: (i) work in a private business in a clear subordinate relationship and (ii) the worker does not receive coverage of the social security system (Spanish acronym).

FIGURE A1. Flowchart of questions to classify workers into the main occupational categories (Employed, Unemployed)



Source: Author's estimates, based on data from ENOE 2005–2025.

FIGURE A2. Flowchart of Questions to Classify Workers into the types of workers and formality status



Source: Author's estimates, based on data from ENOE 2005–2025.

Q1: Did you work for at least an hour during last week?

1. Yes
2. No

Q2: Regardless of your previous answer, did you spend at least an hour

1. doing any remunerated activity?
2. helping with the plot or in a family's or other person's business?
3. did not work during last week

Q3: Though you told me you did not work during last week, do you have any job, business, or were you self-employed?

1. doing any remunerated activity?
2. helping with the plot or in a family's or other person's business?
3. did not work during last week

Q4: What is the reason why you did not work during last week.

1. Strike
2. Technical shutdown
3. Temporal suspension of activities (wage-earners)
4. Attending training courses (continues in the next column)
5. Vacations
6. On leave, sickness, taking care of personal matters
7. Lack of vehicle or machine not working

8. Lack of materials, financing, or clients

9. Bad weather or natural event

10. End of season

11. Will start a job or new business

12. Other reasons

99. Don't know

Q5: During this period, do you earn a salary or profit?

1. Yes
2. No
3. Don't know

Q6: How long will it take to come back to this same job?

1. I already came back this week
2. In four weeks or so
3. In more than four weeks
4. It is not certain whether I will come back or resume activities
5. Will not come back
9. Don't know

Q7: Have you tried to?

1. look for work in another country or make arrangements to cross the border?
2. look for work here in the country?
3. start a business or do any activity in your own account without even starting?
4. Then, don't you have try to look for work?
9. Don't know.

Q8: How long have you been looking for work?

1. Up to 1 month
2. Between 1 and 2 months
3. Between 2 and 3 months
4. More than 3 months
9. Don't know

Q9: Were you willing to work last week?

1. Yes
2. No
9. Don't know

Classification of employed informal or formal

Q10: In your job do you have a supervisor?

1. Yes
2. No
9. Don't know

Q11: Do you have your own business?

1. Yes
2. No
9. Don't know

Q12: In this job...

1. Do you receive a payment?
2. Are you a family non-paid worker?
3. Are you a non-paid worker?

Q13: Do you have employees?

1. Yes
2. No
9. Don't know

Q14: Do they receive a payment?

1. Yes
2. No
9. Don't know

Q15: Type of industry

1. Agriculture.
2. Education or health.
3. Public or non-profit.
4. Private business.
5. Undetermined.

Q16: Due you have access to health insurance in

1. Mexican Institute of Social Security (IMSS, Spanish acronym).
2. Special regime (e.g Military).
3. Institute for Public Workers' Security and Social Services (ISSSTE, spanish acronym).
4. Institute for Workers' Security and Social Services at the State Level (ISSSTE State, Spanish acronym).
5. Other medical institutions.
6. Does not receive medical coverage.
7. Don't know.

TABLE B1. Labour-market state composition by gender, age range and sample

	Full sample				Stock sample			
	All ages		Prime 20–60		All ages		Prime 20–60	
	Men	Women	Men	Women	Men	Women	Men	Women
Formal	21.1	11.1	27.5	14.6	20.5	10.5	26.7	13.7
Informal	22.1	7.6	24.6	8.7	22.3	7.3	24.9	8.4
Self-employed	16.8	10	18.2	11.6	17.2	10	18.6	11.5
Public employee	6.9	6.1	8.9	8.1	7.1	6.1	9.2	8.2
Employer	4.7	1.1	5.5	1.4	4.6	1	5.4	1.3
Non-paid	2.7	3.3	2	3.5	3	3.6	2.1	3.7
Unemployed	3.1	1.8	3.5	2.2	3.2	1.8	3.5	2.2
Inactive	22.5	59	9.8	49.9	22.1	59.7	9.6	51.1

Source: Author's estimates based on the pooled data from ENOE 2005-2025.

Appendix B. Job transitions

B.1. Composition of employment and non-employment by gender

B.2. Computation of tenure

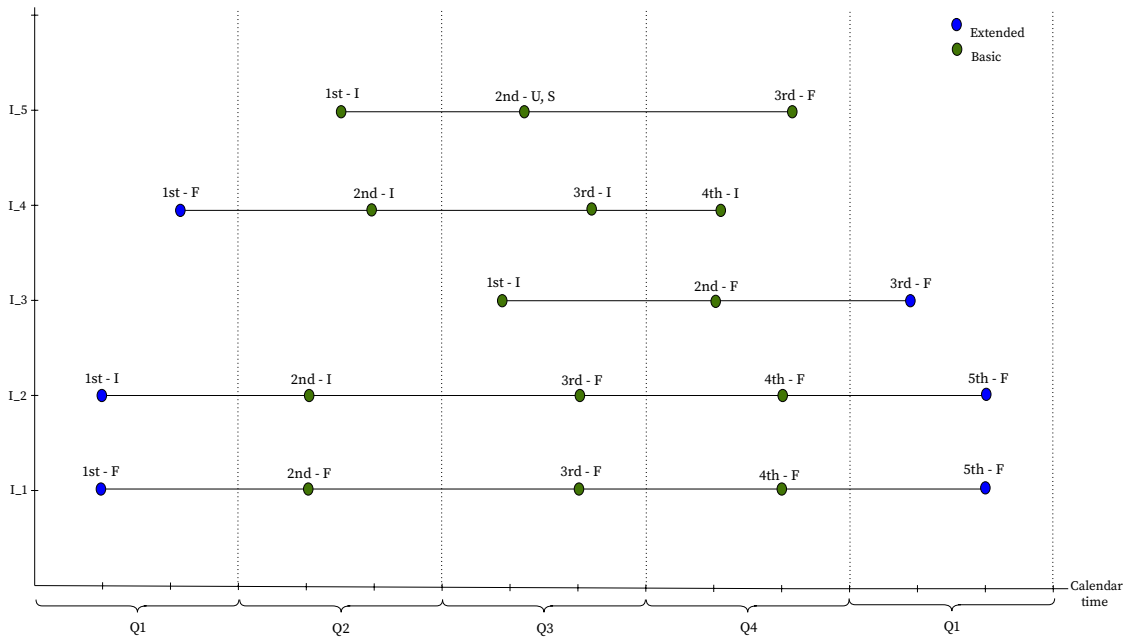
Tenure measurement plays a central role to understanding worker transitions between formal and informal employment. Tenure is constructed from question 3r of the extended questionnaire, which for subordinate workers reads “*¿En qué año [nombre] entró a trabajar por primera vez para su actual empresa, institución o patrón?*” — in what year did the worker first start working for their current firm, institution or employer.⁴² The question therefore records a date of entry rather than an elapsed duration. Respondents report the job-entry time in three categories: (a) *current year* (9.7%), (b) *last year* (15.8%), and (c) *before last year* (74.1%). If the answer is (a) or (b), the survey additionally records the month of entry; otherwise only the entry year is recorded. Approximately 2% of answers in (a) and (b) are missing the month. This pattern of collection is fortunate for the classification used here: the month is observed precisely in the range where the one-quarter threshold separating within-firm from between-firm transitions can bind, and where it is not observed the worker entered before last year, so tenure exceeds the threshold whatever the month. The coarseness of the year-only answers therefore affects the measured level of tenure but not the within-versus-between split.

⁴²A parallel branch of the same question asks independent workers when they started or took over their current business; it is not used here, since the sample is restricted to private wage employment.

I define tenure as the elapsed time between the interview date and the reported job-entry date. The ENOE survey design presents a key challenge for tenure measurement: the survey includes both a basic and an extended questionnaire, and entry-date information is collected only in the extended version. The extended questionnaire is used in the first quarter of each year since 2007 (and in all quarters prior to 2007). Therefore, a worker observed for up to five consecutive quarterly interviews typically has at most two tenure records, while social security status is observed in every quarter. Figure B1 summarizes the interview structure and illustrates that, for workers I_1 and I_2 , the job-entry date is recorded twice (two extended interviews), whereas for workers I_3 and I_4 it is recorded once and for worker I_5 it is never recorded. Overall, extended-questionnaire observations represent 68.5% of the sample.

A key measurement challenge arises from interval censoring of entry dates for workers reporting category (c). For answers (a) and (b), tenure can be computed in months because both year and month of entry are observed. For answer (c), only the entry year is observed, implying that the entry month is interval-censored within a 12-month window. Figure B2 illustrates the three alternatives for worker I_1 .

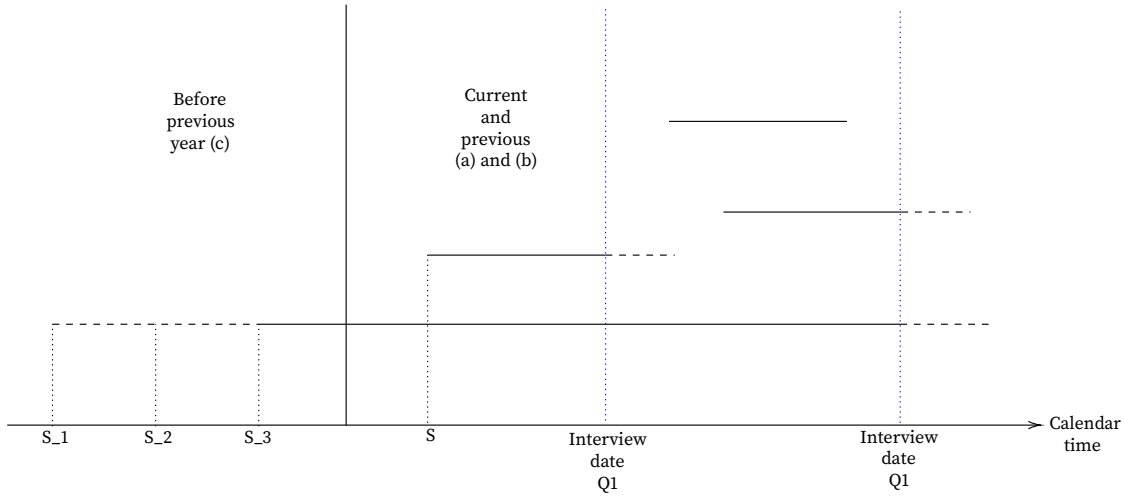
FIGURE B1. Overview of the questionnaires, interviews and states in a year



Source: Author's elaboration. Note: This figure depicts interview structure done after 2007. Before this year the extended version was used in different quarters.

Two sampling schemes are relevant (Lancaster, 1990). The *stock sample* refers to workers observed at the time of the extended interview; these workers are employed in their cur-

FIGURE B2. Stock and flow sampling with interval censored starting time



Source: Author's elaboration based on [Babington P. and Cano-Urbina, 2018](#).

rent jobs at the interview date and tenure is observed. The *flow sample* refers to quarterly transitions across labor-market states observed in the basic questionnaire (e.g., unemployment to formal employment), but without entry-date information, so tenure is not directly observed for those transitions (e.g., worker I_5 in Figure B1).

Table B2 reports average raw tenure by entry-date category. As expected, categories (a) and (b) correspond to short tenures (around half a year), while category (c) corresponds to substantially longer tenures across employment types; in particular, formal and informal workers report, on average, seven and five years of tenure while other occupations ten years.

To obtain a quarterly tenure measure, I exploit the rotating panel dimension. For each worker with at least one observed entry date, I construct tenure in quarters and fill missing quarters by adding (or subtracting) one quarter between consecutive interviews, provided the worker remains in the same job spell. Concretely, if worker i reports tenure of 20 years in the last extended interview and 19 years in the first extended interview, the intervening quarters are imputed by increments of three months. Similarly, for workers with a single extended interview, I backcast tenure to earlier quarters by three-month steps as long as the worker does not change job spell, for example, to self employment or unemployment.

I define a job spell using the ENOE establishment/craft tenure question. A worker is classified as a mover between $t-1$ and t if inferred tenure is inconsistent with continuation of the same job spell (i.e., tenure resets or declines beyond a tolerance band). For workers lacking tenure information in the relevant waves, mover status is treated as missing. This

TABLE B2. Average years of tenure by type of employment

	Formal	Informal	Other	Total
Answer: (a) and (b)				
Current and previous years	0.531 (0.380)	0.396 (0.370)	0.526 (0.398)	0.464 (0.385)
Answer: (c) Before previous year				
Beginning of period	7.429 (8.024)	5.199 (7.630)	12.42 (11.99)	8.780 (10.20)
Mid of period	7.112 (7.940)	4.952 (7.525)	12.06 (11.94)	8.465 (10.12)
End of period	6.731 (7.843)	4.655 (7.405)	11.62 (11.87)	8.088 (10.03)

Source: Author's estimates based on the pooled data from ENOE 2005-2019. Other includes self-employment with and without employees, public employee and non-paid work. The question used to compute tenure is For how long have you been working in this establishment/institution/craft? and possible answers are (a) current year, (b) last year and (c) before last year.

allows me to distinguish within-spell changes in formality status from transitions accompanied by job-spell changes.

Following [Babington P. and Cano-Urbina 2018](#), I impute the entry month using the mid-point of the reported year (June), which minimises bias in Monte Carlo evidence relative to beginning-of-year (January) and end-of-year (December) imputations. To ensure data quality, I apply several consistency checks and drop observations that fail to satisfy basic plausibility conditions:

1. Workers with five interviews may have two tenure records. When the implied tenure growth between the two extended interviews is inconsistent with calendar time (approximately one year) and there is no intervening non-employment or self-employment, I compute tenure using alternatively the first or the last reported entry date and assess robustness.
2. A small fraction of workers (about 4%) report long tenure in an extended interview but exhibit a state transition (e.g., unemployment to formal employment) within the same year. Because this pattern is inconsistent with a single ongoing job spell, I drop these observations.
3. Conditional on remaining in the same job spell, tenure should increase by at most one quarter between consecutive quarterly interviews.
4. I drop observations with implied tenure exceeding the worker's age.

B.3. Job transitions

TABLE B3. Quarterly transitions probabilities

	<i>Next period state, %</i>					
	Formal (Within)	Formal (Between)	Informal (Within)	Informal (Between)	Unemployed	Shares
Panel A. Aggregate						
Formal	75.2	7.2	6.7	1.8	9.1	52.7
Informal	11	3.3	51.2	19.3	15.2	31.9
Unemployed		27.8		34.4	37.8	15.4
Panel B. By characteristics						
Female and High Skill						
Formal	75.3	8.6	5.6	1.7	8.8	60.4
Informal	17.1	6	41.5	18.3	17.2	21.2
Unemployed		29.3		21.8	48.9	18.4
Female and Low Skill						
Formal	76.2	9	5.5	1.8	7.4	56.7
Informal	10.4	3.8	50.1	23.6	12.2	30.8
Unemployed		31.5		31.9	36.7	12.6
Male and High Skill						
Formal	75.7	6.8	5.5	1.6	10.3	59.6
Informal	15	4.5	45.4	16.5	18.6	22.1
Unemployed		29.5		24.7	45.8	18.3
Male and Low Skill						
Formal	74.3	6	8.6	2	9.1	44.7
Informal	8.9	2.3	54.7	19.3	14.8	41.5
Unemployed		24.6		47.8	27.6	13.8

Source: Authors' estimates based on data from ENOE 2005–2025. The sample includes workers aged 20 to 60.

TABLE B4. Quarterly transitions probabilities — six-month tenure threshold

	<i>Next period state, %</i>					
	Formal		Informal		Not em- ployed	Shares
	<i>(Within)</i>	<i>(Between)</i>	<i>(Within)</i>	<i>(Between)</i>		
Panel A. Aggregate						
Formal	61.8	9.8	5.5	2	20.9	15.5
Informal	7.2	3	34.1	16.3	39.5	11.5
Not employed		3.9		6.6	89.5	73
Panel B. By characteristics						
Female and High Skill						
Formal	55.6	10.5	4.1	1.7	28.2	13.5
Informal	8.5	4.1	20.3	12.7	54.3	6.8
Not employed		4.3		4.8	91	79.8
Female and Low Skill						
Formal	54.7	9.9	4	1.7	29.8	6.1
Informal	4.6	2.2	22.3	12.9	58	5.3
Not employed		1.8		3.5	94.7	88.6
Male and High Skill						
Formal	65.6	10	4.8	1.9	17.8	32.1
Informal	10.3	4.3	31.6	14.9	38.9	14.6
Not employed		9.6		11.9	78.5	53.3
Male and Low Skill						
Formal	66.3	9.2	7.6	2.5	14.5	31.3
Informal	6.8	2.5	43	19.1	28.6	32.8
Not employed		10.2		27.4	62.3	35.9

Source: Author's estimates based on data from ENOE 2005–2025. The sample includes workers aged 20 to 60. Mirror of the main-text transitions table with the within-firm/between-firm classification recomputed under a six-month tenure threshold instead of the baseline one quarter: a transition counts as within-firm if tenure at the transition exceeds six months. The employment shares in the last column do not depend on the threshold.

TABLE B5. Industry change across formal–informal transitions

		2-digit		4-digit	
	<i>Change</i>	<i>No change</i>	<i>Change</i>	<i>No change</i>	
Informal to formal					
<i>between</i>	46.1	53.9	59	41	
<i>within</i>	28.8	71.2	38.9	61.1	
Formal to informal					
<i>between</i>	45.6	54.4	59.2	40.8	
<i>within</i>	29.2	70.8	39.5	60.5	
Informal to informal					
<i>between</i>	20.4	79.6	31.9	68.1	
<i>within</i>	20.3	79.7	30.5	69.5	
Formal to formal					
<i>between</i>	26	74	36.1	63.9	
<i>within</i>	22.3	77.7	31	69	

Source: Authors' estimates based on ENOE 2005–2025, workers aged 20–60. Each cell is the share of transitions in which the reported industry changes across a formal–informal transition; the informal-stayer (I–I) and formal-stayer (F–F) rows are no-status-change benchmarks, so their change rate is the measurement-noise floor. ENOE codes industry with the household version of SCIAN; only the 2-digit sector (20 sectors) is SCIAN-comparable and is the meaningful measure, since a firm rarely changes sector. The 4-digit code adds a survey-specific disaggregation that records the activity description rather than the firm's sector (for example, within construction, sector 23, the same firm may be coded as residential building, 2361, or stone work and installations, 2382), so within-firm changes at this level are not exactly industry reallocations. The 4-digit column excludes pairs straddling the 2021Q3 SCIAN-2007 to SCIAN-2018 change; the 2-digit sector is stable across it. Within and between follow the tenure rule of the main transition table.

TABLE B6. Occupation change across formal–informal transitions

	4-digit		11-group		Manual/Cognitive	
	<i>Change</i>	<i>No change</i>	<i>Change</i>	<i>No change</i>	<i>Change</i>	<i>No change</i>
Informal to formal						
<i>between</i>	62.8	37.2	37.4	62.6	24.8	75.2
<i>within</i>	51.4	48.6	28.5	71.5	16.8	83.2
Formal to informal						
<i>between</i>	62.6	37.4	37.1	62.9	25	75
<i>within</i>	52	48	28.8	71.2	17.5	82.5
Informal to informal						
<i>between</i>	43.2	56.8	19.4	80.6	18.5	81.5
<i>within</i>	43.5	56.5	21.2	78.8	16.6	83.4
Formal to formal						
<i>between</i>	49.2	50.8	25.5	74.5	14	86
<i>within</i>	47.1	52.9	25	75	13.1	86.9

Source: Authors' estimates based on ENOE 2005–2025, workers aged 20–60. Each cell is the share of transitions in which the occupation changed across a formal–informal transition; the informal-stayer (I–I) and formal-stayer (F–F) rows are no-status-change benchmarks, so their change rate is the measurement-noise floor. The 4-digit SINCO unit group identifies the product rather than the skill content of the job (for example growing maize, 6111, versus vegetables, 6112; or bricklaying, 7121, versus house painting, 7135), so movement across these cells is lateral and, against the stayer floor, largely coding noise; it excludes pairs straddling the 2012Q3 (CMO-2009 to SINCO-2011) and 2020Q3 (SINCO-2011 to SINCO-2019) coding changes. The 11-group is the survey's harmonised occupational statistic; Manual/Cognitive classifies as manual agricultural workers, machinery operators, artisans and elementary occupations, and as cognitive managers, professionals, technicians, traders and scientists. Within and between follow the tenure rule used in the main transition table.

B.4. Institutional overview

The social security contributions are split between employees and employers. Contribution rates are 2.7% for the employee and 21% for the employer. Separate regimes cover public-sector workers (ISSSTE) and state enterprises such as PEMEX; the self-employed need not enrol but must register any subordinate workers they hire. Penalties for informal hiring range from 20 to 250 daily minimum wages per worker in past-due contributions and fines (up to 350% of the average wage), rising to fraud charges punishable by up to nine years; fines can be waived if the employer voluntarily formalises the worker.

The labour law cover all employment relations and guarantees additional benefits but working informality reduces enforceability of the law. The workweek cap is 48 hours; paid annual leave is at least 12 days after one year, rising with tenure; the year-end bonus is at least 15 days' salary; profit-sharing is 10% of the firm's taxable income; and severance for wrongful dismissal is three months' gross wage plus 20 days per year of service. [Kaplan and Joyce 2008](#) show that Mexico's labour courts are slow and costly, and workers in small firms routinely settle for amounts below their legal entitlement or receive no severance at all.

Income taxation also influences labour supply. The lowest statutory bracket applies a 1.92% marginal rate, rising through brackets at MXN 5,470 and 9,614 to 16%; the tax credit reached about MXN 407 per month in 2019 for workers earning between one and two and a half minimum wages. Mexico's net personal average tax rate, including social security contributions net of cash benefits, stood at 13.2% of the gross wage for a single worker at average earnings in 2025 ([OECD, 2026](#)). The 21% employer contribution adds a firm-cost wedge that does not directly affect take-home pay.

TABLE B7. Costs and benefits: contributory and non-contributory arrangements

	Contributory	Non-contributory
A. Social protection		
Cost/Contribution rate	23.7%	0%
<i>Paid by worker</i>	2.7%	0%
<i>Paid by employer</i>	21%	0%
Benefits Social security institute (IMSS)		
<i>Old-age pension, health insurance, childcare</i>	✓	✓
<i>Occupational hazard, housing credit</i>	✓	×
<i>Survival pension, maternity leave</i>	✓	×
<i>Paternity leave</i>	×	×
<i>Benefit transfer to partner and children</i>	✓	×
B. Labour laws	✓	×
Benefits		
<i>Written contract, minimum wage</i>	✓	×
<i>Severance and wrongful dismissal payment</i>	✓	×
<i>Paid overtime and holidays</i>	✓	×
<i>Statutory year-end bonus</i>	✓	×
<i>Profit sharing scheme</i>	✓	×
C. Taxes		
Personal income taxes		
<i>Statutory rate</i>	1.9% – 35%	0%
<i>Fixed Fee</i>	Bracket	0%
Benefits		
<i>Allowances and discounts</i>	✓	×
<i>Tax credits/subsidies</i>	✓	×

Source: Authors' compilation based on Mexican labour legislation and ENOE definitions. The table summarizes key distinctions between formal and informal employment regarding social security, taxation, and labour law coverage.

Appendix C. Job transitions and wages

FIGURE C1. Distribution of hours of work per week

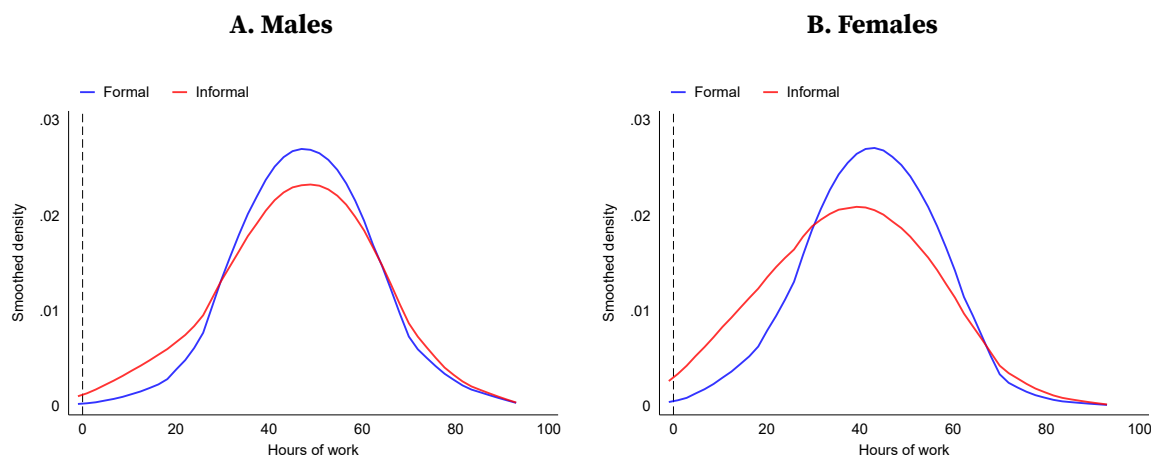
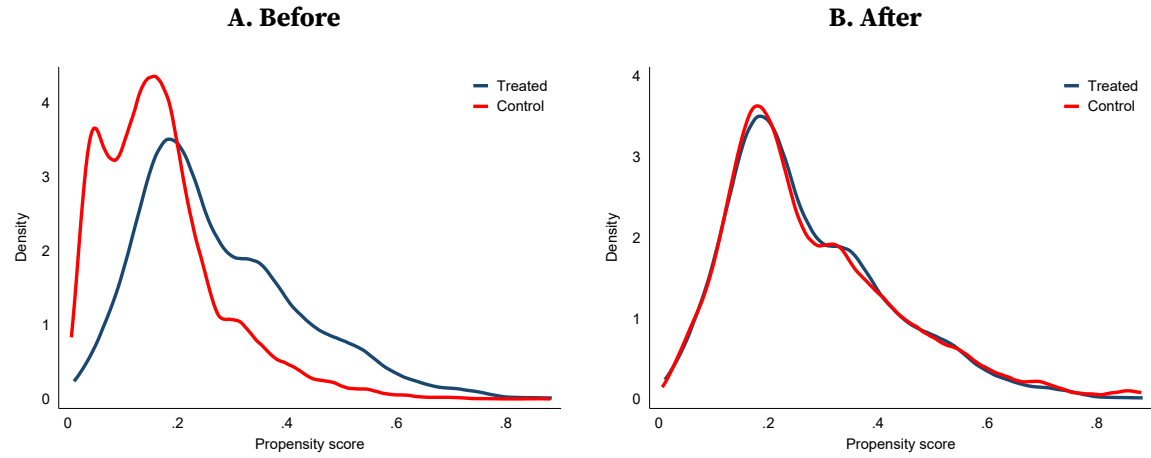


TABLE C1. Group coverage by event time

k	Pre-treatment				Post-treatment				Share, (%)	
	-4	-3	-2	-1	0	1	2	3	$I \rightarrow F$	$F \rightarrow I$
<i>Group g</i>										
2	-	-	-	✓	✓	✓	✓	✓	26.7	27.2
3	-	-	✓	✓	✓	✓	✓	-	26.6	26.5
4	-	✓	✓	✓	✓	✓	-	-	24.4	24.1
5	✓	✓	✓	✓	✓	-	-	-	22.2	22.2
$N(k)$	1	2	3	4	4	3	2	1		

Source: Author's elaboration. Notes: Each worker transitioning is assigned to an interview transition group g which evolves over $t = \{1, \dots, 5\}$. The checkmark (✓) indicates the number of periods available for group g in the relative event time window $k = t - e$. $k = 0$ is the time of the event, in this case formalisation or informalisation, so that $k < 0$ are pre-treatment periods, $k = 0$ is the period of the event, and $k > 0$ are post-treatment periods. Notice that a total of four possible treatment interview groups, $g = \{2, 3, 4, 5\}$, can be aggregated to compute the Aggregate Average Treatment Effect across group weights. The row $N(k)$ is the vertical sum of check marks and depicts how many groups contribute to computing the event design in that particular period. For example, a worker formalising in the second interview belongs to group $g = 2$ but contributes only one pre-treatment observation ($k = -1$), while a worker formalising at the last interview belongs to $g = 5$ but does not contribute any post-treatment observation beyond impact ($k = 0$).

FIGURE C2. Density overlap of propensity score of treated and control groups



Source: Author's estimates based on data from ENOE 2005–2025.

TABLE C2. Sample composition

	N	N × T	Share (%)
<i>Setting 1: Informal to Formal</i>			
Total treated	111,817	454,676	10.38
<i>of which</i>			
Reversible	55,385	260,003	49.53
1 switch	55,282		
> 1 switch	8,578		
Permanent	56,432	194,673	50.47
<i>of which</i>			
Between	19,411	32,055	17.36
Within	32,508	132,852	29.07
Always informal (Control)	965,722	1,455,631	89.62
Total (Treatment + Control)	1,077,539	1,910,307	100
<i>Setting 2: Formal to Informal</i>			
Total treated	104,278	425,098	6.19
<i>of which</i>			
Reversible	55,385	260,003	53.11
1 switch	55,765		
> 1 switch	8,095		
Permanent	48,893	165,095	46.89
<i>of which</i>			
Between	17,612	25,215	16.89
Within	27,689	111,067	26.55
Always formal (Control)	1,580,079	2,940,050	93.81
Total (Treatment + Control)	1,684,357	3,365,148	100

Source: Author's estimates based on data from ENOE 2005–2025. Shares of Reversible, Permanent (and its Between/Within decomposition) are expressed as a percentage of total treated workers (Reversible + Permanent); the share of the control group is expressed as a percentage of the grand total (Treated + Control). The sum of between and within sample does not equal the total permanent since not all worker report tenure. The two lines below each Reversible row (one switch / more than one switch) are computed on the reversibility-robustness sample of Appendix C.8: 63,860 workers who reverse formality status, split by the number of transitions in the relevant direction. They are individual counts (*N*) from that estimation sample and therefore do not sum to the Reversible row above, which is defined on the main sample as workers with exactly one transition in each direction.

C.1. Sensitivity to violations of parallel trends

This appendix formalises the sensitivity analysis proposed by [Rambachan and Roth, 2023](#) and implemented in the main text. The objective is to assess how robust the estimated treatment effects remain to economically plausible departures from the parallel trends assumption. Let the event-study coefficients from equation (1) be collected in the vector

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_{\text{pre}} \\ \hat{\beta}_{\text{post}} \end{pmatrix},$$

with estimated covariance matrix $\hat{\Sigma}$, such that

$$\hat{\beta} \sim \mathcal{N}(\beta, \hat{\Sigma}).$$

The event-study coefficients admit the decomposition

$$(C1) \quad \beta = \tau + \delta,$$

where τ denotes the dynamic causal effects of formalisation and δ represents deviations from parallel trends, namely the evolution of untreated potential outcomes for treated workers relative to the comparison group. In the present application, such deviations may arise from gradual employer learning, structured promotion paths, or other forms of dynamic selection within firms. Under the standard no-anticipation assumption,

$$\tau_{\text{pre}} = 0,$$

so that the pre-treatment coefficients identify the pre-treatment deviation from parallel trends,

$$\beta_{\text{pre}} = \delta_{\text{pre}},$$

whereas the post-treatment coefficients combine the causal treatment effect and any remaining bias,

$$\beta_{\text{post}} = \tau_{\text{post}} + \delta_{\text{post}}.$$

Under exact parallel trends,

$$\delta = 0,$$

implying

$$\tau_{\text{post}} = \beta_{\text{post}}.$$

However, the observed data contain no information about δ_{post} , implying that exact parallel trends is an identifying assumption rather than a testable implication of the model. Rather than imposing $\delta = 0$, Rambachan and Roth (2023) assume that post-treatment deviations belong to a restricted set,

$$\delta \in \Delta,$$

where Δ contains departures from parallel trends considered economically plausible. For a scalar parameter of interest

$$\theta = \ell' \tau_{\text{post}},$$

where ℓ selects the contemporaneous treatment effect ($k = 0$ in this application), the identified set is

$$(C2) \quad \mathcal{S}(\beta, \Delta) = \{ \ell' (\beta_{\text{post}} - \delta_{\text{post}}) : \delta \in \Delta, \delta_{\text{pre}} = \beta_{\text{pre}} \},$$

for which Rambachan and Roth (2023) construct confidence intervals that remain uniformly valid over every admissible deviation satisfying $\delta \in \Delta$.⁴³ As the restriction set Δ becomes larger, increasingly severe departures from parallel trends are admitted, resulting in wider confidence intervals. Consequently, the estimated treatment effect is considered robust only if it remains statistically distinguishable from zero under economically plausible restrictions. Two alternative restriction sets are considered. The first is the *relative-magnitude restriction*,

$$(C3) \quad \Delta^{RM}(\bar{M}) = \left\{ \delta : |\delta_{k+1} - \delta_k| \leq \bar{M} \max_{s < 0} |\delta_{s+1} - \delta_s|, \quad \forall k \geq 0 \right\},$$

which limits post-treatment changes in the differential trend relative to the largest change observed before treatment. When $\bar{M} = 1$, the largest post-treatment change in the trend is constrained to be no larger than the largest change observed during the pre-treatment period. Larger values of $\bar{M}$ allow progressively larger departures from parallel trends. This parameter corresponds to the horizontal axis of Figures 6A–6C. The second restriction is the *smoothness restriction*,

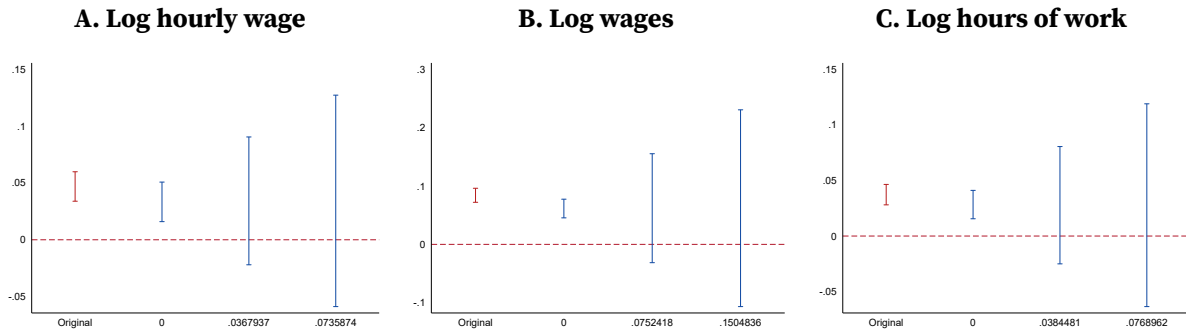
⁴³The empirical application uses fixed-length confidence intervals computed from the event-study coefficient vector and covariance matrix obtained from the within-firm doubly robust event study.

$$|\delta_{k+1} - 2\delta_k + \delta_{k-1}| \leq M,$$

which bounds the curvature of the differential trend. The parameter M is calibrated using the maximum observed curvature during the pre-treatment period. The limiting case $M = 0$ imposes a perfectly linear differential trend and corresponds to the linear extrapolation exercise presented in the main text.

For each restriction set, the sensitivity parameter is progressively increased until the robust confidence interval first includes zero. Following [Rambachan and Roth, 2023](#), this threshold is referred to as the breakdown value. Intuitively, it measures the magnitude of the departure from parallel trends required for the estimated treatment effect to become statistically indistinguishable from zero. In the empirical application, the estimated hourly wage effect remains robust for deviations up to approximately $\bar{M} = 1$, the monthly wage effect up to approximately $\bar{M} = 1.5$, whereas the estimated effect on hours worked becomes insignificant under substantially smaller departures from parallel trends.

FIGURE C3. Honest-DiD sensitivity (pre-trend-based restriction)

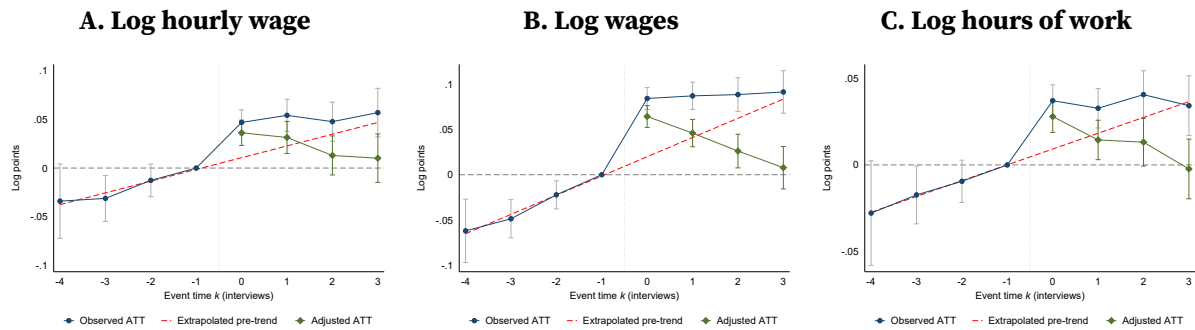


Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; vertical bars are 95% confidence intervals from a wild-cluster bootstrap with 999 replications.

C.2. Linear extrapolation of trend

Lastly, I linearly extrapolate the estimated pre-treatment trend into the post-treatment period and net it out, regressing the pre-treatment coefficients on a linear event-time trend, projecting the fitted trend forward, and subtracting it from the estimates (Figures C4A–C4C). Both deliver the same split as the main text: a sizable, discrete effect on the hourly wage remains at the time of formalisation, while much of the increase in hours is explained by continuation of the pre-trend, so formalisation itself is associated with a discrete hourly gain that the pre-existing trajectory cannot fully account for.

FIGURE C4. Linear pre-trend extrapolation



Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; vertical bars are 95% confidence intervals from a wild-cluster bootstrap with 999 replications.

C.3. Derivation of the Group Average Treatment Effect (GATT)

Define D_i^g as a dummy variable equal to one if worker i is eventually treated in group $g \in \{2, 3, 4, 5\}$. Let $y_{it}(0)$ denote the untreated potential outcome path of worker i at interview t if the worker were never treated; for groups g , $y_{it}(g)$ denotes the potential outcome that worker i would experience at time t if they become formal in interview g . The observed outcome can be written:

$$(C4) \quad y_{it} = y_{it}(0) + \sum_{g=2}^T (y_{it}(g) - y_{it}(0)) D_i^g$$

A worker who is not treated in group g has $D_i^g = 0, \forall g \in \{2, 3, 4, 5\}$ and the outcome is $y_{it} = y_{it}(0)$. Note that $D_i^g = 0$ does not uniquely identify never-treated workers: a worker treated at group $g' \neq g$ also satisfies $D_i^g = 0$. If $\sum_g D_i^g = 0$, this refers to workers never treated in any group. A worker treated in interview 2 belongs to group $g = 2$ and has outcome path $y_{i1}(0), y_{i2}(2), y_{i3}(2), y_{i4}(2), y_{i5}(2)$. The Group Average Treatment Effect on the Treated is:

$$(C5) \quad \text{ATT}(g, t) = \mathbb{E} \left[y_t(g) - y_t(0) \mid D_i^g = 1 \right].$$

Identification of (C5) relies on three assumptions.

Assumption 1. Irreversibility of treatment: $D_{t=1} = 0$ almost surely. For $t = 2, \dots, T$, $D_{t1} = 1$ implies that $D_t = 1$ almost surely. This means that no one is treated at time $t = 1$, and that once a unit becomes treated, that unit will remain treated in the next period.

Assumption 2 Limited Treatment Anticipation. Let $\ell \geq 0$ be the anticipation period:

$$\mathbb{E} [y_{it}(1) \mid X, D = 1] = \mathbb{E} [y_{it}(0) \mid X, D = 1] \quad \forall t < \ell.$$

When $\ell = 0$, this is a strict no-anticipation condition; for $\ell > 0$ workers anticipate the treatment ℓ periods in advance.

Assumption 3 Conditional Parallel Trends, Never-Treated Comparison: Let C denote a dummy equal to one if the worker is never treated, this is equivalent to $\sum_g D_i^g = 0$. For each $t \geq \ell$,

$$\mathbb{E} [y_{it}(0) - y_{i,t-1}(0) \mid X, D = 1] = \mathbb{E} [y_{it}(0) - y_{i,t-1}(0) \mid X, C = 1].$$

Conditional on covariates X , treated and never-treated workers would have followed paral-

lel paths absent treatment.

Assumption 4 Overlap: For each t , there exists $\epsilon > 0$ such that $\epsilon < P(D = 1 \mid X) < 1 - \epsilon$.

The estimation procedure consist in using two by two differences-in-differences designs for each group, holding pre-treatment characteristics constant across designs. Consider two periods $t = 0, 1$, and D_t the binary treatment indicator at time t that equals one if treated and zero. I abstract from the subscript i and consider the outcome $y_t(d)$, thus, the Average Treatment Effect on the Treated (ATT) in time $t = 1$,

$$ATT = \mathbb{E}[y_1(1) - y_1(0) \mid D = 1]$$

$$ATT = \mathbb{E}[y_1(1) \mid D = 1] - \mathbb{E}[y_1(0) \mid D = 1]$$

The key is to estimate the counterfactual $\mathbb{E}[y_1(0) \mid D = 1]$ correctly.

The doubly robust estimator relies in two strategies. The outcome regression and the inverse probability weighing.

C.4. Outcome regression

In the [Heckman, 1997](#) approach the counterfactual change in the comparison group is replaced by a fitted outcome regression. Let $\hat{m}(X, t) = \hat{\mathbb{E}}[y_t \mid X, D = 0]$ denote the outcome regression of y_t on X fit on the never-treated. In a two-period the estimand of the ATT is:

$$\widehat{ATT}_{or} = \mathbb{E}[y_1 - y_0 \mid D = 1] - \mathbb{E}[\hat{m}(X, 1) - \hat{m}(X, 0) \mid D = 1] .$$

The first term is the observed pre-to-post change for treated workers; the second is the model-implied counterfactual change, evaluated on the covariate distribution of the treated. The procedure is two-stage: (i) fit $\hat{m}(X, t)$ on the never-treated; (ii) average the fitted within-treated difference across treated workers and subtract from the observed treated change. The OR estimator is consistent when $\hat{m}(X, t)$ is correctly specified but can be biased under misspecification of the outcome model.

C.5. Inverse probability weighing

The [Abadie, 2005](#) in the context of a 2x2 panel data design gives the following differences:

$$\Delta y(1) = y_1(1) - y_0(1) \quad \text{for } D = 1$$

$$\Delta y(0) = y_1(0) - y_0(0) \quad \text{for } D = 0$$

The average differences in outcomes are $\mathbb{E}[\Delta y(1) \mid D = 1]$ and $\mathbb{E}[\Delta y(0) \mid D = 0]$. The

weighted average change in outcomes for the control group using propensity scores:

$$\mathbb{E}[\Delta y(0) \mid D = 0] = \mathbb{E} \left[\frac{p(X)}{1 - p(X)} \Delta y(0) \mid D = 0 \right]$$

Where $p(X) = P(D = 1 \mid X)$ is the propensity score. Finally, the ATT using [Abadie, 2005](#) estimator in a 2x2 design is:

$$ATT = \mathbb{E}[\Delta y(1) \mid D = 1] - \mathbb{E} \left[\frac{p(X)}{1 - p(X)} (y_1(0) - y_0(0)) \mid D = 0 \right]$$

To ensure the correct reweighting and normalisation, the expectation is adjusted by the overall probability of treatment:

$$ATT = \frac{1}{\mathbb{E}[D]} \mathbb{E} \left[D(y_1 - y_0) - \frac{(1 - D)p(X)}{1 - p(X)} (y_1 - y_0) \right]$$

Simplifying the expression gives the final form for the ATT:

$$ATT = \frac{1}{\mathbb{E}[D]} \mathbb{E} \left[\frac{D - p(X)}{1 - p(X)} (y_1 - y_0) \right]$$

The term $\frac{(1-D)p(X)}{1-p(X)}$ adjusts for the propensity score, ensuring that the treated and control groups are balanced. When $D = 1$ the term becomes one but if $D = 0$, then the term is $\frac{-p(X)}{1-p(X)}$. Therefore, the ATT effectively balances the treated and control groups, allowing to isolate the treatment effect on the treated units. Note that the baseline period here is period $t = 0$ and thus the propensity score is computed at this baseline period. So we are balancing two similar workers before the treatment, and then compute the effect considering the propensity of each worker to be treated. The IPW estimator is consistent when $p(X)$ is correctly specified but can be biased under misspecification of the propensity score.

C.6. Doubly-robust estimator and worked examples

The outcome-regression and probability estimands rely on correct specification of different pieces of the data-generating process: the outcome regression requires a correct conditional outcome model $\hat{m}(X, t)$, while the inverse probability requires a correct propensity score $\hat{p}(X)$. The doubly-robust (DR) estimator of [Sant'Anna and Zhao, 2020](#) combines both and is consistent for the ATT if either the outcome regression or the propensity score is correctly specified — but not necessarily both.

In the 2×2 panel setting (one cohort, one reference period $t = 0$, one post-period $t = 1$),

the DR estimand is:

$$(C6) \quad \widehat{ATT}_{dr} = \mathbb{E} \left[\left(\frac{D}{\mathbb{E}[D]} - \frac{\frac{p(X)(1-D)}{1-p(X)}}{\mathbb{E} \left[\frac{p(X)(1-D)}{1-p(X)} \right]} \right) ((y_1 - y_0) - (\hat{m}(X, 1) - \hat{m}(X, 0))) \right].$$

The first factor in parentheses is the reweighting term; the second is the residual of the observed change against the outcome-regression prediction. If $p(X)$ is correct, the reweight identifies the ATT directly and the outcome-regression residual averages to zero in the limit; if $\hat{m}(X, t)$ is correct, the residual identifies the ATT and the reweight only affects efficiency. Either way, the estimand recovers the ATT. The multi-period extension follows [Callaway and Sant'Anna, 2021](#). For group g and post-treatment period $t \geq g - \delta$, the doubly-robust group-time ATT under the never-treated comparison is:

$$(C7) \quad ATT_{dr}^{nev}(g, t; \delta) = \mathbb{E} \left[\left(\frac{D^g}{\mathbb{E}[D^g]} - \frac{\frac{p_g(X) C}{1-p_g(X)}}{\mathbb{E} \left[\frac{p_g(X) C}{1-p_g(X)} \right]} \right) ((y_t - y_{g-\delta-1}) - m_{g,t,\delta}^{nev}(X)) \right],$$

where $C = \mathbb{1}\{\sum_g D^g = 0\}$ flags never-treated workers, $p_g(X) = P(D^g = 1 \mid X, D^g + C = 1)$ is the group-specific propensity score, and $m_{g,t,\delta}^{nev}(X) = \mathbb{E}[y_t - y_{g-\delta-1} \mid X, C = 1]$ is the outcome regression of the change between t and $g - \delta - 1$ fit on the never-treated. Setting $\delta = 0$ (no anticipation) gives the baseline estimand used in the main text; $\delta > 0$ corresponds to the limited-anticipation robustness check.

C.7. Not-yet-treated comparison

The baseline (Assumption 3) uses never-treated workers as the comparison group, $C = \mathbb{1}\{\sum_g D^g = 0\}$. The not-yet-treated comparison instead uses, at each period t , every worker who has not yet formalised by t , that is the never-treated together with the cohorts that formalise after t . Let $C_t^{ny} = \mathbb{1}\{F_{g'} > t\}$ flag a worker whose first-treatment interview $F_{g'}$ exceeds t . The doubly-robust group-time ATT under the not-yet-treated comparison replaces C , the propensity score, and the outcome regression in (C7) with their not-yet-treated counterparts:

$$(C8) \quad ATT_{dr}^{ny}(g, t) = \mathbb{E} \left[\left(\frac{D^g}{\mathbb{E}[D^g]} - \frac{\frac{p_{g,t}^{ny}(X) C_t^{ny}}{1-p_{g,t}^{ny}(X)}}{\mathbb{E} \left[\frac{p_{g,t}^{ny}(X) C_t^{ny}}{1-p_{g,t}^{ny}(X)} \right]} \right) ((y_t - y_{g-1}) - m_{g,t}^{ny}(X)) \right],$$

where $p_{g,t}^{ny}(X) = P(D^g = 1 \mid X, D^g + C_t^{ny} = 1)$ is the propensity score estimated on cohort g against the workers not yet treated at t , and $m_{g,t}^{ny}(X) = \mathbb{E}[y_t - y_{g-1} \mid X, C_t^{ny} = 1]$ is the

outcome regression fit on those workers. The comparison group is larger than under the never-treated design, which raises precision, at the cost of requiring conditional parallel trends to hold for the not-yet-treated cohorts as well as for the never-treated.

C.8. Reversible treatment

The baseline conditions on permanent transitions (Assumption 1, irreversibility). To relax it I use the estimator of [Chaisemartin and D’Haultfoeuille, 2020](#), defined for a binary treatment $D_{it} \in \{0, 1\}$ that may switch on and off across interviews. A worker is a *switcher* at t if $D_{it} \neq D_{i,t-1}$: a *joiner* (informal to formal) if $D_{i,t-1} = 0$ and $D_{it} = 1$, and a *leaver* (formal to informal) if $D_{i,t-1} = 1$ and $D_{it} = 0$. Identification replaces irreversibility with parallel trends on both potential outcomes and the existence, at each switching date, of *stable* groups whose treatment does not change. This implies that the comparison groups isn’t the never treated, not yet treated its those workers that did not transition in a particular period.

Assumption R1 (Parallel trends, both potential outcomes). For each t , the expected change in each untreated and treated potential outcome is the same for switchers and for stable groups with the same period-one treatment.

Assumption R2 (Stable groups). At each switching date there is at least one group whose treatment has not changed since the start of the panel: a stably untreated group for joiners and a stably treated group for leavers.

For each period t , the estimator compares the outcome change of switchers to that of the stable groups,

$$(C9) \quad DID_{+,t} = \Delta \bar{y}(\text{joiners at } t) - \Delta \bar{y}(\text{stably untreated}),$$

$$(C10) \quad DID_{-,t} = \Delta \bar{y}(\text{stably treated}) - \Delta \bar{y}(\text{leavers at } t),$$

and aggregates them with weights proportional to the number of switchers,

$$(C11) \quad DID_M = \sum_t \left[\frac{N_{1,0,t}}{N_S} DID_{+,t} + \frac{N_{0,1,t}}{N_S} DID_{-,t} \right],$$

where $N_{1,0,t}$ and $N_{0,1,t}$ count joiners and leavers at t and N_S the total number of switchers. DID_M is unbiased for the average treatment effect across the cells that switch between $t-1$ and t , with non-negative weights, so it is robust to heterogeneous effects across workers and time. The dynamic version reports effects ℓ periods after a switch and placebos ℓ periods before; the placebos test the parallel-trends assumption. I estimate it on the

reversible sample, workers observed in both formal and informal wage employment with reversals allowed, so that joiners identify the informal-to-formal effect and leavers the formal-to-informal effect.

The estimator reports each event-study coefficient as the effect of moving to the *higher* treatment dose. For joiners, whose dose rises from zero to one, the reported effect is $DID_{+,t}$ directly. For leavers, whose dose falls from one to zero, the raw comparison $DID_{-,t}$ in (C11) is defined with the stably-treated group's change entering first, so a positive value already measures the amount by which formality raises the outcome rather than the loss from leaving it. Joiner and leaver event studies are therefore expressed on a common scale on which a positive coefficient denotes the gain from being formal, the same value-of-formality convention used for the baseline estimates in the main text. This is why, in the comparison figures, the baseline Callaway–Sant’Anna formal-to-informal coefficient, which is reported in its natural “effect of leaving formality” (negative) sign, is re-oriented before being overlaid.

TABLE C3. Characteristics at baseline — Formalizers (I→F) vs. Always Informal (I→I)

	All			Between-firm			Within-firm		
	I→F	I→I	Diff.	I→F	I→I	Diff.	I→F	I→I	Diff.
Panel A. Before reweighting									
<i>Demographic:</i>									
Age (years)	32.62	34.33	1.72***	30.18	34.33	4.15***	33.25	34.33	1.08***
Education (years)	9.70	7.90	-1.80***	9.74	7.90	-1.85***	9.69	7.90	-1.79***
Female (share)	0.29	0.20	-0.09***	0.25	0.20	-0.05***	0.30	0.20	-0.10***
Married (share)	0.61	0.67	0.06***	0.60	0.67	0.07***	0.61	0.67	0.06***
Child under 5 in household (share)	0.40	0.43	0.04***	0.44	0.43	-0.01	0.39	0.43	0.05***
Children 5–17 only in household (share)	0.34	0.35	0.01*	0.31	0.35	0.04***	0.35	0.35	0.00
Household size	4.53	4.71	0.18***	4.47	4.71	0.24***	4.55	4.71	0.17***
<i>Occupations:</i>									
High-skill	0.14	0.07	-0.07***	0.11	0.07	-0.04***	0.15	0.07	-0.08***
Medium-skill	0.74	0.71	-0.03***	0.73	0.71	-0.02	0.74	0.71	-0.03***
Low-skill	0.12	0.22	0.10***	0.16	0.22	0.06***	0.11	0.22	0.11***
<i>Industry:</i>									
Agriculture	0.07	0.21	0.14***	0.07	0.21	0.14***	0.07	0.21	0.14***
Manufacturing	0.33	0.31	-0.02***	0.36	0.31	-0.05***	0.33	0.31	-0.01*
Services	0.59	0.48	-0.12***	0.57	0.48	-0.09***	0.60	0.48	-0.12***
North region (share)	0.22	0.11	-0.12***	0.25	0.11	-0.15***	0.21	0.11	-0.11***
<i>Firm size:</i>									
Small firm (2–10)	0.61	0.84	0.24***	0.73	0.84	0.12***	0.58	0.84	0.27***
Medium firm (11–50)	0.26	0.12	-0.14***	0.19	0.12	-0.06***	0.28	0.12	-0.16***
Large firm (> 50)	0.02	0.00	-0.02***	0.01	0.00	-0.01**	0.03	0.00	-0.02***
<i>Outcomes:</i>									
Weekly hours worked	48.33	48.22	-0.11	46.68	48.22	1.53***	48.76	48.22	-0.54***
Monthly wage (MXN)	4694.53	3953.16	-741.37***	4117.24	3953.16	-164.08**	4845.50	3953.16	-892.34***
Observations	79924			70888			77561		
Panel B. After reweighting									
<i>Demographic:</i>									
Age (years)	32.62	33.27	-0.65***	30.18	30.19	-0.00	33.25	33.51	-0.25*
Education (years)	9.70	9.82	-0.13*	9.74	10.03	-0.29***	9.69	9.95	-0.27***
Female (share)	0.29	0.32	-0.04***	0.25	0.29	-0.04***	0.30	0.34	-0.04***
Married (share)	0.61	0.58	0.03***	0.60	0.59	0.01	0.61	0.60	0.01*
Child under 5 in household (share)	0.40	0.37	0.03***	0.44	0.44	0.00	0.39	0.37	0.01*
Children 5–17 only in household (share)	0.34	0.36	-0.02**	0.31	0.31	0.01	0.35	0.34	0.00
Household size	4.53	4.57	-0.04	4.47	4.40	0.07	4.55	4.46	0.09***
<i>Occupations:</i>									
High-skill	0.14	0.15	-0.01	0.11	0.11	0.00	0.15	0.16	-0.01
Medium-skill	0.74	0.75	-0.01	0.73	0.73	-0.00	0.74	0.74	0.00
Low-skill	0.12	0.10	0.02***	0.16	0.16	0.00	0.11	0.10	0.01
<i>Industry:</i>									
Agriculture	0.07	0.09	-0.01***	0.07	0.08	-0.01	0.07	0.08	-0.00
Manufacturing	0.33	0.30	0.03***	0.36	0.32	0.04***	0.33	0.29	0.04***
Services	0.59	0.61	-0.02**	0.57	0.60	-0.03**	0.60	0.63	-0.03***
North region (share)	0.22	0.22	0.00	0.25	0.26	-0.01	0.21	0.22	-0.01
<i>Firm size:</i>									
Small firm (2–10)	0.61	0.67	-0.07***	0.73	0.76	-0.03**	0.58	0.65	-0.07***
Medium firm (11–50)	0.26	0.27	-0.01	0.19	0.20	-0.01	0.28	0.29	-0.01
Large firm (> 50)	0.02	0.02	-0.00	0.01	0.01	-0.00	0.03	0.03	-0.00
<i>Outcomes:</i>									
Weekly hours worked	48.33	46.86	1.47***	46.68	47.05	-0.37	48.76	46.74	2.02***
Monthly wage (MXN)	4694.53	4461.05	233.48***	4117.24	4403.95	-286.72***	4845.50	4570.60	274.90***
Observations	79924			70888			77561		

Source: Authors' calculations using ENOE 2005–2025. Notes: Means at $k = -1$ (last pre-treatment interview). Panel B control means are IPW-reweighted (logit propensity score, ATT).*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE C4. Characteristics at baseline — Informalizers (F→I) vs. Always Formal (F→F)

	All			Between-firm			Within-firm		
	F→I	F→F	Diff.	F→I	F→F	Diff.	F→I	F→F	Diff.
Panel A. Before reweighting									
<i>Demographic:</i>									
Age (years)	33.05	35.66	2.62***	31.41	35.66	4.26***	33.49	35.66	2.18***
Education (years)	9.46	10.55	1.09***	9.26	10.55	1.29***	9.52	10.55	1.04***
Female (share)	0.24	0.34	0.09***	0.20	0.34	0.14***	0.26	0.34	0.08***
Married (share)	0.63	0.68	0.05***	0.64	0.68	0.04***	0.63	0.68	0.05***
Child under 5 in household (share)	0.42	0.39	-0.03***	0.45	0.39	-0.06***	0.41	0.39	-0.02***
Children 5–17 only in household (share)	0.32	0.37	0.04***	0.30	0.37	0.06***	0.33	0.37	0.04***
Household size	4.54	4.34	-0.20***	4.49	4.34	-0.15***	4.56	4.34	-0.22***
<i>Occupations:</i>									
High-skill	0.13	0.17	0.04***	0.08	0.17	0.08***	0.14	0.17	0.03***
Medium-skill	0.79	0.77	-0.01**	0.83	0.77	-0.05***	0.78	0.77	-0.00
Low-skill	0.09	0.06	-0.02***	0.09	0.06	-0.03***	0.08	0.06	-0.02***
<i>Industry:</i>									
Agriculture	0.06	0.02	-0.04***	0.04	0.02	-0.02***	0.06	0.02	-0.04***
Manufacturing	0.38	0.43	0.05***	0.41	0.43	0.02*	0.37	0.43	0.06***
Services	0.56	0.55	-0.01**	0.55	0.55	-0.00	0.56	0.55	-0.02**
North region (share)	0.20	0.34	0.13***	0.21	0.34	0.13***	0.20	0.34	0.13***
<i>Firm size:</i>									
Small firm (2–10)	0.33	0.15	-0.18***	0.24	0.15	-0.10***	0.35	0.15	-0.20***
Medium firm (11–50)	0.36	0.29	-0.07***	0.37	0.29	-0.08***	0.36	0.29	-0.07***
Large firm (> 50)	0.07	0.22	0.15***	0.10	0.22	0.11***	0.06	0.22	0.16***
<i>Outcomes:</i>									
Weekly hours worked	50.23	49.26	-0.97***	51.13	49.26	-1.87***	49.98	49.26	-0.72***
Monthly wage (MXN)	5245.98	5978.25	732.27***	4887.23	5978.25	1091.02***	5342.35	5978.25	635.90***
Observations	192510			184250			190291		
Panel B. After reweighting									
<i>Demographic:</i>									
Age (years)	33.05	33.65	-0.60***	31.41	31.33	0.07	33.49	33.63	-0.14
Education (years)	9.46	9.86	-0.39***	9.26	9.45	-0.18*	9.52	9.77	-0.25***
Female (share)	0.24	0.26	-0.02***	0.20	0.22	-0.02**	0.26	0.28	-0.03***
Married (share)	0.63	0.62	0.01**	0.64	0.63	0.01	0.63	0.62	0.01
Child under 5 in household (share)	0.42	0.40	0.01*	0.45	0.43	0.01	0.41	0.40	0.01
Children 5–17 only in household (share)	0.32	0.33	-0.01	0.30	0.31	-0.01	0.33	0.33	0.00
Household size	4.54	4.56	-0.02	4.49	4.44	0.05	4.56	4.49	0.06**
<i>Occupations:</i>									
High-skill	0.13	0.13	-0.01	0.08	0.09	-0.00	0.14	0.14	-0.00
Medium-skill	0.79	0.80	-0.02***	0.83	0.83	-0.00	0.78	0.78	-0.00
Low-skill	0.09	0.06	0.02***	0.09	0.09	0.00	0.08	0.08	0.01
<i>Industry:</i>									
Agriculture	0.06	0.05	0.01**	0.04	0.05	-0.00	0.06	0.07	-0.00
Manufacturing	0.38	0.36	0.02***	0.41	0.38	0.03**	0.37	0.35	0.02***
Services	0.56	0.59	-0.03***	0.55	0.57	-0.03*	0.56	0.58	-0.02***
North region (share)	0.20	0.21	-0.01	0.21	0.21	-0.00	0.20	0.20	-0.00
<i>Firm size:</i>									
Small firm (2–10)	0.33	0.17	0.16***	0.24	0.16	0.09***	0.35	0.17	0.18***
Medium firm (11–50)	0.36	0.36	-0.00	0.37	0.37	-0.00	0.36	0.36	-0.00
Large firm (> 50)	0.07	0.08	-0.01**	0.10	0.11	-0.00	0.06	0.06	-0.00
<i>Outcomes:</i>									
Weekly hours worked	50.23	49.82	0.40**	51.13	50.01	1.12***	49.98	49.74	0.25
Monthly wage (MXN)	5245.98	5385.95	-139.97***	4887.23	5221.96	-334.73***	5342.35	5430.04	-87.69*
Observations	192510			184250			190291		

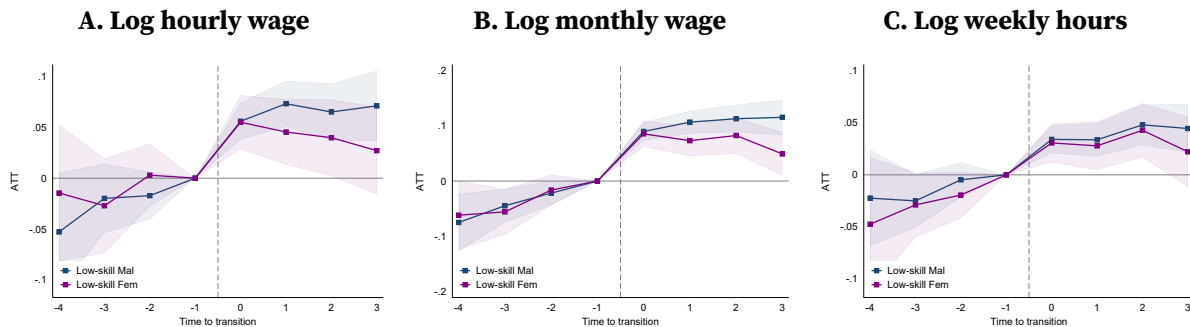
Source: Authors' calculations using ENOE 2005–2025. Notes: Means at $k = -1$ (last pre-treatment interview). Panel B control means are IPW-reweighted (logit propensity score, ATT).*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C.9. Heterogeneity by gender and skill

This appendix reports the formalisation and informalisation event studies disaggregated by gender and skill; the aggregate estimates appear in the main text. The within-firm formalisation premium is concentrated in low-skilled males. They experience a 5% hourly wage gain on impact that remains positive and significant throughout the post-treatment window. Low-skilled females begin at a comparable 5% on impact, but the effect fades over time and is statistically insignificant at 2.0% after three periods. High-skilled workers, both genders, show no statistically significant hourly change at any horizon. The take-home premium for low-skilled males is net of the $\sim 11\%$ employee-side tax wedge, so the implied gross-wage increase for this group is approximately 18%.

No demographic subgroup experiences a clean hourly wage gain from between-firm formalisation; the hourly effect is statistically insignificant on impact for every group. Monthly wages rise by 15–17% and weekly hours by 14–15% across groups, consistent with workers moving from a flexible informal schedule, often well below or above 48 hours per week, to the standard 48-hour formal workweek.

FIGURE C5. ATET of informal-to-formal transition ($I \rightarrow F$) within firms — Low-skilled

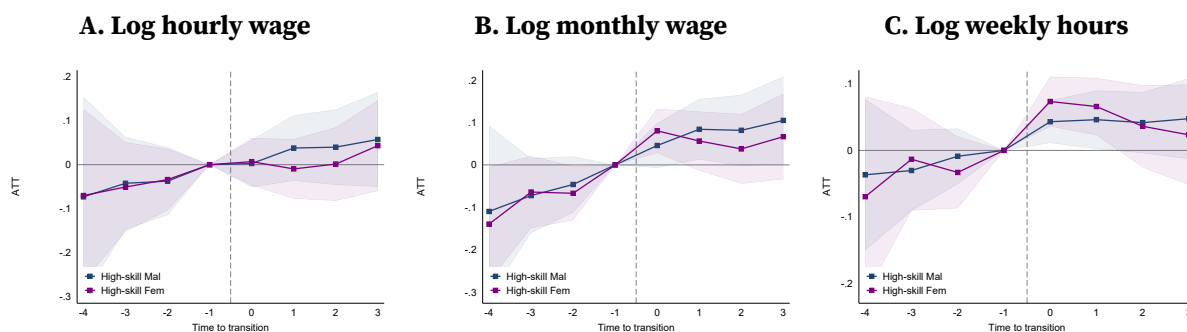


Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; shaded areas are 95% confidence intervals from bootstrapping standard errors.

Informalisation by gender and skill

The informalisation event studies by gender and skill are reported below. As in the aggregate, the formal-to-informal transition does not mirror a simple flexibility recovery: within-firm informalisations show hours and wages falling together rather than the hours increase that a voluntary move toward a preferred schedule would imply, consistent with firm-driven adjustment in which the firm withdraws formal registration and lowers both the assigned wage and scheduled hours. Between-firm informalisation remains ambiguous between a voluntary compensating-differential move, with the partial hours recovery

FIGURE C6. ATET of informal-to-formal transition ($I \rightarrow F$) within firms — High-skilled



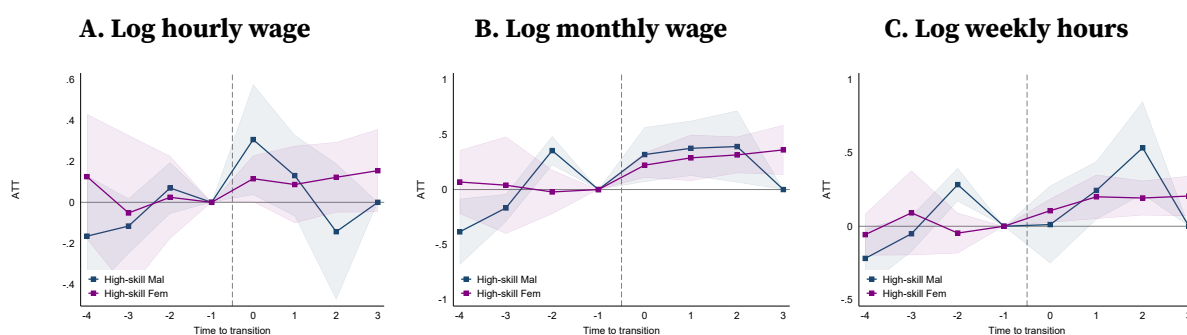
Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; shaded areas are 95% confidence intervals from bootstrapping standard errors.

FIGURE C7. ATET of informal-to-formal transition ($I \rightarrow F$) between firms — Low-skilled



Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; shaded areas are 95% confidence intervals from bootstrapping standard errors.

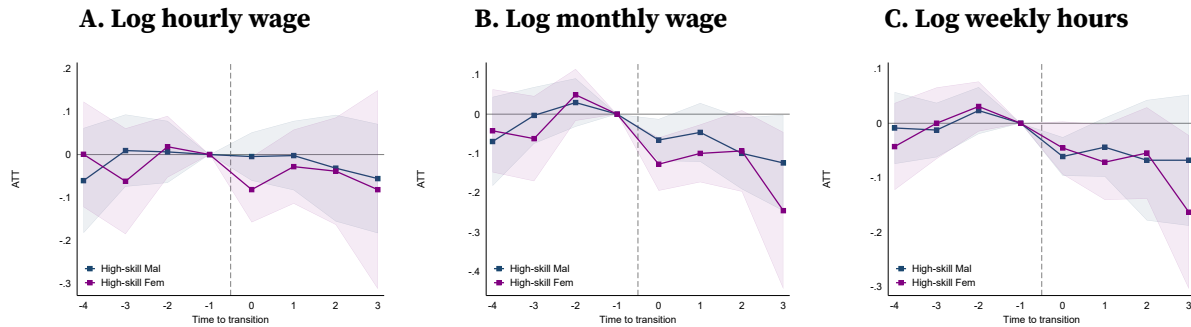
FIGURE C8. ATET of informal-to-formal transition ($I \rightarrow F$) between firms — High-skilled



Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust estimates; shaded areas are 95% confidence intervals from bootstrapping standard errors.

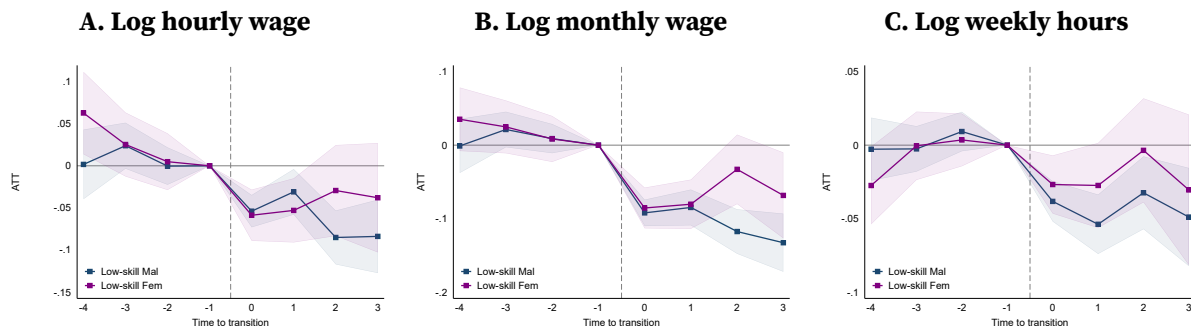
reflecting regained schedule choice, and involuntary displacement to a worse informal match.

FIGURE C9. ATET of formal-to-informal transition ($F \rightarrow I$) within firms — High-skilled



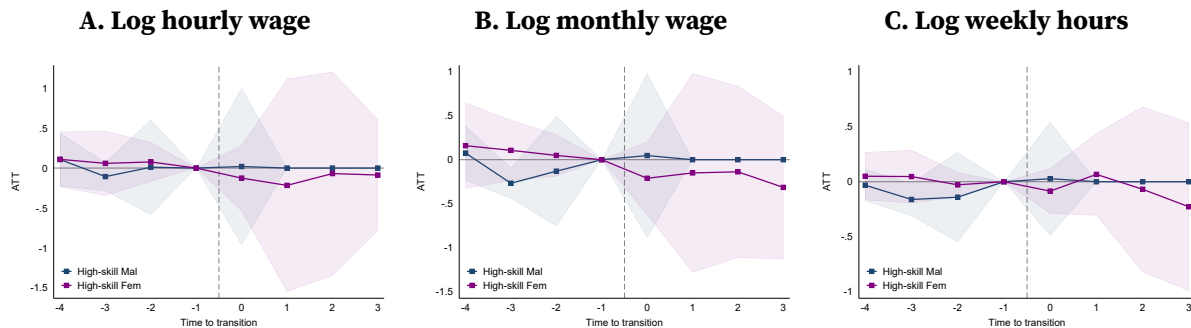
Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust IPW estimates; shaded areas are 95% confidence intervals from analytical standard errors clustered at the individual level.

FIGURE C10. ATET of formal-to-informal transition ($F \rightarrow I$) within firms — Low-skilled



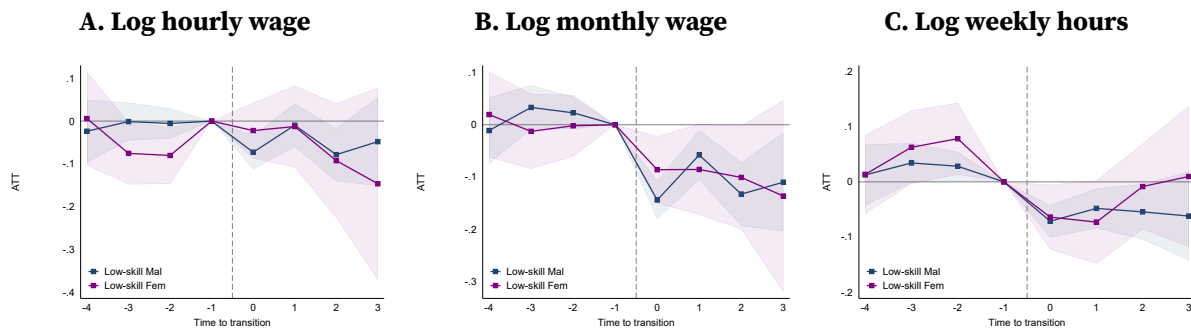
Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust IPW estimates; shaded areas are 95% confidence intervals from analytical standard errors clustered at the individual level.

FIGURE C11. ATET of formal-to-informal transition ($F \rightarrow I$) between firms — High-skilled



Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust IPW estimates; shaded areas are 95% confidence intervals from analytical standard errors clustered at the individual level.

FIGURE C12. ATET of formal-to-informal transition ($F \rightarrow I$) between firms — Low-skilled



Source: Author's estimates based on data from ENOE 2005–2025. Sample covers workers aged 20–60 years old. Doubly-robust IPW estimates; shaded areas are 95% confidence intervals from analytical standard errors clustered at the individual level.

C.10. Wage cuts over the distribution

Appendix D. Model derivations and diagnostics

This appendix collects the closed-form expressions derived from the Bellman equations of Section 3, the symmetric flow equation and companion expression that close the steady-state system, and the full set of theoretical transition rates whose canonical structure is illustrated in Section 3.2. It also reports the transition-probability diagnostics evaluated at the estimated parameters.

D.1. Closed-form reservation wages

The formal-to-informal reservation wage $R_{fi}(w)$ is the symmetric counterpart:

$$(D1) \quad \begin{aligned} R_{fi}(w) = & \frac{(r + \delta_i) a_f}{r + \delta_f} + \frac{(r + \delta_i) w}{r + \delta_f} - \frac{r(\delta_i - \delta_f)}{r + \delta_f} V_n \\ & + \left[\frac{r + \delta_i}{r + \delta_f} \lambda_{ff} - (\lambda_{if} + \eta_{if}) \right] \int_w^\infty V'_f(x) [1 - F_f(x)] dx \\ & + \left[\frac{r + \delta_i}{r + \delta_f} (\lambda_{fi} + \eta_{fi}) - \lambda_{ii} \right] \int_{R_{fi}(w)}^\infty V'_i(x) [1 - F_i(x)] dx. \end{aligned}$$

The reservation wages from non-employment follow from the indifference $V_f(R_{nf}) = V_n$ and $V_i(R_{ni}) = V_n$. The formal threshold is:

$$(D2) \quad \begin{aligned} R_{nf} = & -a_f + b + (\lambda_{nf} - \lambda_{ff}) \int_{R_{nf}}^\infty V'_f(x) [1 - F_f(x)] dx \\ & + (\lambda_{ni} - \lambda_{fi} - \eta_{fi}) \int_{R_{ni}}^\infty V'_i(x) [1 - F_i(x)] dx, \end{aligned}$$

and the informal threshold is the symmetric counterpart, differing only by the absence of the amenity term:

$$(D3) \quad \begin{aligned} R_{ni} = & b + (\lambda_{ni} - \lambda_{ii}) \int_{R_{ni}}^\infty V'_i(x) [1 - F_i(x)] dx \\ & + (\lambda_{nf} - \lambda_{if} - \eta_{if}) \int_{R_{nf}}^\infty V'_f(x) [1 - F_f(x)] dx. \end{aligned}$$

The marginal value functions appearing in the integrands are:

$$V'_i(x) = \frac{1}{r + \delta_i + \lambda_{ii} \bar{F}_i(x) + (\lambda_{if} + \eta_{if}) \bar{F}_f(R_{if}(x))}, \quad V'_f(x) = \frac{1}{r + \delta_f + \lambda_{ff} \bar{F}_f(x) + (\lambda_{fi} + \eta_{fi}) \bar{F}_i(R_{fi}(x))}.$$

Inverting equations (D2) and (D3) for a_f and b gives the recovery expressions used in Section 3.2.

D.2. Formal flow equation and companion expression

The flow equation for the formal accepted-wage distribution is the symmetric counterpart of equation (12); at every wage $w \geq R_{nf}$:

$$\begin{aligned}
 (D4) \quad & \underbrace{N_n \lambda_{nf} F_f(w)}_{\text{from } n} + \underbrace{N_i(\lambda_{if} + \eta_{if}) \int_{R_{ni}}^{R_{fi}(w)} [F_f(w) - F_f(R_{if}(x))] dG_i(x)}_{\text{from informal employment}} \\
 & = \underbrace{N_f \delta_f G_f(w)}_{\text{to } n} + \underbrace{N_f(\lambda_{fi} + \eta_{fi}) \int_{R_{nf}}^w \bar{F}_i(R_{fi}(x)) dG_f(x)}_{\text{to informal job}} + \underbrace{N_f \lambda_{ff} \bar{F}_f(w) G_f(w)}_{\text{to formal job}}.
 \end{aligned}$$

Evaluating equation (12) and rearranging for the term involving G_f at $R_{if}(w)$ produces the companion expression needed to close the system in equation (13):

$$(D5) \quad G_f(R_{if}(w)) = \frac{N_n [\lambda_{nf} F_f(R_{if}(w)) + \lambda_{ni} F_i(w)] - G_i(w) \cdot N_i [\delta_i + (\lambda_{if} + \eta_{if}) \bar{F}_f(R_{if}(w)) + \lambda_{ii} \bar{F}_i(w)]}{N_f [\delta_f + (\lambda_{fi} + \eta_{fi}) \bar{F}_i(w) + \lambda_{ff} \bar{F}_f(R_{if}(w))]}$$

The formal accepted-wage distribution G_f then follows from the inverse identity $R_{if}(R_{fi}(w)) = w$ applied to equation (D5).

D.3. Theoretical transition rates

The canonical hazard form developed in Section 3.2 for π_{ii} extends to the remaining employment-origin transition probabilities. For an informal worker, the between- and within-firm formalisation rates and the destruction rate to non-employment are:

$$\begin{aligned}
 \pi_{if}^{\text{between}} &= \int_{R_{ni}}^{\infty} \frac{\lambda_{if} \bar{F}_f(R_{if}(w))}{h_i(w)} (1 - e^{-h_i(w)}) g_i(w) dw, \\
 \pi_{if}^{\text{within}} &= \int_{R_{ni}}^{\infty} \frac{\eta_{if} \bar{F}_f(R_{if}(w))}{h_i(w)} (1 - e^{-h_i(w)}) g_i(w) dw, \\
 \pi_{in} &= \int_{R_{ni}}^{\infty} \frac{\delta_i}{h_i(w)} (1 - e^{-h_i(w)}) g_i(w) dw.
 \end{aligned}$$

For a formal worker, the total hazard is $h_f(w) = \delta_f + \lambda_{ff}\bar{F}_f(w) + (\lambda_{fi} + \eta_{fi})\bar{F}_i(R_{fi}(w))$, and the formal-to-formal ladder rate, the formal-to-informal between- and within-firm rates, and the destruction rate are:

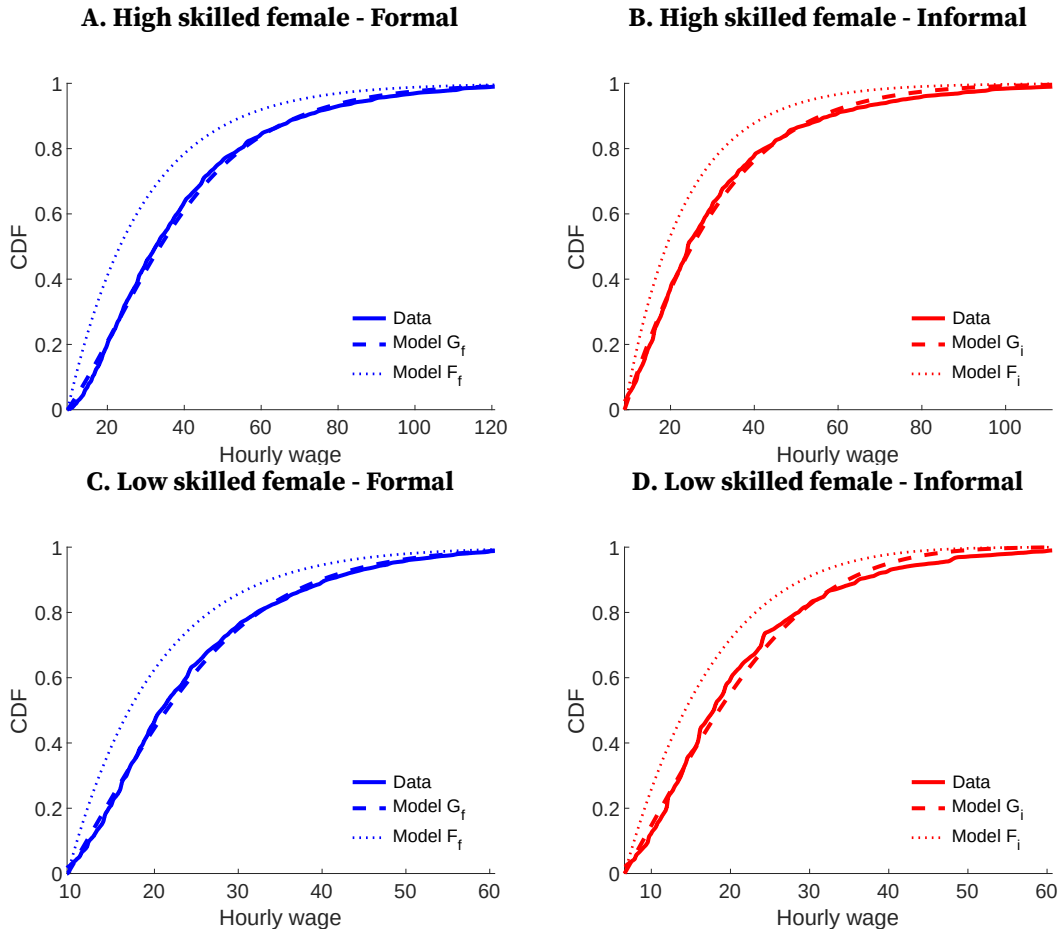
$$\pi_{ff} = \int_{R_{nf}}^{\infty} \frac{\lambda_{ff}\bar{F}_f(w)}{h_f(w)} (1 - e^{-h_f(w)}) g_f(w) dw,$$

$$\pi_{fi}^{\text{between}} = \int_{R_{nf}}^{\infty} \frac{\lambda_{fi}\bar{F}_i(R_{fi}(w))}{h_f(w)} (1 - e^{-h_f(w)}) g_f(w) dw,$$

$$\pi_{fi}^{\text{within}} = \int_{R_{nf}}^{\infty} \frac{\eta_{fi}\bar{F}_i(R_{fi}(w))}{h_f(w)} (1 - e^{-h_f(w)}) g_f(w) dw,$$

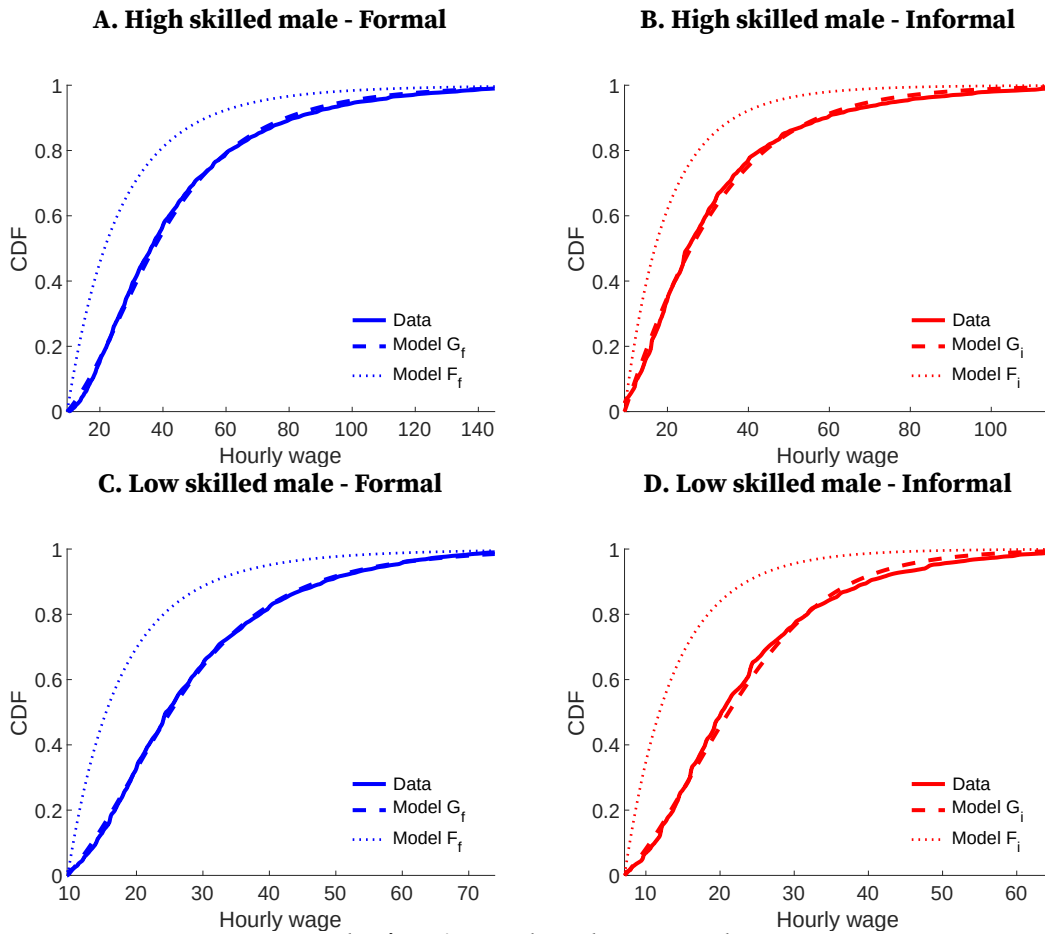
$$\pi_{fn} = \int_{R_{nf}}^{\infty} \frac{\delta_f}{h_f(w)} (1 - e^{-h_f(w)}) g_f(w) dw.$$

FIGURE D1. Model fit quantiles - Female



Source: Author's estimates based on ENOE data 2005-2025.

FIGURE D2. Model fit quantiles - Male



Source: Author's estimates based on ENOE data 2005-2025.

TABLE C5. Distribution of wage and hours changes at the transition (median)

	Median	Cut	Raise	% cut
Panel A. Informal to formal, within firm				
Log hourly wage	6.7	-20.3	26.0	41
Log monthly wage	8.4	-16.0	23.9	38
Log hours	0.0	-12.4	13.1	45
<i>Control: informal stayers (I→I)</i>				
Log hourly wage	2.2	-18.9	19.8	47
Log monthly wage	1.5	-17.7	18.0	46
Log hours	0.0	-14.3	14.4	49
Panel B. Informal to formal, between firms				
Log hourly wage	6.8	-23.2	27.8	43
Log monthly wage	13.5	-16.6	29.9	34
Log hours	3.2	-14.1	20.7	40
<i>Control: informal stayers (I→I)</i>				
Log hourly wage	2.2	-18.9	19.8	47
Log monthly wage	1.5	-17.7	18.0	46
Log hours	0.0	-14.3	14.4	49
Panel C. Formal to informal, within firm				
Log hourly wage	-3.0	-24.0	22.5	54
Log monthly wage	-4.1	-22.8	17.5	57
Log hours	-0.2	-14.0	11.8	50
<i>Control: formal stayers (F→F)</i>				
Log hourly wage	2.9	-17.1	18.4	45
Log monthly wage	2.4	-14.2	15.1	44
Log hours	0.0	-8.5	8.5	46
Panel D. Formal to informal, between firms				
Log hourly wage	-3.0	-25.1	26.5	53
Log monthly wage	-7.1	-27.7	22.8	58
Log hours	-0.6	-16.1	13.8	51
<i>Control: formal stayers (F→F)</i>				
Log hourly wage	2.9	-17.1	18.4	45
Log monthly wage	2.4	-14.2	15.1	44
Log hours	0.0	-8.5	8.5	46

Source: Author's estimates based on data from ENOE 2005–2025. The sample includes wage workers aged 20 to 60. Changes are 100 times the difference in the log outcome between the interviews after and before the transition (approximately percent). Median is the median change over the sample; the % cut column is the share of workers with a negative change; Cut and Raise are the median change among workers with a decrease and with an increase, respectively. Each panel also reports the corresponding event-study control group – informal stayers for formalisation and formal stayers for informalisation – as a benchmark for background wage volatility. Wage and hours levels are trimmed at the 1st and 99th percentiles. Within- and between-firm are defined by tenure at the establishment. Estimates are survey-weighted.

TABLE C6. Distribution of wage and hours changes at the transition (mean)

	Mean	Cut	Raise	% cut
Panel A. Informal to formal, within firm				
Log hourly wage	7.3	-29.8	33.4	41
Log monthly wage	11.7	-23.8	33.1	38
Log hours	3.0	-17.0	22.0	45
<i>Control: informal stayers (I→I)</i>				
Log hourly wage	2.1	-26.1	26.9	47
Log monthly wage	2.1	-26.0	26.0	46
Log hours	-0.1	-21.7	21.7	49
Panel B. Informal to formal, between firms				
Log hourly wage	6.5	-32.8	36.5	43
Log monthly wage	18.4	-24.7	40.9	34
Log hours	11.1	-19.2	34.2	40
<i>Control: informal stayers (I→I)</i>				
Log hourly wage	2.1	-26.1	26.9	47
Log monthly wage	2.1	-26.0	26.0	46
Log hours	-0.1	-21.7	21.7	49
Panel C. Formal to informal, within firm				
Log hourly wage	-2.9	-31.9	30.5	54
Log monthly wage	-7.9	-32.1	24.2	57
Log hours	-3.8	-22.3	17.0	50
<i>Control: formal stayers (F→F)</i>				
Log hourly wage	2.7	-24.1	24.8	45
Log monthly wage	2.6	-21.0	21.3	44
Log hours	-0.0	-12.0	12.0	46
Panel D. Formal to informal, between firms				
Log hourly wage	-1.8	-34.9	35.4	53
Log monthly wage	-9.1	-36.4	29.0	58
Log hours	-6.1	-27.3	18.4	51
<i>Control: formal stayers (F→F)</i>				
Log hourly wage	2.7	-24.1	24.8	45
Log monthly wage	2.6	-21.0	21.3	44
Log hours	-0.0	-12.0	12.0	46

Source: Author's estimates based on data from ENOE 2005–2025. The sample includes wage workers aged 20 to 60. Changes are 100 times the difference in the log outcome between the interviews after and before the transition (approximately percent). Mean is the mean change over the sample; the % cut column is the share of workers with a negative change; Cut and Raise are the mean change among workers with a decrease and with an increase, respectively. Each panel also reports the corresponding event-study control group – informal stayers for formalisation and formal stayers for informalisation – as a benchmark for background wage volatility. Wage and hours levels are trimmed at the 1st and 99th percentiles. Within- and between-firm are defined by tenure at the establishment. Estimates are survey-weighted.

TABLE D1. Model fit: Theoretical rates and data π

	Female				Male			
	Skilled		Unskilled		Skilled		Unskilled	
	Data	Model	Data	Model	Data	Model	Data	Model
Panel A. Transitions rates (π)								
$f \rightarrow f$	0.068	0.068	0.068	0.068	0.062	0.062	0.057	0.057
$f \rightarrow i$	0.013	0.013	0.014	0.014	0.015	0.015	0.019	0.019
$f \rightarrow i$ (<i>within</i>)	0.044	0.044	0.042	0.043	0.051	0.051	0.081	0.081
$f \rightarrow u$	0.277	0.277	0.295	0.295	0.175	0.175	0.144	0.144
$i \rightarrow i$	0.101	0.101	0.112	0.112	0.123	0.123	0.160	0.160
$i \rightarrow f$	0.033	0.033	0.018	0.018	0.034	0.034	0.019	0.019
$i \rightarrow f$ (<i>within</i>)	0.094	0.094	0.050	0.050	0.113	0.113	0.074	0.074
$i \rightarrow u$	0.544	0.544	0.581	0.581	0.390	0.390	0.290	0.290
$u \rightarrow i$	0.048	0.048	0.035	0.035	0.119	0.119	0.274	0.274
$u \rightarrow f$	0.042	0.042	0.017	0.017	0.096	0.096	0.102	0.102
Panel B. Employment shares								
Formal	13.5	11.5	6.1	4.6	32.1	31.7	31.3	30.5
Informal	6.8	3.9	5.3	3.0	14.6	11.2	32.8	30.8
Non-employment	79.8	84.6	88.6	92.4	53.3	57.1	35.9	38.7

Source: Author's estimates based on data from ENOE 2005–2025.

D.4. Computation of home production and amenity

The model has four reservation wages. The two between-sector thresholds, $R_{if}(w)$ and $R_{fi}(w)$, depend on the worker's current wage and solve the differential equation (10). The other two are the wages at which a non-employed worker accepts a formal or an informal offer, R_{nf} and R_{ni} . I estimate both jointly with the offer distributions, as the location parameters of F_f and F_i respectively: each is the lowest wage in its sector that a worker accepts on leaving non-employment. Both are therefore known once the model is estimated.

The flow value of non-employment b and the amenity a_f do not enter the transition moments, so the estimation step does not identify them. I recover them afterwards from the two reservation-wage equations of non-employment, (D2) and (D3), which form a system of two equations in the two unknowns.

$$(D6) \quad \begin{aligned} b = R_{ni} &- (\lambda_{ni} - \lambda_{ii}) \int_{R_{ni}}^{\infty} V'_i(x) [1 - F_i(x)] dx \\ &- (\lambda_{nf} - \lambda_{if} - \eta_{if}) \int_{R_{nf}}^{\infty} V'_f(x) [1 - F_f(x)] dx. \end{aligned}$$

$$(D7) \quad \begin{aligned} a_f = -R_{nf} &+ b + (\lambda_{nf} - \lambda_{ff}) \int_{R_{nf}}^{\infty} V'_f(x) [1 - F_f(x)] dx \\ &+ (\lambda_{ni} - \lambda_{fi} - \eta_{fi}) \int_{R_{ni}}^{\infty} V'_i(x) [1 - F_i(x)] dx, \end{aligned}$$

The informal threshold carries no amenity, so (D6) gives b directly; substituting b into the formal threshold (D7) then gives a_f . These are the expressions reported as Equations (15) and (16) in the main text. I solve the integrals numerically with the trapezoidal rule on a grid over the support $[R_{ns}, \bar{w}_s]$, for $s = \{f, i\}$.

D.5. How much the amenity depends on the formal reservation wage

The amenity is a residual, and a small one. Equation (D7) recovers it as $a_f = -R_{nf} + b +$ (option-value integrals), so R_{nf} enters with a coefficient of minus one and the recovered a_f is a difference between quantities several times its own size. Whatever pins down R_{nf} therefore deserves to be examined rather than assumed, and this appendix asks how far the amenity moves when it is treated differently.

The main text estimates R_{nf} as a free parameter: it is the location parameter of the formal offer distribution F_f , identified off the lower formal wage quantiles exactly as the informal location parameter already is. The alternative, followed by [Meghir et al., 2015](#) and adopted in an earlier version of this paper, is to pin it at the legal minimum wage, here 9.65 pesos per hour, the full-sample median of the statutory floor in real hourly terms. That convention is not innocuous: it imposes the level of the very object from which b and a_f are then backed out, so the reservation wage is treated as an optimal decision in the derivation while being set by assumption in the estimation. Table D2 reports the estimates under both, not to offer the reader a choice between them but to bound how much of the amenity is a consequence of the convention. Both panels come from a common earlier estimation run, so their levels differ slightly from Table 5, which reports the current one, and they are normalised by the median formal wage rather than by the median informal wage used there. What the appendix is for is the comparison between the panels, which is internally consistent because they share a vintage, rather than the levels themselves. Confidence intervals are not reported here; those for the estimated specification are in Table 5.

Three results follow. First, the estimated formal reservation wage lies *at or above* the legal floor in every group, between 9.67 and 11.21 pesos per hour, rather than below it. For skilled men it is indistinguishable from the floor (9.67); for the remaining three groups it sits 12 to 16% above. The data therefore do not reject the pinned value so much as mildly overshoot it. Second, the flow value of non-employment is essentially unchanged, moving by less than 1.4 pesos per hour in every group and by less than two percentage points of the median formal wage in three of the four.

Third, and this is what the exercise is for, the ordering of the amenity survives but its level does not. Under both specifications a_f is ordered skilled men > skilled women > 0 > unskilled men > unskilled women. What moves is the magnitude, and for unskilled men it moves a long way: from -10.9% of the median formal wage under the pinned floor to -2.7% when R_{nf} is estimated. That is the difference between a group that places a substantial negative value on the formal contract and one whose valuation cannot be told apart from zero. The reading taken in the main text, that the sign of the low-skilled male amenity is not identified by these data, is therefore a conclusion about the fragility of that estimate rather

than a claim to have measured it precisely. The three amenities that are far from zero, both skilled groups and unskilled women, are insensitive to the convention by comparison.

The practical implication is that the pinned specification was close to correct in level but for the wrong reason. It imposed a binding statutory floor on a labour market in which, as Section 3.2 discusses, the minimum wage functions largely as a wage-setting norm rather than as a constraint that binds formal hiring. The free specification recovers a reservation wage near the floor because that is where workers' outside options place it, not because the law puts it there.

TABLE D2. Sensitivity of the amenity to the formal reservation wage

	Female		Male	
	Skilled	Unskilled	Skilled	Unskilled
Median formal hourly wage (reference)				
<i>Pesos per hour</i>	29.82	19.39	32.09	22.36
Panel A. Estimated R_{nf} (formal reservation wage)				
Home production (b)				
<i>Pesos per hour</i>	26.910	14.049	32.822	19.757
<i>% of formal median</i>	90.2%	72.4%	102.3%	88.4%
Amenity (a_f)				
<i>Pesos per hour</i>	3.389	-3.612	8.555	-0.600
<i>% of formal median</i>	11.4%	-18.6%	26.7%	-2.7%
Reservation wage (R_{nf})				
<i>Pesos per hour</i>	11.213	10.844	9.666	10.815
<i>% of formal median</i>	37.6%	55.9%	30.1%	48.4%
Reservation wage (R_{ni})				
<i>Pesos per hour</i>	8.681	5.989	8.836	7.651
<i>% of formal median</i>	29.1%	30.9%	27.5%	34.2%
Panel B. Fixed R_{nf} at the minimum wage				
Home production (b)				
<i>Pesos per hour</i>	26.370	14.216	32.744	21.147
<i>% of formal median</i>	88.4%	73.3%	102.1%	94.6%
Amenity (a_f)				
<i>Pesos per hour</i>	4.096	-2.652	8.592	-2.446
<i>% of formal median</i>	13.7%	-13.7%	26.8%	-10.9%
Reservation wage (R_{nf})				
<i>Pesos per hour</i>	9.650	9.650	9.650	9.650
<i>% of formal median</i>	32.4%	49.8%	30.1%	43.2%
Reservation wage (R_{ni})				
<i>Pesos per hour</i>	8.989	6.460	9.248	6.734
<i>% of formal median</i>	30.1%	33.3%	28.8%	30.1%

Source: Author's estimates based on data from ENOE 2005 – 2025. Panel A reports the specification used throughout the main text, in which the formal reservation wage is estimated jointly with the offer distributions as the location parameter of the formal offer distribution. Panel B fixes it at 9.65 pesos per hour, the full-sample median of the statutory minimum wage in real hourly terms. Home production and the amenity are recovered as residuals from the two non-employment reservation-wage conditions in both panels; all other elements of the estimation are identical. The median formal hourly wage is a feature of the data and is common to both panels.

D.6. Multiple shocks

The waiting times reported are constructed holding all other shocks constant, that is, they reflect the expected time until a single Poisson clock fires in isolation, $1/(4\lambda)$. However, a worker faces multiple competing shocks simultaneously. For an informal worker at wage w , the total hazard aggregates all competing Poisson shocks:

$$(D8) \quad h_i(w) = \delta_i + \lambda_{ii}\bar{F}_i(w) + (\lambda_{if} + \eta_{if})\bar{F}_f(R_{if}(w)).$$

Since the minimum of independent exponential random variables is itself exponential with rate equal to the sum of individual rates, the time until the next transition of any kind is exponentially distributed with rate $h_i(w)$. The expected spell length conditional on current wage w is therefore:

$$(D9) \quad \mathbf{E}(T \mid w) = \frac{1}{h_i(w)},$$

which is strictly shorter than any single-shock waiting time $1/\lambda$ since $h_i(w) > \lambda$ for every component λ . Integrating over the distribution of accepted wages yields the unconditional expected spell length:

$$(D10) \quad \mathbf{E}(T) = \int_{R_{ni}}^{\infty} \frac{1}{h_i(w)} g_i(w) dw.$$

An analogous expression holds for formal workers, replacing $h_i(w)$ with $h_f(w) = \delta_f + \lambda_{ff}\bar{F}_f(w) + (\lambda_{fi} + \eta_{fi})\bar{F}_i(R_{fi}(w))$. For non-employed workers the hazard is wage-independent, $h_n = \lambda_{nf} + \lambda_{ni}$, so the expected non-employment spell is $\mathbf{E}(T_n) = 1/(4h_n)$ years, which is directly computable from Table 4 without integration.

This quantity can be computed numerically given the estimated parameters: for each quadrature point w in the support, $h_s(w)$ is computed, then its inverse, and integrated against $g_s(w)$. The overall quarterly exit probability from informal employment satisfies the adding-up restriction:

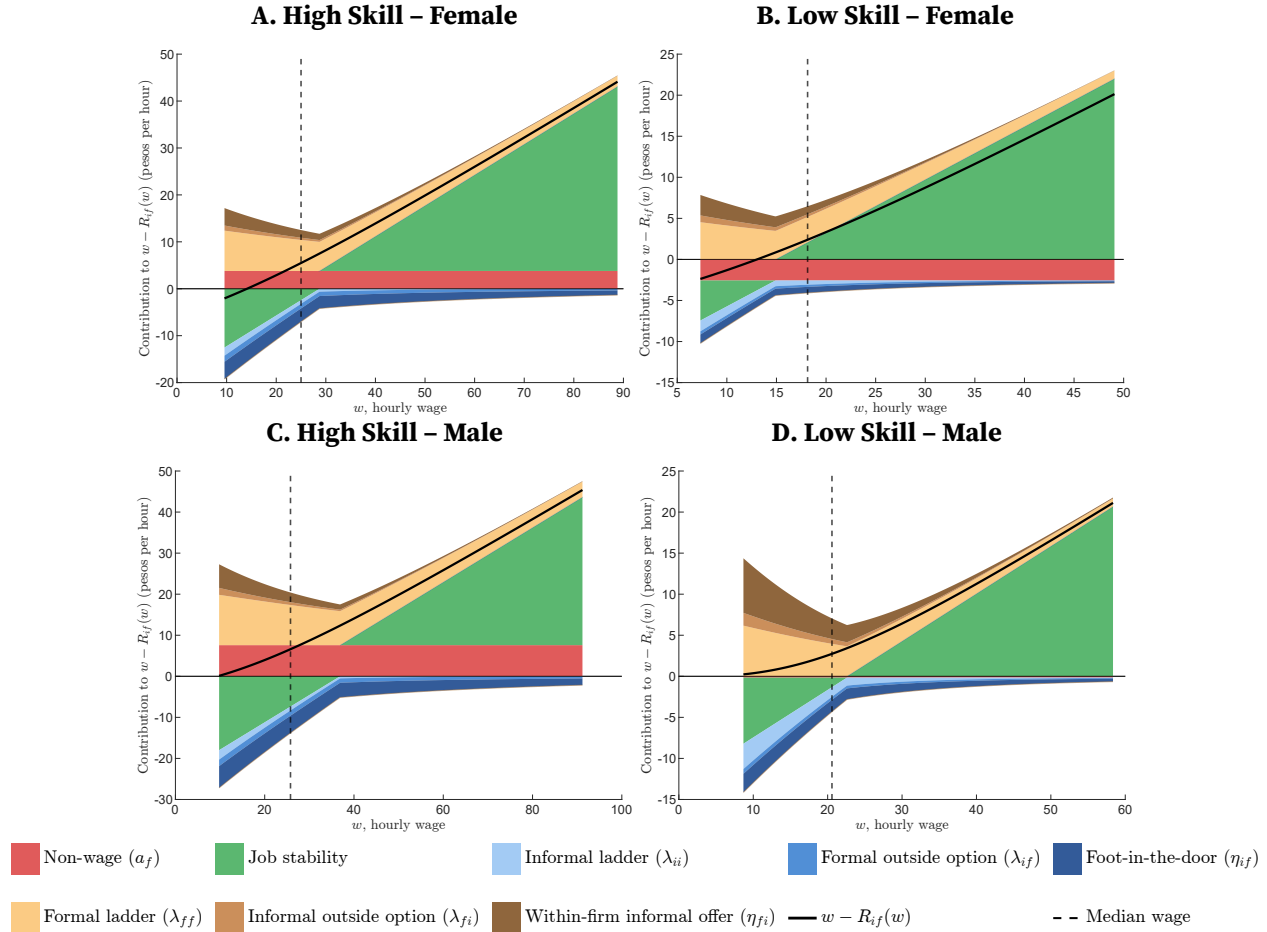
$$(D11) \quad P(\text{exit} \mid i) = \int_{R_{ni}}^{\infty} \left(1 - e^{-h_i(w)}\right) g_i(w) dw = \pi_{ii} + \pi_{if}^{\text{between}} + \pi_{if}^{\text{within}} + \pi_{in},$$

and analogously for formal employment. These restrictions provide a numerical check on the estimated parameters, and the implied average spell length serves as a summary measure of job stability that accounts for all competing exit routes simultaneously.

D.7. Full decomposition of the value of a formal job

Table D3 and Figure D3 report the value-of-formality gap $w - R_{if}(w)$ split into its full eight channels. The main-text Figure 12 aggregates the six ladder pieces into two: the *informal job ladder* sums the own-sector term ($-\lambda_{ii}$), the between-firm informal outside option (λ_{fi}) and the within-firm informal offer (η_{fi}); the *formal job ladder* sums the between-firm formal outside option ($-\lambda_{if}$), the within-firm foot-in-the-door ($-\eta_{if}$) and the own-sector formal term (λ_{ff}). The within-firm components η_{if} and η_{fi} are reported separately here.

FIGURE D3. Full decomposition of $w - R_{if}(w)$



Source: Author's estimates based on ENOE 2005–2025. Eight-channel version of the main-text Figure 12; each panel decomposes $w - R_{if}(w)$ across the informal wage support, with the six ladder pieces shown separately. The dashed line marks the median informal wage.

TABLE D3. Full decomposition of the value of a formal job $w - R_{if}(w)$

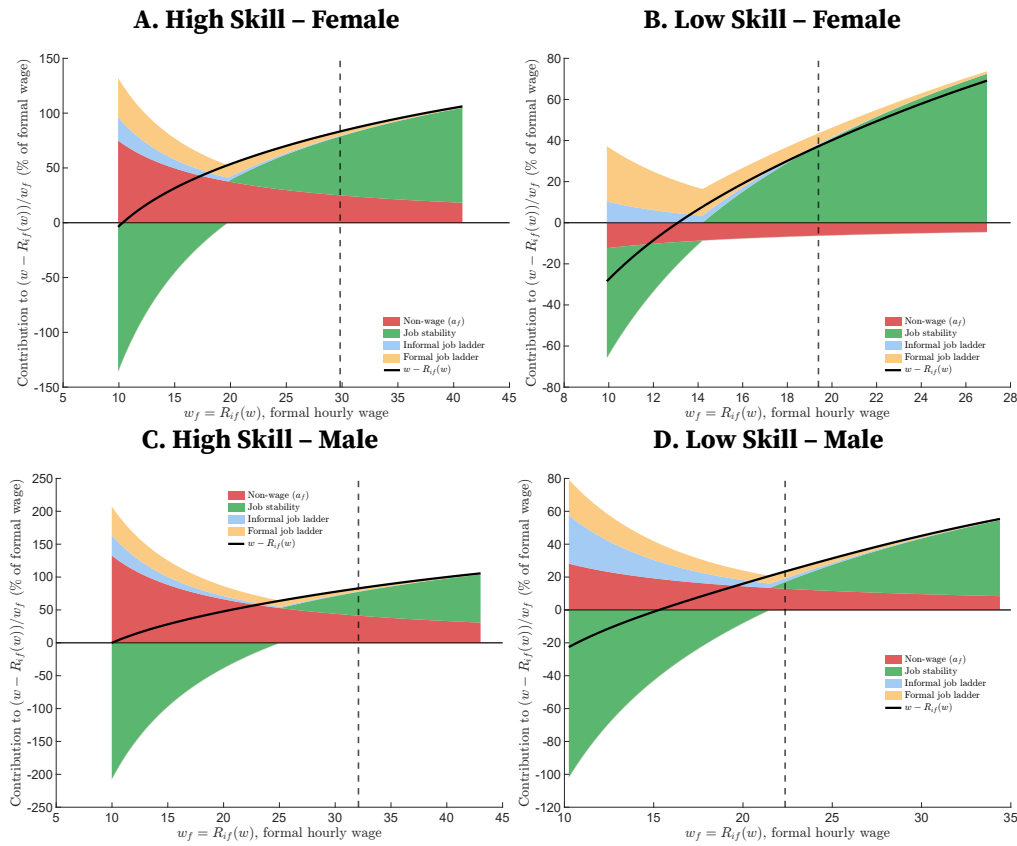
	Female		Male	
	Skilled	Unskilled	Skilled	Unskilled
<i>Median informal wage (reference)</i>				
<i>Pesos per hour</i>	25.00	18.14	25.78	20.58
<i>$R_{if}(w)$ at the median wage</i>				
<i>Pesos per hour</i>	19.52	15.74	19.17	17.84
<i>Value of formality $w - R_{if}(w)$</i>				
<i>Pesos per hour</i>	5.48	2.40	6.60	2.74
<i>% of informal median</i>	21.9%	13.2%	25.6%	13.3%
<i>Non-wage (a_f)</i>				
<i>Pesos per hour</i>	3.79	-2.56	7.59	-0.16
<i>% of informal median</i>	15.2%	-14.1%	29.4%	-0.8%
<i>Job stability</i>				
<i>Pesos per hour</i>	-2.44	2.08	-7.38	-1.17
<i>% of informal median</i>	-9.8%	11.5%	-28.6%	-5.7%
<i>Informal ladder (λ_{ij})</i>				
<i>Pesos per hour</i>	-0.76	-0.52	-0.94	-1.16
<i>% of informal median</i>	-3.0%	-2.9%	-3.6%	-5.7%
<i>Formal outside option (λ_{if})</i>				
<i>Pesos per hour</i>	-1.00	-0.26	-1.27	-0.37
<i>% of informal median</i>	-4.0%	-1.5%	-4.9%	-1.8%
<i>Foot-in-the-door (η_{if})</i>				
<i>Pesos per hour</i>	-2.84	-0.74	-4.22	-1.45
<i>% of informal median</i>	-11.3%	-4.1%	-16.4%	-7.1%
<i>Formal ladder (λ_{ff})</i>				
<i>Pesos per hour</i>	6.60	3.09	9.77	3.95
<i>% of informal median</i>	26.4%	17.0%	37.9%	19.2%
<i>Informal outside option (λ_{fi})</i>				
<i>Pesos per hour</i>	0.48	0.33	0.69	0.59
<i>% of informal median</i>	1.9%	1.8%	2.7%	2.9%
<i>Within-firm informal offer (η_{fi})</i>				
<i>Pesos per hour</i>	1.63	0.98	2.36	2.51
<i>% of informal median</i>	6.5%	5.4%	9.2%	12.2%

Source: Author's estimates based on data from ENOE 2005–2025. Components evaluated at the median informal wage, in pesos per hour and as a share of the wage. Eight-channel version of the four-channel aggregation plotted in the main-text Figure 12.

D.8. Value of a formal job under an alternative normalisation

Every percentage reported in the paper is expressed as a share of the median informal wage, the wage a formalising worker gives up. Table D4 and Figure D4 report the same decomposition under the alternative normalisation, dividing instead by the median hourly wage over formal and informal jobs pooled. The channels in pesos are unchanged; only the denominator differs, so the ordering and the sign of every channel are the same and only the scale of the percentages moves. The figure plots the channels as a share of that pooled median, over the formal wage the worker obtains, $R_{if}(w)$.

FIGURE D4. Decomposition of $w - R_{if}(w)$ over the pooled median wage



Source: Author's estimates based on ENOE 2005–2025. Formal-wage counterpart of the main-text Figure 12: each panel decomposes $w - R_{if}(w)$ as a share of the median hourly wage over formal and informal jobs pooled, plotted over the formal wage the worker obtains, $R_{if}(w)$; the dashed line marks that pooled median.

TABLE D4. Decomposition of $w - R_{if}(w)$ as a share of the pooled median wage

	Female		Male	
	Skilled	Unskilled	Skilled	Unskilled
Median hourly wage, formal and informal (reference)				
<i>Pesos per hour</i>	29.69	19.78	31.88	22.47
Value of formality $w - R_{if}(w)$				
<i>Pesos per hour</i>	5.48	2.40	6.60	2.74
	[4.61, 5.60]	[1.81, 2.58]	[6.30, 6.81]	[2.63, 2.91]
<i>% of median wage</i>	18.5%	12.1%	20.7%	12.2%
Non-wage (a_f)				
<i>Pesos per hour</i>	3.79	-2.56	7.59	-0.16
	[2.87, 4.34]	[-3.09, -2.28]	[6.62, 8.30]	[-0.28, 0.33]
<i>% of median wage</i>	12.8%	-13.0%	23.8%	-0.7%
Job stability (non-employment)				
<i>Pesos per hour</i>	-2.44	2.08	-7.38	-1.17
	[-2.99, -2.19]	[2.01, 2.36]	[-8.07, -7.10]	[-1.36, -1.16]
<i>% of median wage</i>	-8.2%	10.5%	-23.1%	-5.2%
Informal job option				
<i>Pesos per hour</i>	1.36	0.79	2.12	1.94
	[1.26, 1.49]	[0.65, 0.81]	[1.99, 2.22]	[1.74, 2.00]
<i>% of median wage</i>	4.6%	4.0%	6.6%	8.6%
Formal job option				
<i>Pesos per hour</i>	2.77	2.09	4.28	2.13
	[2.45, 3.06]	[1.89, 2.18]	[3.94, 4.89]	[2.03, 2.26]
<i>% of median wage</i>	9.3%	10.6%	13.4%	9.5%

Source: Author's estimates based on data from ENOE 2005–2025. Same channels and pesos as Table D3 (evaluated at the median informal wage), but each expressed as a share of the median hourly wage over formal and informal jobs pooled, rather than of the median informal wage used in Table D3 and throughout the main text. Brackets report 95% bootstrap percentile confidence intervals from the 100 replications described in Table 4, propagated through the reservation-wage solution and taken over the draws after the channels are aggregated, so each interval is the interval on that channel rather than a sum of bounds. The job-stability channel has no separately stored draw and is recovered per draw as the residual of the additive decomposition.